KLT FANO VARIETIES WHOSE ALPHA INVARIANTS ARE NOT COMPUTED BY DIVISORS

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Abstract. We disprove the conjecture of Cheltsov and Shramov that the infimum defining the alpha invariant of a klt Fano variety is a minimum, and answer negatively a question of Zhuang whether the alpha invariant of a polarized klt pair is always computed by a prime divisor. Our counterexample is a klt Fano variety whose alpha invariant is irrational and is not computed by any prime divisor over the variety, but is computed by a quasi-monomial valuation of rational rank two.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Tian introduced the alpha invariant of a Fano manifold [Tia87] in the study of Kähler–Einstein metrics. For a klt Fano variety X, the alpha invariant of X, also called the global log canonical threshold of X [CS08, Definition 1.2], is

$$\alpha(X) = \inf\{\mathrm{lct}(X; D) \mid D \geq 0 \text{ is a } \mathbb{Q}\text{-divisor on } X \text{ such that } D \sim_{\mathbb{Q}} -K_X\},$$

where $\mathrm{lct}(X; D)$ is the log canonical threshold of D with respect to X (see Definition 2.4). When X is smooth, $\alpha(X)$ is equal to the analytic invariant of Tian [CS08, Theorem A.3].

For a Fano manifold X, Tian defined analytic invariants of X through the sections of $-mK_X$ for positive integers m, and asked whether they are equal to $\alpha(X)$ for any sufficiently large m [Tia90, Question 1]. [Bir21, Section 1] recalls Tian's question as asking whether the infimum defining $\alpha(X)$ is a minimum. Cheltsov and Shramov formulated the question for klt Fano varieties, together with the rationality of $\alpha(X)$, as follows.

Conjecture 1.1 ([CS08, Conjecture 1.12])**.** *Let X be a klt Fano variety. Then there exists a $\mathbb{Q}$-divisor $D \geq 0$ on X such that $D \sim_{\mathbb{Q}} -K_X$ and $\alpha(X) = \mathrm{lct}(X; D) \in \mathbb{Q}$.*

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Tian also considered the alpha invariant of polarized manifolds. Let (X, B) be a projective klt pair and L an ample line bundle on X, viewed as a Cartier divisor. For any positive integer m, let

$$\alpha_m(X, B; L) := \inf\{\mathrm{lct}(X, B; \tfrac{1}{m}D) \mid D \in |mL|\}$$

[Zhu20, Definition 2.1]. In [Zhu20, Definition 2.1] (the case $k = 1$), the infimum is taken over all linear subsystems $M \subset |mL|$, which is equal to the infimum above, since the multiplicity of M along any prime divisor over X is the minimum of the multiplicities of the members of M, and each member of $|mL|$ is a linear subsystem. For any $\mathbb{Q}$-divisor $D \geq 0$ such that $D \sim_{\mathbb{Q}} L$, there exists a positive integer m such that $mD \in |mL|$. Thus

$$\alpha(X, B; L) = \inf_{m \geq 1} \alpha_m(X, B; L),$$

where $\alpha(X, B; L)$ is as in Definition 2.4. Tian's stabilization conjecture [Tia12, Conjecture 5.3] predicts that, for any polarized manifold (X, L), the infimum $\inf_{m\geq 1} \alpha_m(X, 0; L)$ is a minimum. Zhuang formulated the conjecture for klt pairs: there exists a positive integer m such that $\alpha(X, B; L) = \alpha_m(X, B; L)$ [Zhu20, Conjecture A.1].

Zhuang also observed that [Zhu20, Conjecture A.1] does not hold for arbitrary polarized klt pairs [Zhu20, Remark A.7, Example A.8]. In [Zhu20, Example A.8], $B = 0$ and $X = \mathbb{P}_C(\mathcal{E})$, where C is a curve of genus at least 2 and $\mathcal{E}$ is a stable vector bundle of rank 2 and degree 0 on C such that $L_0 = \mathcal{O}_{\mathbb{P}_C(\mathcal{E})}(1)$ is pseudo-effective but not numerically equivalent to any $\mathbb{Q}$-divisor $G \geq 0$, and $L = L_0 + aF$, where F is a fiber and $a \geq 2$ is an integer. It is shown there that $\alpha(X, 0; L) = 1/a$ is computed by a prime divisor over X.

Let $D \geq 0$ be a $\mathbb{Q}$-divisor such that $D \equiv L$. For any fiber F, writing $D = cF + D_0$ with F not contained in $\operatorname{Supp} D_0$, we have $c < a$ since L_0 is not numerically equivalent to any $\mathbb{Q}$-divisor $G \geq 0$, and since $(D_0 \cdot F) = (L \cdot F) = 1$ and $a \geq 2$, inversion of adjunction shows that (X, tD) is lc near F for $t = \min\{1, 1/c\} > 1/a$ (with $t = 1$ when $c = 0$). Since $c > 0$ for only finitely many fibers F, we have $\mathrm{lct}(X; D) > 1/a$ [Zhu20, Example A.8]. Thus [Zhu20, Example A.8] shows that $\alpha(X, 0; L)$ need not be attained by a $\mathbb{Q}$-divisor $D \geq 0$ such that $D \equiv L$, while it is still computed by a prime divisor over X. Moreover, in neither [Zhu20, Remark A.7] nor [Zhu20, Example A.8] is L a multiple of $-K_X$. Zhuang then asked the following question.

Question 1.2 ([Zhu20, Question A.9])**.** Let (X, B) be a projective klt pair such that B is a $\mathbb{Q}$-divisor. Let L be an ample line bundle on X. Does there exist a prime divisor E over X such that $\alpha(X, B; L) = a(E, X, B)/T_L(E)$?

Here $a(E, X, B)$ is the log discrepancy of E with respect to (X, B), and $\alpha(X, B; L)$ and the maximal vanishing order $T_L(E)$ of L along E are as in Definition 2.4. By Lemma 2.5(3), $T_L(E)$ is equal to the pseudo-effective threshold of L with respect to E used in [Zhu20], computed on any normal $\mathbb{Q}$-factorial projective higher model of X on which E is a divisor, for example a log resolution of X with projective source on which E is a divisor.

Conjecture 1.1 is known in the following cases.

(1) [Bir21, Theorem 1.7] proved Conjecture 1.1 when $\alpha(X) \leq 1$, and also the analogous statement for klt log Fano pairs.

(2) [Che08, Theorem 1.7] computes $\alpha(X)$ for any smooth del Pezzo surface X, and the paragraph after it deduces [Che08, Conjecture 1.6] for such X and the trivial group, which implies Conjecture 1.1 for such X, since the values of $\alpha(X)$ in [Che08, Theorem 1.7] are rational, and numerical and $\mathbb{Q}$-linear equivalence coincide on X by [Laz04, Remark 1.1.20], as $H^1(X, \mathcal{O}_X) = 0$.

(3) [Zhu20, Theorem A.2] proved Conjecture 1.1 for smooth Fano complete intersections whose dimension is large compared with the codimension ([Zhu20, Theorem A.2] gives $\alpha(X, 0; H) = \alpha_m(X, 0; H)$ for the hyperplane class H and some positive integer m, which is attained by a member of $|mH|$ by [Amb16, Theorem 3.6], and $\alpha(X) = \alpha(X, 0; H)/(-s)$ for $K_X \sim sH$ by Lemma 2.5(4), so $\alpha(X)$ is the log canonical threshold of a $\mathbb{Q}$-divisor, which is rational [KM98, Lemma 2.30 and Corollary 2.31(3)]).

See also [Jin24, Theorem 1.4], which gives an ample line bundle L on a smooth projective surface X such that the sequence $\alpha_m(X,0;L)$ is not eventually constant, although $\alpha_m(X,0;L) = \alpha(X,0;L)$ for any even m [Jin24, Theorem 6.1], and leaves open the case where X is a Fano manifold and $L = -K_X$, [ACS18, Theorems 1.2 and 1.3] for the stronger [Tia12, Conjecture 5.4], and [LZ21, Theorems 1.3 and 1.4] and [JR25, Theorems 1.4 and 1.6] for the torus-equivariant version of Tian's stabilization conjecture on toric Fano manifolds, which considers the invariants $\alpha_{k,G}$ and $\alpha_{k,m,G}$ for $L = -K_X$ and a group G containing the compact torus $(S^1)^n$, not the invariants $\alpha_m(X,0;L)$ above.

We now explain why the method of [Bir21, Theorem 1.7] does not reach the case $\alpha(X) > 1$. Let X be a klt Fano variety, $D \geq 0$ an $\mathbb{R}$-divisor such that $D \sim_{\mathbb{R}} -K_X$, and $t = \mathrm{lct}(X;D)$. When $t < 1$, the argument for [Bir21, Proposition 3.4] chooses a general $\mathbb{R}$-divisor $H \geq 0$ such that $H \sim_{\mathbb{R}} -K_X$, so that $(X, tD + (1-t)H)$ is an lc pair and

$$K_X + tD + (1-t)H \sim_{\mathbb{R}} 0,$$

and then runs an MMP on a birational model of X extracting an lc place of (X, tD). The argument needs $(1-t)H \geq 0$ (for the case $t = 1$, see [Bir21, Lemma 3.3]). When $\alpha(X) > 1$, any $\mathbb{R}$-divisor $D \geq 0$ such that $D \sim_{\mathbb{R}} -K_X$ satisfies

$$K_X + \alpha(X)D \sim_{\mathbb{R}} (\alpha(X) - 1)(-K_X),$$

whose right-hand side is ample, so there is no lc pair $(X, \alpha(X)D + G)$ with $G \geq 0$ and $K_X + \alpha(X)D + G \sim_{\mathbb{R}} 0$, and the method does not apply. For any integer $n \geq 2$, [Tot24, Theorems 6.1 and 7.1] give a klt Fano variety X of dimension n such that $\alpha(X) > 1$.

We now turn to valuations over X (see Definition 2.2). By [BJ20, Theorem C and Remark 4.5],

$$\alpha(X) = \inf_E \frac{a(E,X,0)}{T_{-K_X}(E)},$$

where E runs over all prime divisors over X (see Lemma 2.5(1)). By [BJ20, Theorem E], applied to $\mathcal{O}_X(-rK_X)$ for a positive integer r such that rK_X is Cartier, and [BJ20, Remark 4.5], the infimum is attained by a valuation over X, and together with [Xu20, Theorem 1.1] via [BJ20, Proposition 4.8(i)] (cf. [KL26, Section 1]), it follows that a quasi-monomial valuation over X computes $\alpha(X)$. [Pen25, Theorem 1.2] extends the existence of a quasi-monomial valuation computing the alpha invariant to big divisors, and [KL26, Theorem 1.1] shows that, for any projective klt pair (X,B) and any big $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor L on X, a quasi-monomial valuation over X computes $\alpha(X,B;L)$. None of [BJ20, Xu20, Pen25, KL26] provides a computing valuation of rational rank 1, that is, a prime divisor over X computing the alpha invariant. See also [LXZ22, Theorem 1.2], which shows that the stability threshold of a log Fano pair of dimension n is rational and computed by a prime divisor over the variety when it is less than $(n+1)/n$, and [AZ22, Theorem 1.4], which gives a log Fano pair of dimension 2 whose stability threshold is irrational.

We now explain the case of Fano manifolds. Let X be a Fano manifold. If $H^0(X,-K_X) \neq 0$, then any $D \in |-K_X|$ satisfies $D \geq 0$ and $D \neq 0$, so $\alpha(X) \leq \mathrm{lct}(X;D) \leq 1$, and Conjecture 1.1 holds for X by [Bir21, Theorem 1.7]. Thus Conjecture 1.1 can fail for a Fano manifold X only if $H^0(X,-K_X) = 0$. By [HS20, Proposition 1.4], $H^0(X,-K_X) \neq 0$ when $\dim X \leq 5$. The Ambro–Kawamata effective non-vanishing conjecture [Kaw00, Conjecture 2.1] predicts that $H^0(X,D) \neq 0$ for any klt pair (X,B) with X complete and any nef Cartier divisor D on X such that $D - (K_X + B)$ is nef and big. For a Fano manifold X, the boundary $B = 0$ and the Cartier divisor $D = -K_X$ satisfy the assumptions of [Kaw00, Conjecture 2.1], since $-K_X - K_X = -2K_X$ is ample, so whether there exists a Fano manifold X such that $H^0(X,-K_X) = 0$ is a case of [Kaw00, Conjecture 2.1]. [Aji26] constructs counterexamples to the Ambro–Kawamata effective non-vanishing conjecture, but they do not include the anti-canonical case on Fano manifolds, and [Aji26, Section 5.6] lists the Fano case as an open problem.

The goal of this paper is to show that Conjecture 1.1 does not hold, and that the answer to Question 1.2 is negative even when $B = 0$ and L is a multiple of $-K_X$.

Our main result is the following.

Theorem 1.3. *For any sufficiently large positive integer n, there exists a klt Fano variety X of dimension $10^8 n$ satisfying the following.*

(1) $\alpha(X)$ is an irrational number greater than 1.
(2) No prime divisor over X computes $\alpha(X)$ (see Definition 2.4).
(3) There exists a quasi-monomial valuation over X of rational rank 2 (see Definition 2.2) which computes $\alpha(X)$.

Theorem 1.3(2) implies that the infimum defining $\alpha(X)$ is not attained, even if $\mathbb{R}$-divisors are allowed.

Corollary 1.4. *Let X be a klt Fano variety. Assume that no prime divisor over X computes $\alpha(X)$. Then $\operatorname{lct}(X; D) > \alpha(X)$ for any $\mathbb{R}$-divisor $D \geq 0$ on X such that $D \sim_{\mathbb{R}} -K_X$.*

The log canonical threshold of a $\mathbb{Q}$-divisor is computed on a log resolution, so it is a rational number. Thus Theorem 1.3(1) alone only implies that $\operatorname{lct}(X; D) \neq \alpha(X)$ for any $\mathbb{Q}$-divisor $D \geq 0$ such that $D \sim_{\mathbb{Q}} -K_X$.

In the following five paragraphs, X denotes a klt Fano variety satisfying (1), (2), and (3) of Theorem 1.3.

First, X does not satisfy Conjecture 1.1, and in fact both conclusions of Conjecture 1.1 fail for X. By Corollary 1.4, no $\mathbb{Q}$-divisor $D \geq 0$ such that $D \sim_{\mathbb{Q}} -K_X$ satisfies $\operatorname{lct}(X; D) = \alpha(X)$. By Theorem 1.3(1), $\alpha(X) \notin \mathbb{Q}$.

Second, let I be a positive integer such that $-IK_X$ is Cartier. Then Question 1.2 has a negative answer for the klt pair $(X, 0)$ and the ample line bundle $\mathcal{O}_X(-IK_X)$. For any $\mathbb{Q}$-divisor $D \geq 0$ on X, we have $D \sim_{\mathbb{Q}} -K_X$ if and only if $ID \sim_{\mathbb{Q}} -IK_X$, $\operatorname{lct}(X; ID) = \operatorname{lct}(X; D)/I$, and $\operatorname{mult}_E(ID) = I \operatorname{mult}_E D$ for any prime divisor E over X. Thus

$$\alpha(X, 0; -IK_X) = \frac{\alpha(X)}{I} \quad \text{and} \quad T_{-IK_X}(E) = IT_{-K_X}(E)$$

for any prime divisor E over X. Hence a prime divisor E over X satisfies $\alpha(X, 0; -IK_X) = a(E, X, 0)/T_{-IK_X}(E)$ if and only if E computes $\alpha(X)$, which is impossible by Theorem 1.3(2).

Third, Tian's stabilization conjecture in the form [Zhu20, Conjecture A.1] fails for the klt pair $(X, 0)$ and the ample line bundle $\mathcal{O}_X(-IK_X)$, where I is as in the previous paragraph. Recall that $\alpha_m(X, 0; -IK_X)$ is the infimum of $\operatorname{lct}(X; \frac{1}{m}D)$ over all $D \in |-mIK_X|$. If $|-mIK_X| = \emptyset$, then $\alpha_m(X, 0; -IK_X) = +\infty \neq \alpha(X, 0; -IK_X)$. Since X is projective and klt, by [Amb16, Theorem 3.6], for any m such that $|-mIK_X| \neq \emptyset$, the function $D \to \operatorname{lct}(X; D)$ on $|-mIK_X|$ takes only finitely many values, so $\alpha_m(X, 0; -IK_X) = \operatorname{lct}(X; \frac{1}{m}D_m)$ for some $D_m \in |-mIK_X|$. If $\alpha(X, 0; -IK_X) = \alpha_m(X, 0; -IK_X)$ for some m, then $|-mIK_X| \neq \emptyset$, $\frac{1}{mI}D_m \sim_{\mathbb{Q}} -K_X$ and

$$\operatorname{lct}\left(X; \tfrac{1}{mI}D_m\right) = I \operatorname{lct}\left(X; \tfrac{1}{m}D_m\right) = I\alpha(X, 0; -IK_X) = \alpha(X),$$

which contradicts Corollary 1.4.

Fourth, the assumption in [Bir21, Theorem 1.7] that the threshold is at most 1 cannot be removed. Let $\operatorname{lct}(X, 0, |-K_X|_{\mathbb{R}})$ be the infimum of $\operatorname{lct}(X; D)$ over all $\mathbb{R}$-divisors $D \geq 0$ such that $D \sim_{\mathbb{R}} -K_X$ [Bir21, Section 1]. Since any $\mathbb{Q}$-divisor is an $\mathbb{R}$-divisor, $\operatorname{lct}(X, 0, |-K_X|_{\mathbb{R}}) \leq \alpha(X)$. By Corollary 1.4, $\operatorname{lct}(X; D) > \alpha(X)$ for any $\mathbb{R}$-divisor $D \geq 0$ such that $D \sim_{\mathbb{R}} -K_X$. Thus

$$\operatorname{lct}(X, 0, |-K_X|_{\mathbb{R}}) = \alpha(X) > 1,$$

and the infimum is not attained. The pair $(X, 0)$ is projective and klt, and $-K_X$ is ample. Hence $(X, 0)$ satisfies all hypotheses of [Bir21, Theorem 1.7] except the bound $\operatorname{lct}(X, 0, |-K_X|_{\mathbb{R}}) \leq 1$, while neither conclusion of [Bir21, Theorem 1.7] holds: no $\mathbb{R}$-divisor $D \geq 0$ such that $D \sim_{\mathbb{R}} -K_X$ satisfies $\operatorname{lct}(X; D) = \operatorname{lct}(X, 0, |-K_X|_{\mathbb{R}})$, and $\operatorname{lct}(X, 0, |-K_X|_{\mathbb{R}})$ is irrational by Theorem 1.3(1).

Fifth, [KL26, Theorem 1.1] cannot be strengthened to provide a quasi-monomial valuation of rational rank 1 computing $\alpha(X)$. By [JM12, Remark 3.9], a quasi-monomial valuation v over X of rational rank 1 is equal to c times the valuation given by a prime divisor E over X for some real number $c > 0$. Both $a(\cdot, X, 0)$ and T_{-K_X} are homogeneous of degree one under multiplication of valuations by positive real numbers, so $a(v, X, 0) = ca(E, X, 0)$ and $T_{-K_X}(v) = cT_{-K_X}(E)$. Thus if v computes $\alpha(X)$, then E computes $\alpha(X)$, which contradicts Theorem 1.3(2). In particular, the rational rank 2 in Theorem 1.3(3) is the smallest possible.

Sketch of the proof. Theorem 1.3 follows from Theorem 7.1, which shows

$$\alpha(X) = \alpha_0 = \frac{r}{8\ell_1} \approx 2126.9,$$

where ℓ_1, ℓ_2, and r are the constants of Set-up 4.2(10)(11), and α_0 is irrational by Lemma 4.6(3). Its proof has three parts.

(i) The pair (Z, B) of Construction-Proposition 4.4, where $\pi : Z \to \mathbb{P}^2$ is the blow-up at a point p with exceptional curve E, H is the pullback of a line, and $B = C/N$ for a reduced divisor C and an integer $N \geq 3$ fixed in (c), satisfies $a(F, Z, B) > \alpha_0 T_L(F)$ for $L = -(K_Z + B) \sim_{\mathbb{Q}} \ell_1 H - \ell_2 E$ and any prime divisor F over Z, with equality for w_{r^2-1} (Set-up 4.2(5)) of rational rank 2 (Proposition 4.7).

(ii) Construction-Theorem 5.2 turns (Z, B) into a klt Fano variety X of dimension N, and Proposition 5.3 lifts w_{r^2-1} to X preserving rational rank, log discrepancy, and T, so $\alpha(X) \leq \alpha_0$.

(iii) If $X \dashrightarrow Z$ maps a prime divisor P over X onto a prime divisor F over Z and the pullback of F has multiplicity k along P ($k = 0$ if P dominates Z), then $a(P, X, 0) \geq ka(F, Z, B)$ and

$$T_{-K_X}(P) - kT_L(F) \leq \frac{\gamma}{\delta_N}\big(a(P, X, 0) - ka(F, Z, B)\big)$$

(Proposition 6.7), where $\gamma = \ell_1/(3 - \ell_1)$ and $\delta_N = (N - 2)/N$. As

$$\alpha_0\gamma = \frac{r}{8(3 - \ell_1)} < \frac{1}{4} < \delta_N$$

(Lemma 4.6(4)), the bound and (i) imply $a(P, X, 0) > \alpha_0 T_{-K_X}(P)$, so $\alpha(X) = \alpha_0$, computed by the lift in (ii).

The key new step is Lemma 6.6 in (iii). The comparison in (i) is tight: $a(w_t, Z, B) - \alpha_0 T_L(w_t) = (\sqrt{t+1} - r)^2/16$ near $t = r^2 - 1$ (Lemma 4.6(5)), and (iii) keeps $1 - \alpha_0\gamma/\delta_N \approx 7/8$ of $a(P, X, 0) - ka(F, Z, B)$.

Ideas (a) and (b) explain (i), (c) explains (ii), and (d) explains (iii).

(a) An irrational minimizer on the blow-up. By Lemma 3.2, based on the valuative tree of [FJ04], it suffices to consider valuations monomial in formal coordinates at a point. Along a segment of such valuations, such as $\{w_t\}$ of Set-up 4.2(5), $a(w_t, Z, B)$ is piecewise affine in t with rational breakpoints. Thus, if $T_L(w_t)$ were the maximum of finitely many $w_t(G)$ with $0 \leq G \sim_{\mathbb{Q}} L$ near the minimizer of $a(w_t, Z, B)/T_L(w_t)$, the minimizer would be a rational t, that is, a prime divisor. For very general local parameters at $p \in \mathbb{P}^2$, $\mu(s) = T_{\mathcal{O}_{\mathbb{P}^2}(1)}(v_s)$ with $v_{t+1} = w_t$ (Set-up 4.2(2)(5)) equals $\sqrt{s}$ for $9 < s \leq 1127/125$ (Lemma 4.1(2)), and no plane curve attains it [Tev26, Theorem C]. On $\mathbb{P}^2$, the critical point of an affine function of s divided by $\sqrt{s}$ would still be rational. On Z, since $w_t(E) = 1$ and $E \not\subset \operatorname{Supp} B$, both $a(w_t, Z, B)$ and $T_L(w_t) = \ell_1\sqrt{t+1} - \ell_2$ (Lemma 4.5(2)) are lower by ℓ_2 than $a(v_{t+1}, \mathbb{P}^2, \pi_* B)$ and $T_{\pi_* L}(v_{t+1}) = \ell_1\mu(t+1)$, and the critical point becomes $t = r^2 - 1$ with r irrational (Lemma 4.6(3)).

(b) Two families of cuspidal curves. If $a(w_t, Z, B)$ were affine in t for $1 \leq t \leq t_h$, then, as $a(w_t, Z, B) - \alpha_0 T_L(w_t) \geq 0$ vanishes at $t = r^2 - 1$, it would be the tangent line of $\alpha_0 T_L(w_t)$ there, which is less than $\alpha_0 T_L(w_1)$ at $t = 1$ because $r^2 + 2 < 4r$ and $T_L(w_1) \geq 2\ell_1 - \ell_2$.

Instead, C contains the strict transforms of plane curves whose branches at p have first Puiseux pairs $(k_1, k_2) = (125, 1127)$ or $(500, 4507)$ (Lemma 4.3), so they leave the segment at

$t_h = 1002/125$ and $t_l = 4007/500$ (Construction-Proposition 4.4(3)). They come from very ample linear systems on weighted blow-ups of $\mathbb{P}^2$ at p, whose nef cones [Tev26, Corollary 5.1] yield curves of degree at most 3.0028 times their multiplicity at p, barely above the bound $\sqrt{k_2/k_1}$ of [Tev26, Theorem C]. Thus B has multiplicity $q = 4993/5000$ at the point $o \in E$ of Set-up 4.2(4) while L stays ample, which needs $q\sqrt{t_l+1} < 3$, so $t_l + 1$ is close to 9. Then $a(w_t, Z, B) = 1 + (1-q)t$ for $t \le t_l$, which keeps $a(w_t, Z, B) > \alpha_0 T_L(w_t)$ for $1 \le t \le 8$, where $T_L(w_t)$ is bounded above only through $\mu(8) = 17/6$ [DHK+17, Table 5.1 (D2)] and $\mu(9) = 3$. For $t_l \le t \le t_h$, $a(w_t, Z, B)$ has slope $1/16$, as Lemma 4.6(5) needs.

(c) Removing the boundary. A cyclic cover branched along C cannot replace (Z, B): $[C]/N \notin \mathrm{Pic}(Z)$, and when such a cover $Y \to Z$ of degree N exists, K_Y is the pullback of $K_Z + (1-1/N)C$, not of $K_Z + B$. Instead, Construction-Theorem 5.2 adjoins to the Cox ring of Z variables $U_1, \dots, U_{N-1}$ of degree $[C]/N$ with the single relation $F_C + \sum_i U_i^N = 0$, where F_C defines C. The ring R so obtained is graded by $\mathrm{Pic}(Z) + \mathbb{Z}[C]/N$, and X is the quotient of the L-semistable open subset $\widehat{X}$ of the hypersurface $\overline{X} = \operatorname{Spec} R$ by a two-dimensional torus. By adjunction, K_X corresponds to the character $[C] - (3H - E) - (N-1)[C]/N$, which is the class of $K_Z + B = -L$, so $H^0(X, -jK_X) \supset H^0(Z, jL)$.

After a log resolution of (Z, C), toric resolutions of the polynomials $z_1^{m_1} z_2^{m_2} + \sum_i U_i^N$, which are Newton non-degenerate in the sense of [AGM24, Definition 1.2] (cf. [ELW26, Definition 3.2]), show that $\overline{X}$ is klt along $\widehat{X}$, and X is klt by [BGLM24, Theorem 2]. If a quasi-monomial valuation over Z has value c on a local equation of C, its lift (Proposition 5.3) assigns each U_i the weight c/N, so F_C and each U_i^N have value c, and the log discrepancy drops by c/N, the value of B. Since $B = C/N$ with C reduced must realize the multiplicities β_h, β_l and the class $\ell_1 H - \ell_2 E$ of Set-up 4.2(10), whose denominators divide 10^8, we take $N = 10^8 n$ with n large for Lemma 4.3, while (iii) only needs $N \ge 3$.

(d) Prime divisors over X. The pushforward to Z of the valuation defined by P does not control $T_{-K_X}(P)$, as T_{-K_X} is positive on prime divisors dominating Z. Lemma 6.4 replaces P by a torus-invariant valuation over $\overline{X}$, preserving log discrepancy and orders of sections. Its pushforward to the spectrum $\mathbb{A}^4$ of the Cox ring of Z is determined by the pushforward to Z and an additive function λ on $\mathrm{Pic}(Z)_{\mathbb{Q}}$ (Lemma 6.5), and $\sigma = \lambda(L)$ satisfies $0 \le \sigma \le \gamma\zeta$ for the number $\zeta \ge 0$ of Lemma 6.6 (proof of Proposition 6.7). A section of $-jK_X$ is an element of the degree jL part R_{jL} of R (Construction-Theorem 5.2(4)). In the normal form of Construction-Theorem 5.2(6), its monomials of degree d in the U_i have coefficients in $H^0(Z, jL - d[C]/N)$, so $((jL - d[C]/N) \cdot H) = j\ell_1 - d(3 - \ell_1) \ge 0$, that is, $d \le j\gamma$.

Lemma 6.6 makes a finite base change along the center, a divisor on a smooth model of $\mathbb{A}^4$, and rescales the U_i, after which a finite cover of $\overline{X}$ is birational to a product with the Fermat hypersurface $V_N \subset \mathbb{P}^{N-1}$ of degree N, and a section of $-jK_X$ becomes a power of an N-th root s_0 of a local equation of the center times a section of $\mathcal{O}_{V_N}(d)$ with non-zero restrictions to general fibers. Since the section is arbitrary, we need the bound $\mathrm{lct}(V_N; D) \ge \delta_N$ for any $\mathbb{R}$-divisor $D \ge 0$ numerically equivalent to a hyperplane section [Che01, Theorem 1.3]. With inversion of adjunction [Amb16, Lemma 2.7], it bounds the order of the section along P by the smallest value among its terms plus d/δ_N times $\eta = a(P, X, 0) - ka(F, Z, B) - \sigma \ge 0$ (Lemma 6.6(1) and the proof of Proposition 6.7), minus $d\zeta$ (Lemma 6.6(2)). As $\eta \ge \zeta$, $\delta_N < 1$, and $d \le j\gamma$, we have $(d/\delta_N)\eta - d\zeta \le j\gamma(\eta/\delta_N - \zeta)$, and $j\gamma\zeta$ absorbs the $j\sigma$ in the smallest value since $\sigma \le \gamma\zeta$. Hence the slope in (iii) is γ/δ_N.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce quasi-monomial valuations, Gauss extensions of valuations, alpha invariants, the valuative tree of a smooth surface, and the Cox ring of the blow-up of the plane at a point.

In Section 3 we show that, for a klt pair on a smooth projective surface and an ample $\mathbb{Q}$-divisor on the surface, any quasi-monomial valuation that is centered at a closed point and is not

monomial in any formal coordinates at that point has a strictly larger ratio of log discrepancy to maximal vanishing order than some prime divisor over the surface (Lemma 3.2). In Section 4 we construct a klt log del Pezzo pair on the blow-up of $\mathbb{P}^2$ at a point, and show that its alpha invariant with respect to the anti-log canonical divisor is computed by a quasi-monomial valuation of rational rank 2 and is not computed by any prime divisor over the surface (Proposition 4.7).

In Section 5, starting from a klt log del Pezzo pair whose underlying surface is the blow-up of $\mathbb{P}^2$ at a point, whose boundary is $1/N$ times a reduced divisor for an integer $N \geq 3$, and whose anti-log canonical divisor satisfies the inequality of Set-up 5.1(3), we construct a klt Fano variety X of dimension N, and lift quasi-monomial valuations over the blow-up of $\mathbb{P}^2$ to quasi-monomial valuations over X, preserving rational rank, log discrepancy, and maximal vanishing order (Construction-Theorem 5.2 and Proposition 5.3). In Section 6 we bound the maximal vanishing order of $-K_X$ along a prime divisor P over X in terms of the log discrepancy of P and the image of P on the blow-up of $\mathbb{P}^2$ (Proposition 6.7). In Section 7 we apply Sections 5 and 6 to the pair of Section 4, and prove Theorem 1.3 and Corollary 1.4.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs, valuations, and alpha invariants.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c \operatorname{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Definition 2.2. Let X be a normal variety. A *valuation over* X is a function v from the set of non-zero rational functions on X to $\mathbb{R}$ such that $v(\phi\psi) = v(\phi)+v(\psi)$, $v(\phi+\psi) \geq \min\{v(\phi), v(\psi)\}$ whenever $\phi + \psi \neq 0$, $v(c) = 0$ for any $c \in \mathbb{C}^*$, and there exists a point $\eta \in X$ such that $v(\phi) \geq 0$ for any $0 \neq \phi \in \mathcal{O}_{X,\eta}$ and $v(\phi) > 0$ for any non-zero element ϕ of the maximal ideal of $\mathcal{O}_{X,\eta}$. We denote by $\operatorname{center}_X v$ the closure of η and call it the *center* of v on X. For any $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $D = \sum_i r_i D_i$ on X with $r_i \in \mathbb{R}$ and D_i Cartier, we let $v(D) := \sum_i r_i v(\phi_i)$, where ϕ_i is a local equation of D_i at η (this is well defined, since the map from Cartier divisors to Weil divisors on the normal variety X stays injective after tensoring with $\mathbb{R}$). For any dominant rational map $f : X \dashrightarrow X'$ to a normal variety X' such that f is a morphism or X' is projective, we denote by f_*v the valuation over X' defined by $(f_*v)(\phi) := v(f^*\phi)$ for any non-zero rational function ϕ on X'.

A valuation v over X is *quasi-monomial* if there exist a proper birational morphism $h : Y \to X$ from a smooth variety Y, a closed point $y \in Y$, a regular system of parameters $y_1, \ldots, y_k$ of $\mathcal{O}_{Y,y}$,

and real numbers $t_1, \dots, t_k \geq 0$, such that

$$v(g) = \min\Big\{ \sum_{i=1}^{k} t_i \beta_i \Big| c_\beta \neq 0 \Big\}$$

for any $0 \neq g \in \mathcal{O}_{Y,y}$, written as $g = \sum_\beta c_\beta y_1^{\beta_1} \cdots y_k^{\beta_k}$ in $\widehat{\mathcal{O}}_{Y,y} = \mathbb{C}[[y_1, \dots, y_k]]$, where β runs over the k-tuples $(\beta_1, \dots, \beta_k) \in \mathbb{Z}^k_{\geq 0}$ and $c_\beta \in \mathbb{C}$. The *rational rank* of such v is $\dim_{\mathbb{Q}}(\mathbb{Q}t_1 + \cdots + \mathbb{Q}t_k)$ (this is the rational rank of the value group $\mathbb{Z}t_1 + \cdots + \mathbb{Z}t_k$ of v, so it does not depend on the representation). If moreover (X, B) is a pair, h is a log resolution of (X, B), and $\operatorname{Supp}(h_*^{-1}B) \cup \operatorname{Exc}(h)$ is contained in the zero locus of $y_1 \cdots y_k$ near y, then we let

$$a(v, X, B) := \sum_{i=1}^{k} t_i a(E_i, X, B),$$

where E_i is the prime divisor on Y defined by y_i near y. Any quasi-monomial valuation over X admits such a representation for some log resolution h of (X, B) [JM12, Remark 3.4 and Lemma 3.6], and $a(v, X, B)$ does not depend on the choice of h, y, and $y_1, \dots, y_k$: indeed, $a(v, X, B) = A_Y(v) + v(K_Y - h^*(K_X + B))$, where A_Y is the log discrepancy on the smooth variety Y of [JM12, Proposition 5.1], the right-hand side does not change when Y is replaced by a higher log resolution by [JM12, Proposition 5.1(iii)], and any two log resolutions of (X, B) are dominated by a third one.

The quasi-monomial valuations of Definition 2.2 are those of [JM12, Section 3.1], where the point y is allowed to be non-closed: one replaces such y by a general closed point of the closure of y, completes $y_1, \dots, y_k$ to a regular system of parameters there, and assigns weight 0 to the new coordinates [JM12, Proposition 3.1(ii)]. The valuation f_*v has a center on X': if f is a morphism, then $\operatorname{center}_{X'}(f_*v)$ is the closure of $f(\eta)$, and if X' is projective, then the center exists by the valuative criterion of properness [Har77, Theorem II.4.7]. By [JM12, Remark 3.9], a quasi-monomial valuation over X has rational rank 1 if and only if it is a positive real multiple c of the valuation defined by a prime divisor E over X, and taking coordinates at a general closed point of E on a log resolution of (X, B), with weights $(c, 0, \dots, 0)$, shows that the log discrepancies of Definitions 2.1 and 2.2 coincide in this case.

Lemma 2.3. *Let X be a smooth quasi-projective variety and $B \geq 0$ an $\mathbb{R}$-divisor on X. Then $a(v, X, B) = a(v, X, 0) - v(B)$ for any quasi-monomial valuation v over X.*

Proof. Let $h : Y \to X$, y, $y_1, \dots, y_k$, and $t_1, \dots, t_k$ represent v as in Definition 2.2, where h is a log resolution of (X, B), and let E_i be the prime divisor on Y defined by y_i near y. Then h is also a log resolution of $(X, 0)$, so $a(v, X, 0) = \sum_i t_i a(E_i, X, 0)$. Since X is smooth, $h^*(K_X + B) = h^*K_X + h^*B$, so $a(E_i, X, B) = a(E_i, X, 0) - \operatorname{mult}_{E_i} h^*B$ for any i. Since $\operatorname{Supp} h^*B \subset \operatorname{Supp}(h_*^{-1}B) \cup \operatorname{Exc}(h)$, the pullback of a local equation at $h(y)$ of any component of B, which is also a local equation at the generic point of $\operatorname{center}_X v$ as $h(y) \in \operatorname{center}_X v$, is a unit times a monomial in $y_1, \dots, y_k$ near y, so $v(B) = \sum_i t_i \operatorname{mult}_{E_i} h^*B$. Hence

$$a(v, X, B) = \sum_i t_i a(E_i, X, B) = a(v, X, 0) - v(B).$$

□

Definition 2.4. Let (X, B) be a projective klt pair and L an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X. For any $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $D \geq 0$ on X, we let

$$\operatorname{lct}(X, B; D) := \sup\{t \geq 0 \mid (X, B + tD) \text{ is lc}\}$$

and $\operatorname{lct}(X; D) := \operatorname{lct}(X, 0; D)$. For any valuation v over X, we let

$$T_L(v) := \sup\{v(D) \mid D \geq 0,\ D \sim_{\mathbb{Q}} L\},$$

and for any prime divisor E over X, we let $T_L(E) := \sup\{\mathrm{mult}_E D \mid D \geq 0,\ D \sim_{\mathbb{Q}} L\}$, where $\mathrm{mult}_E D$ denotes the coefficient of E in the pullback of D to a higher model of X on which E is a divisor. We call $T_L(v)$ and $T_L(E)$ the *maximal vanishing orders* of L along v and along E. The *alpha invariant* of (X, B) with respect to L is

$$\alpha(X, B; L) := \inf\{\mathrm{lct}(X, B; D) \mid D \geq 0,\ D \sim_{\mathbb{Q}} L\}.$$

If X is a klt Fano variety, then we let $\alpha(X) := \alpha(X, 0; -K_X)$. A non-trivial quasi-monomial valuation v over X *computes* $\alpha(X, B; L)$ if $a(v, X, B) = \alpha(X, B; L)T_L(v)$. A prime divisor E over X *computes* $\alpha(X, B; L)$ if $a(E, X, B) = \alpha(X, B; L)T_L(E)$.

Since $a(v, X, B)$ and $T_L(v)$ are homogeneous of degree one in v, whether v computes $\alpha(X, B; L)$ depends only on the positive real multiples of v. In particular, a quasi-monomial valuation of rational rank 1 computes $\alpha(X, B; L)$ if and only if the corresponding prime divisor over X computes $\alpha(X, B; L)$.

Lemma 2.5. *Let (X, B) be a projective klt pair and L an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X. Then the following hold.*

(1) $\alpha(X, B; L) = \inf_E a(E, X, B)/T_L(E)$, where E runs over all prime divisors over X.
(2) $\alpha(X, B; L)T_L(v) \leq a(v, X, B)$ for any quasi-monomial valuation v over X.
(3) Let $f : Y \to X$ be a birational morphism from a normal $\mathbb{Q}$-factorial projective variety Y and E a prime divisor on Y. Then

$$T_L(E) = \sup\{t \geq 0 \mid f^*L - tE \text{ is pseudo-effective}\}.$$

In particular, $\mathrm{mult}_E D \leq T_L(E)$ for any $\mathbb{R}$-divisor $D \geq 0$ on X such that $D \sim_{\mathbb{R}} L$.
(4) $\alpha(X, B; cL) = \alpha(X, B; L)/c$ and $T_{cL} = cT_L$ for any rational number $c > 0$.

Proof. (1) For $B = 0$, this is [BJ20, Corollary 4.2 and Remark 4.5], and for pairs it is the divisorial form of the valuative formula of [KL26, Section 2.3]. The divisorial form follows from the definitions by exchanging two infima. For any $D \geq 0$ such that $D \sim_{\mathbb{Q}} L$ and any prime divisor E over X, we have $a(E, X, B + tD) = a(E, X, B) - t\,\mathrm{mult}_E D$, so $\mathrm{lct}(X, B; D) = \inf_E a(E, X, B)/\mathrm{mult}_E D$, where E runs over the prime divisors over X such that $\mathrm{mult}_E D > 0$. Since L is ample, $T_L(E) > 0$ for any E. Thus

$$\alpha(X, B; L) = \inf_D \inf_E \frac{a(E, X, B)}{\mathrm{mult}_E D} = \inf_E \frac{a(E, X, B)}{T_L(E)}.$$

(2) Let $D \geq 0$ be a $\mathbb{Q}$-divisor such that $D \sim_{\mathbb{Q}} L$, and let $\lambda := \mathrm{lct}(X, B; D)$. Then $\lambda \geq \alpha(X, B; L)$, and $(X, B + \lambda D)$ is lc because $a(E, X, B + tD) = a(E, X, B) - t\,\mathrm{mult}_E D$ is continuous in t for any E. Represent v as in Definition 2.2 by $h : Y \to X$, y, $y_1, \ldots, y_k$, and weights $(t_1, \ldots, t_k)$, where h is a log resolution of (X, B). For any $w = (w_1, \ldots, w_k) \in \mathbb{R}^k_{>0}$, let v_w be the quasi-monomial valuation over X represented by h, y, $y_1, \ldots, y_k$, and the weights w [JM12, Proposition 3.1]. Then $a(v_w, X, B) = \sum_i w_i a(E_i, X, B)$ is linear in w. Let n be a positive integer such that nD is Cartier. Let $g \in \mathcal{O}_{Y,y}$ be a local equation of $h^*(nD)$ at y. Then $v_w(D) = v_w(g)/n$. By Dickson's lemma, the set of exponents β with $c_\beta \neq 0$ in the expansion of g has finitely many minimal elements, so $v_w(g)$ is the minimum of finitely many linear functions of w, and $v_w(D)$ is continuous in w. If $w \neq 0$ is rational, then v_w has rational rank 1, so v_w is c times the valuation defined by a prime divisor E_w over X for some real number $c > 0$ [JM12, Remark 3.9], and $a(v_w, X, B) = ca(E_w, X, B)$ as explained after Definition 2.2. Hence

$$a(v_w, X, B) - \lambda v_w(D) = c\big(a(E_w, X, B) - \lambda\,\mathrm{mult}_{E_w} D\big) = ca(E_w, X, B + \lambda D) \geq 0.$$

Since rational points are dense in $\mathbb{R}^k_{\geq 0}$, we have $a(v_w, X, B) - \lambda v_w(D) \geq 0$ for any $w \in \mathbb{R}^k_{\geq 0}$. Taking $w = (t_1, \ldots, t_k)$, we obtain $a(v, X, B) \geq \lambda v(D) \geq \alpha(X, B; L)v(D)$. Taking the supremum over D implies (2).

(3) This is [BJ20, Corollary 3.9(i)], together with [BJ20, Lemma 3.7(ii) and Remark 3.8], where our $T_L(E)$ is the invariant T of [BJ20] at the valuation defined by E, since T is defined in

[BJ20, Section 2.4 and (3.2)] as the supremum of $\mathrm{mult}_E D/m$ over the positive integers m such that mL is Cartier and over $D \in |mL|$, and a $\mathbb{Q}$-divisor $D \geq 0$ satisfies $D \sim_{\mathbb{Q}} L$ if and only if $mD \in |mL|$ for some such m. Since Y is $\mathbb{Q}$-factorial, E is $\mathbb{Q}$-Cartier, so $f^*L - tE$ is an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor and the pseudo-effectivity of $f^*L - tE$ is defined. For the inequality $\mathrm{mult}_E D \leq T_L(E)$, $f^*D - (\mathrm{mult}_E D)E \geq 0$ and $f^*D - (\mathrm{mult}_E D)E \sim_{\mathbb{R}} f^*L - (\mathrm{mult}_E D)E$, so $f^*L - (\mathrm{mult}_E D)E$ is pseudo-effective, and $\mathrm{mult}_E D \leq T_L(E)$.

(4) For any $\mathbb{Q}$-divisor $D \geq 0$, we have $D \sim_{\mathbb{Q}} cL$ if and only if $D/c \sim_{\mathbb{Q}} L$, and $\mathrm{lct}(X, B; D) = \mathrm{lct}(X, B; D/c)/c$ and $v(D) = cv(D/c)$ for any valuation v over X. Thus (4) follows from Definition 2.4. □

Definition 2.6. Let Y be a normal variety, v a valuation over Y, $u_1, \ldots, u_m$ the coordinates of $\mathbb{A}^m$, and $c_1, \ldots, c_m \geq 0$ real numbers. The *Gauss extension* of v with weights $c_1, \ldots, c_m$ is the unique valuation v' over $Y \times \mathbb{A}^m$ such that

$$v'\Big(\sum_{\beta} a_\beta u_1^{\beta_1} \cdots u_m^{\beta_m}\Big) = \min\Big\{v(a_\beta) + \sum_{i=1}^{m} c_i\beta_i \Big| a_\beta \neq 0\Big\}$$

for any non-zero polynomial $\sum_\beta a_\beta u_1^{\beta_1} \cdots u_m^{\beta_m}$ whose coefficients a_β are rational functions on Y, where $\beta = (\beta_1, \ldots, \beta_m)$ runs over $\mathbb{Z}^m_{\geq 0}$ [Bou72, Chapter VI, §10, no. 1, Lemma 1].

The lemma of [Bou72] is the case $m = 1$, and applying it m times yields the existence and uniqueness of v'. Since $c_i \geq 0$ for any i, the valuation v' is centered at the generic point of $\mathrm{center}_Y v \times V$, where $V \subset \mathbb{A}^m$ is the linear subspace defined by $u_i = 0$ for all i such that $c_i > 0$, so v' is a valuation over $Y \times \mathbb{A}^m$. Gauss extensions are used in the proofs of Lemma 6.4 and Lemma 6.5.

2.2. **The valuative tree.** We use the valuative tree of [FJ04] to study quasi-monomial valuations centered at a closed point of a smooth surface.

Definition 2.7. Let S be a smooth surface, $p \in S$ a closed point, and $\mathfrak{m}_p$ the maximal ideal of $\widehat{\mathcal{O}}_{S,p}$. For any $0 \neq f \in \widehat{\mathcal{O}}_{S,p}$, $\mathrm{mult}_p f$ denotes the largest integer k such that $f \in \mathfrak{m}_p^k$. A *normalized valuation* at p is a function $\nu : \widehat{\mathcal{O}}_{S,p} \to [0, +\infty]$ such that $\nu(fg) = \nu(f) + \nu(g)$, $\nu(f + g) \geq \min\{\nu(f), \nu(g)\}$, $\nu(c) = 0$ for any $c \in \mathbb{C}^*$, and $\min\{\nu(f) \mid f \in \mathfrak{m}_p\} = 1$. A quasi-monomial valuation v over S with $\mathrm{center}_S v = p$ and

$$\min\{v(f) \mid 0 \neq f \in \mathfrak{m}_p \cap \mathcal{O}_{S,p}\} = 1$$

extends uniquely to a normalized valuation at p (the extension exists by [FJ04, Proposition E.1] and is unique, as shown below), again denoted by v, and we say that v is *normalized*. For normalized valuations u and ν at p, we write $u \leq \nu$ if $u(f) \leq \nu(f)$ for any $f \in \widehat{\mathcal{O}}_{S,p}$, and we denote by $u \wedge \nu$ the largest normalized valuation ω at p such that $\omega \leq u$ and $\omega \leq \nu$ [FJ04, Section 3.2]. For any irreducible element $\phi \in \mathfrak{m}_p$, the *curve valuation* ν_ϕ is the normalized valuation at p defined by $\nu_\phi(f) := (\phi \cdot f)_p / \mathrm{mult}_p \phi$. For any normalized valuation ν at p, the *skewness* of ν is

$$s(\nu) := \sup\{\nu(f)/\mathrm{mult}_p f \mid 0 \neq f \in \mathfrak{m}_p\} \in [1, +\infty],$$

and the *tree multiplicity* of ν is

$$m(\nu) := \min\{\mathrm{mult}_p \phi \mid \phi \in \mathfrak{m}_p \text{ is irreducible and } \nu_\phi \geq \nu\},$$

with $m(\nu) := +\infty$ if no such ϕ exists.

The tree multiplicity measures how far v is from being monomial in some formal coordinates at p, as $m(v) = 1$ if and only if v is monomial in some generators of $\mathfrak{m}_p$ (Lemma 2.9(3)). When computing alpha invariants on a smooth surface, it suffices to consider valuations v such that $m(v) = 1$ (Lemma 3.2).

Lemma 2.8. *Let S be a smooth surface, $p \in S$ a closed point, and v a quasi-monomial valuation over S with $\operatorname{center}_S v = p$ and*

$$\min\{v(f) \mid 0 \neq f \in \mathfrak{m}_p \cap \mathcal{O}_{S,p}\} = 1.$$

Then v extends uniquely to a normalized valuation at p.

Proof. The extension exists by $\mathfrak{m}_p$-adic continuity, since $v(g) \geq n$ for any $g \in \mathfrak{m}_p^n \cap \mathcal{O}_{S,p}$. It is unique since any normalized valuation ν at p extending v satisfies $\nu \geq n$ on $\mathfrak{m}_p^n$ for any positive integer n: any $f \in \widehat{\mathcal{O}}_{S,p}$ can be written as $f = g_1 + g_2$ with $g_1 \in \mathcal{O}_{S,p}$ and $g_2 \in \mathfrak{m}_p^n$, and then $\nu(f) = v(g_1)$ whenever $n > v(g_1)$, while $\nu(f) \geq n$ if $v(g_1) \geq n$. □

Lemma 2.9. *Let S be a smooth surface, $p \in S$ a closed point, and v a normalized quasi-monomial valuation over S with $\operatorname{center}_S v = p$. Then the following hold.*

(1) $v(\phi) = \operatorname{mult}_p \phi \cdot s(v \wedge \nu_\phi)$ for any irreducible $\phi \in \mathfrak{m}_p$.

(2) For any $t \in [1, s(v)]$, there exists a unique normalized valuation $v^{(t)} \leq v$ at p such that $s(v^{(t)}) = t$. Moreover, $v^{(t)}$ is a normalized quasi-monomial valuation, $v^{(1)} = \operatorname{mult}_p$, $v^{(s(v))} = v$, and any normalized valuation $u \leq v$ at p is equal to $v^{(t)}$ for some $t \in [1, s(v)]$.

(3) $m(v) = 1$ if and only if there exist generators x', y' of $\mathfrak{m}_p$ such that

$$v\Big(\sum_{i,j} c_{ij} x'^i y'^j\Big) = \min\{i + s(v)j \mid c_{ij} \neq 0\}$$

for any non-zero $\sum_{i,j} c_{ij} x'^i y'^j \in \widehat{\mathcal{O}}_{S,p}$.

(4) $m(v^{(t)}) \leq m(v^{(t')})$ for any $1 \leq t \leq t' \leq s(v)$. If $m(v) \geq 2$, then there exists a rational number $t_1 \in (1, s(v))$ such that $v^{(t_1)}$ is divisorial, $m(v^{(t)}) = 1$ for any $t \leq t_1$, and $m(v^{(t)}) \geq 2$ for any $t > t_1$.

(5) $a(v, S, 0) = 2 + \int_1^{s(v)} m(v^{(t)})dt$. In particular, $a(v, S, 0) \geq 1 + s(v)$, and equality holds if $m(v) = 1$.

Proof. The valuation v, extended to $\widehat{\mathcal{O}}_{S,p}$, is quasi-monomial in the sense of [FJ04, Section 1.5.4], hence divisorial or irrational, with $s(v) < +\infty$ and $m(v) < +\infty$ by [FJ04, Theorem 3.26 and Proposition 3.37(i)].

(1) This is [FJ04, Proposition 3.25].

(2) The normalized valuations $u \leq v$ at p form a segment with lower end mult_p [FJ04, Theorem 3.14 and Proposition 3.20(i)], and skewness restricts to an increasing bijection from the set of normalized valuations $u \leq v$ at p onto $[1, s(v)]$ [FJ04, Theorem 3.26]. This implies (2) except the quasi-monomiality of $v^{(t)}$. Since $v^{(s(v))} = v$, we may assume that $t < s(v)$. Then $v^{(t)} < v$, so $v^{(t)}$ is not an end of the valuative tree, and by [FJ04, Proposition 3.20(ii)], $v^{(t)}$ is divisorial or irrational in the sense of [FJ04]. Thus the rational rank of the value group of $v^{(t)}$ plus the transcendence degree of the residue field of $v^{(t)}$ is 2 [FJ04, Theorems 2.28 and 2.29]. Let u_t be the valuation over S defined by the values of $v^{(t)}$ on $\mathcal{O}_{S,p}$. Since $v^{(t)} \geq n$ on $\mathfrak{m}_p^n$, for any $f \in \widehat{\mathcal{O}}_{S,p}$ with $v^{(t)}(f) < +\infty$ and any $f_1 \in \mathcal{O}_{S,p}$ such that $f - f_1 \in \mathfrak{m}_p^n$ for some integer $n > v^{(t)}(f)$, we have $v^{(t)}(f_1) = v^{(t)}(f)$. If moreover $g \in \widehat{\mathcal{O}}_{S,p}$ and $g_1 \in \mathcal{O}_{S,p}$ satisfy $v^{(t)}(g) = v^{(t)}(f)$ and $g - g_1 \in \mathfrak{m}_p^n$, then

$$fg_1 - f_1 g = (f - f_1)g_1 + f_1(g_1 - g) \quad \text{and} \quad v^{(t)}(fg_1 - f_1 g) \geq n + v^{(t)}(f) > v^{(t)}(gg_1),$$

so the residue class of f/g is equal to the residue class of f_1/g_1. Thus the value group of u_t is equal to the value group of $v^{(t)}$, and the residue field of u_t is equal to the residue field of $v^{(t)}$, so u_t satisfies equality in Abhyankar's inequality, and u_t is quasi-monomial by [JM12, Proposition 3.7]. Since $\operatorname{center}_S u_t = p$ and $v^{(t)}$ is normalized, $v^{(t)}$ is the normalized quasi-monomial valuation u_t.

(3) This is [FJ04, Proposition 3.37(ii)]. If v is monomial in generators x', y' of $\mathfrak{m}_p$ with $v(x') \leq v(y')$, then $v(x') = 1$ since v is normalized, and $v(y') = s(v)$: indeed, $v(y')/\operatorname{mult}_p y' = v(y')$, and

any $f \in \mathfrak{m}_p$ with $\operatorname{mult}_p f = k$ has a monomial $x'^i y'^j$ with $i+j=k$ in the expansion of f, so $v(f) \le i + v(y')j \le v(y')k$.

(4) By [FJ04, Proposition 3.44], there exist divisorial valuations and integers

$$\operatorname{mult}_p = \nu_0 < \nu_1 < \cdots < \nu_g < \nu_{g+1} = v \qquad \text{and} \qquad 1 = m_1 < m_2 < \cdots < m_{g+1} = m(v)$$

such that $m(u) = m_i$ for any normalized valuation u with $\nu_{i-1} < u \le \nu_i$. This implies that $m(v^{(t)}) \le m(v^{(t')})$ for any $1 \le t \le t' \le s(v)$. If $m(v) \ge 2$, then $g \ge 1$, and $t_1 := s(\nu_1)$ is a rational number as in the second sentence of (4), where t_1 is rational because ν_1 is divisorial [FJ04, Theorem 3.26].

(5) By [FJ04, (3.7)], the thinness of v is equal to $2 + \int_1^{s(v)} m(v^{(t)})dt$, and the thinness is equal to $a(v, S, 0)$ when v is divisorial [FJ04, Remark 3.50]. If v is not divisorial, then v is irrational in the sense of [FJ04], $s(v)$ is irrational, and $v^{(t)}$ is divisorial for any rational $t < s(v)$ by (2) and [FJ04, Theorem 3.26]. Since $(v^{(t)})^{(\tau)} = v^{(\tau)}$ for any $1 \le \tau \le t$, the formula for $v^{(t)}$ reads

$$a(v^{(t)}, S, 0) = 2 + \int_1^t m(v^{(\tau)})d\tau.$$

Both sides are continuous in t near $s(v)$: the right-hand side is an integral in t, and the left-hand side is continuous because, on a log resolution of $(S, 0)$ on which v is represented as in Definition 2.2, the valuations $v^{(t)}$ with t close to $s(v)$ are represented with the coordinates representing v, with weights depending continuously on t, while $a(\cdot, S, 0)$ is linear in the weights by Definition 2.2. Here the normalized valuations monomial in the coordinates representing v form a segment of the valuative tree containing v in the interior of the segment, cf. [FJ04, Lemma 6.28]. Thus the formula holds for v. The inequality $a(v, S, 0) \ge 1 + s(v)$ and the equality for $m(v) = 1$ follow from $m(v^{(t)}) \ge 1$ for any t, and from $m(v^{(t)}) = 1$ for any $t \le s(v)$ when $m(v) = 1$ by (4). □

2.3. **The blow-up of the plane at a point.**

Lemma 2.10. *Let $\pi : Z \to \mathbb{P}^2$ be the blow-up of a closed point $p \in \mathbb{P}^2$, E the exceptional curve, and $H := \pi^*\mathcal{O}_{\mathbb{P}^2}(1)$. Then the following hold.*

(1) $\operatorname{Pic}(Z) = \mathbb{Z}H \oplus \mathbb{Z}E$, $H^2 = 1$, $H \cdot E = 0$, $E^2 = -1$, and $-K_Z \sim 3H - E$. For any real numbers c_1, c_2, the $\mathbb{R}$-divisor $c_1H - c_2E$ is ample if and only if $c_1 > c_2 > 0$.

(2) For any integers $d, k \ge 0$, the pullback induces an isomorphism between the space of homogeneous polynomials g of degree d on $\mathbb{P}^2$ with $\operatorname{mult}_p g \ge k$ and $H^0(Z, \mathcal{O}_Z(dH - kE))$, and the divisor of the section corresponding to g is $\pi^* \operatorname{div}(g) - kE$.

(3) The Cox ring of Z is $\mathcal{S} = \mathbb{C}[x_0, x_1, x_2, e]$, graded by $\operatorname{Pic}(Z)$ with $\deg x_0 = H$, $\deg x_1 = \deg x_2 = H - E$, and $\deg e = E$, and $\mathcal{S}_Q = H^0(Z, \mathcal{O}_Z(Q))$ for any $Q \in \operatorname{Pic}(Z)$, where $\mathcal{S}_Q$ is the degree Q part of $\mathcal{S}$. Moreover, the torus $\mathbb{T}_0 := \operatorname{Hom}(\operatorname{Pic}(Z), \mathbb{C}^*)$ acts freely on $\{(x_1, x_2) \ne (0,0),\ (x_0, e) \ne (0,0)\} \subset \mathbb{A}^4$, and Z is the geometric quotient.

Proof. (1) The description of $\operatorname{Pic}(Z)$ and the intersection numbers are [Har77, Proposition V.3.2], and $-K_Z \sim 3H - E$ follows from $K_Z = \pi^*K_{\mathbb{P}^2} + E$ [Har77, Proposition V.3.3]. The surface Z is a rational ruled surface with negative section E and fiber class $H - E$ [Har77, Example V.2.11.5], so for any integers $b > a > 0$, the divisor $bH - (b-a)E = aE + b(H - E)$ is ample by [Har77, Corollary V.2.18(a)]. In particular, $qH - E$ is an ample $\mathbb{Q}$-divisor for any rational number $q > 1$. If $c_1 > c_2 > 0$, then we choose rational numbers q, q' such that $1 < q < c_1/c_2 < q'$, and

$$c_1H - c_2E = \frac{q'c_2 - c_1}{q' - q}(qH - E) + \frac{c_1 - qc_2}{q' - q}(q'H - E)$$

is a combination of ample $\mathbb{Q}$-divisors with positive coefficients, hence ample. Conversely, if $c_1H - c_2E$ is ample, then

$$((c_1H - c_2E) \cdot E) = c_2 > 0 \quad \text{and} \quad ((c_1H - c_2E) \cdot (H - E)) = c_1 - c_2 > 0,$$

where $H - E$ is the class of the strict transform of a line through p [Har77, Proposition V.3.6].

(2) Since $\pi_*\mathcal{O}_Z = \mathcal{O}_{\mathbb{P}^2}$, the pullback induces an isomorphism between the space of homogeneous polynomials of degree d and $H^0(Z, \mathcal{O}_Z(dH))$. For any such $g \neq 0$, the divisor $\pi^* \operatorname{div}(g)$ is the strict transform of $\operatorname{div}(g)$ plus $(\operatorname{mult}_p g)E$ [Har77, Proposition V.3.6], so $\pi^* \operatorname{div}(g) - kE \geq 0$ if and only if $\operatorname{mult}_p g \geq k$. This implies (2).

(3) Choose homogeneous coordinates on $\mathbb{P}^2$ such that p is a torus-fixed point. Then Z is a smooth projective toric surface, and x_0, x_1, x_2, and e correspond to the strict transforms of the three coordinate lines, the first of which does not pass through p, and to E. The two-dimensional cones of the fan of Z correspond to the pairs $\{x_1, e\}$, $\{e, x_2\}$, $\{x_2, x_0\}$, and $\{x_0, x_1\}$ of rays, so the exceptional set of the fan is $\{x_0 = e = 0\} \cup \{x_1 = x_2 = 0\}$. An element $g \in \mathbb{T}_0$ multiplies x_0, x_1, x_2, and e by $g(H)$, $g(H-E)$, $g(H-E)$, and $g(E)$, respectively. If g fixes a point with $(x_1, x_2) \neq (0,0)$ and $(x_0, e) \neq (0,0)$, then $g(H-E) = 1$, and $g(H) = 1$ or $g(E) = 1$. Since $g(H) = g(H-E)g(E)$, we have $g(H) = g(E) = 1$, and since H and E generate $\operatorname{Pic}(Z)$, g is trivial. Thus (3) follows from [CLS11, Theorem 5.1.11(b) and Proposition 5.3.7(a)]. □

3. Quasi-monomial valuations on smooth surfaces

In this section, we prove two properties of quasi-monomial valuations at a closed point of a smooth surface. In Section 4, Lemmas 3.1 and 3.2 reduce the computation of the alpha invariant of a log del Pezzo pair to valuations of tree multiplicity one, and Lemma 3.1 is used again in the proof of Proposition 4.7.

Lemma 3.1. *Let S be a smooth surface, $p \in S$ a closed point, and $u \leq v$ normalized quasi-monomial valuations over S with center p. Then $s(u)v(D) \leq s(v)u(D)$ for any $\mathbb{R}$-divisor $D \geq 0$ on S. In particular, if S is projective, then $s(u)T_L(v) \leq s(v)T_L(u)$ for any ample $\mathbb{Q}$-divisor L on S.*

Proof. We first show that

$$s(u)v(f) \leq s(v)u(f) \tag{3.1}$$

for any $0 \neq f \in \widehat{\mathcal{O}}_{S,p}$. Since u and v take values in $[0, +\infty]$ and vanish on $\mathbb{C}^*$, they vanish on the units of $\widehat{\mathcal{O}}_{S,p}$. Since $\widehat{\mathcal{O}}_{S,p}$ is a regular local ring, it is a unique factorization domain, so any non-zero nonunit of $\widehat{\mathcal{O}}_{S,p}$ is a unit times a product of irreducible elements of $\mathfrak{m}_p$. Since $u(fg) = u(f) + u(g)$ and $v(fg) = v(f) + v(g)$, we may assume that $f = \phi$ for some irreducible element $\phi \in \mathfrak{m}_p$.

We show that $v^{(\lambda)} \leq v^{(\lambda')}$ for any real numbers $1 \leq \lambda \leq \lambda' \leq s(v)$. By Lemma 2.9(2), $v^{(\lambda')}$ is a normalized quasi-monomial valuation, so Lemma 2.9(2) applied to $v^{(\lambda')}$ implies that there exists a normalized valuation $\omega \leq v^{(\lambda')}$ at p such that $s(\omega) = \lambda$. Since $\omega \leq v^{(\lambda')} \leq v$, the uniqueness in Lemma 2.9(2) implies that $\omega = v^{(\lambda)}$. Thus $v^{(\lambda)} \leq v^{(\lambda')}$.

We show that

$$s(u \wedge \nu_\phi) = \min\{s(u), s(v \wedge \nu_\phi)\}. \tag{3.2}$$

By the definition of skewness (Definition 2.7), $s(\omega) \leq s(\omega')$ for any normalized valuations $\omega \leq \omega'$ at p.

Since $u \leq v$, $v \wedge \nu_\phi \leq v$, and $u \wedge \nu_\phi \leq u \leq v$, Lemma 2.9(2) implies that

$$u = v^{(s(u))}, \quad v \wedge \nu_\phi = v^{(s(v \wedge \nu_\phi))}, \quad \text{and} \quad u \wedge \nu_\phi = v^{(s(u \wedge \nu_\phi))}.$$

Since $u \wedge \nu_\phi \leq v$ and $u \wedge \nu_\phi \leq \nu_\phi$, the maximality of $v \wedge \nu_\phi$ implies that $u \wedge \nu_\phi \leq v \wedge \nu_\phi$. Since moreover $u \wedge \nu_\phi \leq u$, we have $s(u \wedge \nu_\phi) \leq \min\{s(u), s(v \wedge \nu_\phi)\}$. Conversely, let $t_0 := \min\{s(u), s(v \wedge \nu_\phi)\}$. Then

$$v^{(t_0)} \leq v^{(s(u))} = u \quad \text{and} \quad v^{(t_0)} \leq v^{(s(v \wedge \nu_\phi))} = v \wedge \nu_\phi \leq \nu_\phi.$$

By the maximality of $u \wedge \nu_\phi$, $v^{(t_0)} \leq u \wedge \nu_\phi$, hence $t_0 \leq s(u \wedge \nu_\phi)$. This proves (3.2).

Let $t := s(v \wedge \nu_\phi)$. Since $v \wedge \nu_\phi \le v$, we have $t \le s(v)$, and since $u \le v$, we have $s(u) \le s(v)$. By Lemma 2.9(1) applied to v and to u, and by (3.2),

$$v(\phi) = t \operatorname{mult}_p \phi \quad \text{and} \quad u(\phi) = \min\{s(u), t\} \operatorname{mult}_p \phi.$$

If $t \le s(u)$, then

$$s(u)v(\phi) = s(u)t \operatorname{mult}_p \phi \le s(v)t \operatorname{mult}_p \phi = s(v)u(\phi).$$

If $t > s(u)$, then

$$s(u)v(\phi) = s(u)t \operatorname{mult}_p \phi \le s(u)s(v) \operatorname{mult}_p \phi = s(v)u(\phi).$$

This proves (3.1).

Let $D = \sum_i d_i D_i$, where $d_i \ge 0$ are real numbers and D_i are distinct prime divisors, and let ϕ_i be a local equation of D_i at p, which is a unit if $p \notin D_i$. By Definition 2.2, $v(D) = \sum_i d_i v(\phi_i)$ and $u(D) = \sum_i d_i u(\phi_i)$. Applying (3.1) to each ϕ_i and summing with the coefficients $d_i \ge 0$, we obtain $s(u)v(D) \le s(v)u(D)$.

Assume that S is projective. For any $D \ge 0$ such that $D \sim_{\mathbb{Q}} L$, we have

$$s(u)v(D) \le s(v)u(D) \le s(v)T_L(u).$$

Taking the supremum over all such D, we obtain $s(u)T_L(v) \le s(v)T_L(u)$. □

A valuation of tree multiplicity at least two is not monomial in any formal coordinates at the center (Lemma 2.9(3)). The next lemma compares such a valuation v with the divisorial valuation $v^{(t_1)}$ of Lemma 2.9(4), where the tree multiplicity jumps for the first time: from $v^{(t_1)}$ to v, the value of a divisor grows at most proportionally to the skewness (Lemma 3.1), while the log discrepancy with respect to the surface grows with slope one before $v^{(t_1)}$ and with slope at least two after $v^{(t_1)}$ (Lemma 2.9(5)).

Lemma 3.2. *Let S be a smooth projective surface, $p \in S$ a closed point, and $B \ge 0$ a $\mathbb{Q}$-divisor on S such that (S, B) is klt. Let v be a normalized quasi-monomial valuation over S with center p and $m(v) \ge 2$. Then there exist a normalized divisorial valuation $u \le v$ with $m(u) = 1$ and a real number $\kappa \in (0, 1)$ such that*

$$\frac{v(D)}{a(v, S, B)} \le \kappa \frac{u(D)}{a(u, S, B)}$$

for any $\mathbb{R}$-divisor $D \ge 0$ on S. In particular, $a(u, S, B)T_L(v) < a(v, S, B)T_L(u)$ for any ample $\mathbb{Q}$-divisor L on S.

Proof. By Lemma 2.9(4), there exists a rational number $t_1 \in (1, s(v))$ such that $v^{(t_1)}$ is divisorial, $m(v^{(t)}) = 1$ for any $t \le t_1$, and $m(v^{(t)}) \ge 2$ for any $t > t_1$. Let $u := v^{(t_1)}$. By Lemma 2.9(2)(4), u is a normalized divisorial valuation over S with center p, $u \le v$, and $s(u) = t_1$. Moreover, $m(u) = 1$ and $1 < s(u) < s(v)$.

By Lemma 2.9(5) and since $m(u) = 1$, we have $a(u, S, 0) = 1 + s(u)$. By Lemma 2.9(5) again,

$$a(v, S, 0) = 2 + \int_1^{s(u)} m(v^{(t)})dt + \int_{s(u)}^{s(v)} m(v^{(t)})dt \ge 2 + (s(u) - 1) + 2(s(v) - s(u)).$$

Thus

$$\begin{aligned} a(v, S, 0) - \frac{s(v)}{s(u)} a(u, S, 0) &\ge 1 + s(u) + 2(s(v) - s(u)) - \frac{s(v)}{s(u)}(1 + s(u)) \\ &= (s(v) - s(u))\Big(1 - \frac{1}{s(u)}\Big) > 0, \end{aligned} \tag{3.3}$$

where the last inequality follows from $s(v) > s(u) > 1$.

By Lemma 2.3 applied to u and v, by (3.3), and by Lemma 3.1 applied to $D = B$, we have

$$a(v, S, B) - \frac{s(v)}{s(u)} a(u, S, B) = a(v, S, 0) - \frac{s(v)}{s(u)} a(u, S, 0) - \Big(v(B) - \frac{s(v)}{s(u)} u(B)\Big) > 0.$$

Since u is divisorial, u is c times the valuation defined by some prime divisor E over S for some real number $c > 0$, and since (S, B) is klt, $a(u, S, B) = ca(E, S, B) > 0$. Thus $a(v, S, B) > \frac{s(v)}{s(u)} a(u, S, B) > 0$. Let

$$\kappa := \frac{s(v)a(u, S, B)}{s(u)a(v, S, B)}.$$

Then $\kappa \in (0, 1)$. For any $\mathbb{R}$-divisor $D \geq 0$ on S, Lemma 3.1 implies that

$$\frac{v(D)}{a(v, S, B)} \leq \frac{s(v)u(D)}{s(u)a(v, S, B)} = \kappa \frac{u(D)}{a(u, S, B)}.$$

Let L be an ample $\mathbb{Q}$-divisor on S, and let n be a positive integer such that nL is a very ample Cartier divisor. We show that $0 < T_L(u) < +\infty$. Let D_0 be a member of $|nL|$ passing through p. Then $D_0/n \sim_{\mathbb{Q}} L$, and since the center of u is p, $u(D_0/n) > 0$. Thus $T_L(u) > 0$. Let $D = \sum_i d_i D_i \geq 0$ be a $\mathbb{Q}$-divisor such that $D \sim_{\mathbb{Q}} L$, where $d_i > 0$ are rational numbers and D_i are distinct prime divisors, and let ϕ_i be a local equation of D_i at p. In the embedding of S defined by $|nL|$, take a hyperplane through p which contains neither the embedded tangent plane of S at p nor any D_i, and let $A \in |nL|$ be the corresponding member. Then A passes through p, A is smooth at p, and A contains no D_i. Let ψ be a local equation of A at p. Since A is smooth at p, $\mathcal{O}_{S,p}/(\psi)$ is a discrete valuation ring whose maximal ideal is the image of the maximal ideal of $\mathcal{O}_{S,p}$, hence

$$\operatorname{mult}_p \phi_i \leq \dim_{\mathbb{C}} \mathcal{O}_{S,p}/(\phi_i, \psi) = (D_i \cdot A)_p \leq (D_i \cdot A),$$

where the last inequality follows from [Har77, Proposition V.1.4]. By the definition of skewness, $u(\phi_i) \leq s(u) \operatorname{mult}_p \phi_i$ if $p \in D_i$, and $u(\phi_i) = 0$ if $p \notin D_i$. Thus

$$u(D) = \sum_i d_i u(\phi_i) \leq s(u) \sum_i d_i \operatorname{mult}_p \phi_i \leq s(u) \sum_i d_i (D_i \cdot A) = s(u)(D \cdot A) = ns(u)(L^2).$$

Taking the supremum over all such D, we obtain $T_L(u) \leq ns(u)(L^2) < +\infty$.

Taking the supremum of $v(D)/a(v, S, B) \leq \kappa u(D)/a(u, S, B)$ over all $D \geq 0$ such that $D \sim_{\mathbb{Q}} L$, we obtain

$$\frac{T_L(v)}{a(v, S, B)} \leq \kappa \frac{T_L(u)}{a(u, S, B)} < \frac{T_L(u)}{a(u, S, B)},$$

where the strict inequality follows from $\kappa < 1$ and $0 < T_L(u) < +\infty$. Since $a(u, S, B) > 0$ and $a(v, S, B) > 0$, multiplying both sides by $a(u, S, B)a(v, S, B)$ gives $a(u, S, B)T_L(v) < a(v, S, B)T_L(u)$. □

4. A klt log del Pezzo pair

In this section, we construct a klt log del Pezzo pair (Z, B), where Z is the blow-up of $\mathbb{P}^2$ at a point, $B = C/N$, and C is a reduced divisor. In Proposition 4.7, we show that $\alpha(Z, B; L)$ is computed by a quasi-monomial valuation of rational rank 2 and is not computed by any prime divisor over Z. Sections 5–7 remove the boundary while preserving both properties.

4.1. Choice of local parameters.

In this subsection, we choose local parameters x, y of $\mathbb{P}^2$ at a closed point p for which Lemma 4.1(2) determines $\mu(8)$, $\mu(9)$, and $\mu(s)$ for $s \in (9, 1127/125]$ by [Tev26, Theorem C] and [DHK+17]. The values of μ in Lemma 4.1(2) are the input of Proposition 4.7.

Lemma 4.1. *Let $p \in \mathbb{P}^2$ be a closed point. Then there exist algebraic local parameters x, y of $\mathbb{P}^2$ at p satisfying the following, where for any real number $s \geq 1$, v_s denotes the quasi-monomial valuation over $\mathbb{P}^2$ such that*

$$v_s\Big(\sum_{i,j} c_{ij} x^i y^j\Big) = \min\{i + sj \mid c_{ij} \neq 0\}$$

for any non-zero element $\sum_{i,j} c_{ij} x^i y^j$ of $\widehat{\mathcal{O}}_{\mathbb{P}^2,p} = \mathbb{C}[[x, y]]$, and $\mu(s) := T_{\mathcal{O}_{\mathbb{P}^2}(1)}(v_s)$.

(1) $\mu(u) \le \mu(s) \le (s/u)\mu(u)$ *for any real numbers* $1 \le u \le s$.
(2) $\mu(8) = 17/6$, $\mu(9) = 3$, *and* $\mu(s) = \sqrt{s}$ *for any real number* $s \in (9, 1127/125]$.
(3) *For* $(k_1, k_2) \in \{(125, 1127), (500, 4507)\}$, *let* $W \to \mathbb{P}^2$ *be the normalized blow-up of the ideal sheaf which is equal to* (x^{k_2}, y^{k_1}) *near* p *and trivial away from* p, *let* E_W *be the exceptional divisor, and let* H_W *be the pullback of* $\mathcal{O}_{\mathbb{P}^2}(1)$. *Then* E_W *is a prime divisor,* $H_W^2 = 1$, $H_W \cdot E_W = 0$, $E_W^2 = -1/(k_1k_2)$, *and the nef cone of* W *is generated by* H_W *and* $H_W - \sqrt{k_1k_2}E_W$.
(4) *The tangent line* J *of the curve* $\{y = 0\}$ *at* p *satisfies* $(J \cdot \{y = 0\})_p = 2$.

Proof. Fix affine coordinates U, V on an affine plane $\mathbb{A}^2 \subset \mathbb{P}^2$ such that $p = (0, 0)$, and let $\mathfrak{m}_p$ be the maximal ideal of $\mathcal{O}_{\mathbb{P}^2,p}$. For local parameters x, y of $\mathbb{P}^2$ at p, the 9-*jet* of (x, y) is the class of (x, y) in $(\mathfrak{m}_p/\mathfrak{m}_p^{10})^2$. Let $\mathcal{J} \subset (\mathfrak{m}_p/\mathfrak{m}_p^{10})^2$ be the subset of pairs whose linear terms are linearly independent. Then $\mathcal{J}$ is a non-empty open subset of an affine space, hence irreducible. Any pair of polynomials in U, V of degree at most 9 whose class belongs to $\mathcal{J}$ is a pair of algebraic local parameters of $\mathbb{P}^2$ at p with this 9-jet. In this proof, a property holds for a *very general* element of $\mathcal{J}$ if it holds for any element of $\mathcal{J}$ outside a countable union of proper closed subsets of $\mathcal{J}$. For any local parameters x, y of $\mathbb{P}^2$ at p, we define v_s and μ by the formulas in the statement.

We first collect the conditions from [Tev26]. Let $s \in (9, 1127/125]$ be a rational number, and write $s = k_2/k_1$, where k_1, k_2 are coprime positive integers. Then $k_2 > 9k_1$ and $\lfloor k_2/k_1 \rfloor = 9$. By [Tev26, Theorem C] and [Tev26, Section 5, the second paragraph after Theorem C], for a very general element of $\mathcal{J}$ and any local parameters x, y with this 9-jet, we have

$$T_{\mathcal{O}_{\mathbb{P}^2}(1)}(k_1v_s) = \sqrt{k_1k_2},$$

and $k_1v_s(P) < d\sqrt{k_1k_2}$ for any non-zero form P of degree d. Since $(9, 1127/125] \cap \mathbb{Q}$ is countable, a very general element of $\mathcal{J}$ satisfies the conditions above for any rational number $s \in (9, 1127/125]$ simultaneously. Moreover, let $(k_1, k_2) \in \{(125, 1127), (500, 4507)\}$, so that $k_2/k_1 \in (9, 1127/125]$, and let W, E_W, and H_W be as in (3). By [Tev26, Section 5, the paragraph before (5.6), and (5.6)], which hold for any local parameters x, y, E_W is a prime divisor, $H_W^2 = 1$, $H_W \cdot E_W = 0$, and $E_W^2 = -1/(k_1k_2)$. By [Tev26, Corollary 5.1], which holds for local parameters satisfying the conditions above for $s = k_2/k_1$, the nef cone of W is generated by H_W and $H_W - \sqrt{k_1k_2}E_W$.

Next we pull back the condition of [DHK+17] to $\mathcal{J}$. Let $\mathcal{J}^\circ \subset \mathcal{J}$ be the open subset of pairs $(\bar{x}, \bar{y})$ such that the coefficient of V in the linear term of $\bar{y}$ is non-zero. Let x, y be local parameters of $\mathbb{P}^2$ at p whose 9-jet belongs to $\mathcal{J}^\circ$. By the Weierstrass preparation theorem, there exist a power series $\gamma \in U\mathbb{C}[[U]]$ and a unit $\epsilon \in \mathbb{C}[[U, V]]$ such that $y = \epsilon \cdot (V - \gamma(U))$. Since the linear terms of x and y are linearly independent, $x(U, \gamma(U)) = \epsilon' U$ for a unit $\epsilon' \in \mathbb{C}[[U]]$, so

$$x = \epsilon' U + (V - \gamma(U))b$$

for some $b \in \mathbb{C}[[U, V]]$. Let $s > 1$, and let v' be the quasi-monomial valuation over $\mathbb{P}^2$ such that

$$v'\Big(\sum_{i,j} c_{ij}U^i(V - \gamma(U))^j\Big) = \min\{i + sj \mid c_{ij} \neq 0\}.$$

Then

$$v'(x - \epsilon'(0)U) \ge \min\{2, s\} > 1 \quad \text{and} \quad v'(y - \epsilon(0)(V - \gamma(U))) \ge s + 1 > s.$$

Thus, in the graded ring of the filtration defined by v', which is the polynomial ring generated by the classes of U and $V - \gamma(U)$, the classes of x and y are non-zero multiples of the generators. Hence $v'(\sum_{i,j} c_{ij}x^iy^j) = \min\{i + sj \mid c_{ij} \neq 0\}$, that is, $v' = v_s$.

In particular, v_8 is the valuation of weights 1 and 8 in the transverse parameter U and the equation $V - \gamma(U)$ of the smooth germ $\{y = 0\}$. If $\gamma' \in U\mathbb{C}[[U]]$ and $\gamma' - \gamma \in U^9\mathbb{C}[[U]]$, then for $s = 8$ we have $v'(\gamma'(U) - \gamma(U)) \ge 9 > 8$, so $V - \gamma'(U)$ and $V - \gamma(U)$ have equal classes in the graded ring of the filtration defined by v', and hence v_8 is also the valuation of weights 1 and 8 in U and $V - \gamma'(U)$. Similarly, if x', y' are local parameters such that $x' - x, y' - y \in \mathfrak{m}_p^9$, then

$v_8(x'-x) \geq 9 > 1$ and $v_8(y'-y) \geq 9 > 8$, so v_8 is also the valuation of weights 1 and 8 in x' and y'. Thus $\mu(8)$ only depends on the 8-jet of γ.

Let $\mathbb{A}_8$ be the affine space of 8-jets of elements of $U\mathbb{C}[[U]]$. In [DHK+17, Definition 2.2], for $\xi \in U\mathbb{C}[[U]]$, $v(\xi, 8)$ is the valuation of weights 1 and 8 in U and $V - \xi(U)$, so $v(\gamma, 8) = v_8$ by the previous paragraph. By [DHK+17, Proposition 2.17], ξ is very general if the 7-jet of ξ belongs to a countable intersection of non-empty Zariski open subsets of the affine space $\mathbb{A}^7$ of 7-jets, and the preimage of such an intersection under the truncation $\mathbb{A}_8 \to \mathbb{A}^7$ is the complement of a countable union of proper closed subsets of $\mathbb{A}_8$. By [DHK+17, Definition 2.2, Proposition 2.17, Lemma 5.1, Proposition 5.2, Table 5.1 (D2), and Remark 5.8], $\mu(8) = 17/6$ if the germ $\{y = 0\}$ is very general, that is, if the 8-jet of γ is outside a countable union of proper closed subsets of $\mathbb{A}_8$. Let $\Phi : \mathcal{J}^\circ \to \mathbb{A}_8$ send $(\bar{x}, \bar{y})$ to the 8-jet of γ. The coefficients of γ up to degree 8 are polynomials in the coefficients of $\bar{y}$ and the inverse of the coefficient of V, so Φ is a morphism. Moreover, Φ is surjective, because the class of $(U, V - \gamma(U))$ is sent to the 8-jet of γ for any polynomial $\gamma \in U\mathbb{C}[U]$ of degree at most 8. Hence the preimage under Φ of a proper closed subset of $\mathbb{A}_8$ is a proper closed subset of $\mathcal{J}^\circ$. Since $\mathcal{J} \backslash \mathcal{J}^\circ$ is a proper closed subset of $\mathcal{J}$, a very general element of $\mathcal{J}$ satisfies $\mu(8) = 17/6$.

For $(\bar{x}, \bar{y}) \in \mathcal{J}^\circ$, write $\gamma = \gamma_1 U + \gamma_2 U^2 + \cdots$. The tangent line J of $\{y = 0\}$ at p is $\{V = \gamma_1 U\}$, and

$$(J \cdot \{y = 0\})_p = \dim_{\mathbb{C}} \mathbb{C}[[U]]/(\gamma(U) - \gamma_1 U).$$

Thus $(J \cdot \{y = 0\})_p = 2$ if and only if $\gamma_2 \neq 0$, which is a non-empty open condition on $\mathcal{J}^\circ$.

Since $\mathbb{C}$ is uncountable, a non-empty open subset of the irreducible variety $\mathcal{J}$ is not contained in a countable union of proper closed subsets of $\mathcal{J}$. Thus there exists an element of $\mathcal{J}^\circ$ satisfying the conditions from [Tev26] for any rational number $s \in (9, 1127/125]$, $\mu(8) = 17/6$, and $\gamma_2 \neq 0$. Let x, y be polynomials in U, V of degree at most 9 with this 9-jet. Then x, y are algebraic local parameters of $\mathbb{P}^2$ at p, and they satisfy all the conditions above, because the conditions only depend on the 9-jet. We prove (1)–(4) for x, y.

(1) For any real numbers $1 \leq u \leq s$ and integers $i, j \geq 0$, we have $i + uj \leq i + sj \leq (s/u)(i + uj)$. Hence $v_u(\phi) \leq v_s(\phi) \leq (s/u)v_u(\phi)$ for any non-zero $\phi \in \mathbb{C}[[x, y]]$, and (1) follows by taking suprema over $\mathbb{Q}$-divisors $D \geq 0$ with $D \sim_{\mathbb{Q}} \mathcal{O}_{\mathbb{P}^2}(1)$. Moreover, any such D is equal to $\frac{1}{m}\operatorname{div}(P)$ for some form P of degree m, and $v_s(P) \leq s \operatorname{mult}_p P \leq sm$, so $\mu(s) \leq s$. By (1), for any real numbers $1 \leq u \leq s$,

$$0 \leq \mu(s) - \mu(u) \leq \left(\frac{s}{u} - 1\right)\mu(u) \leq s - u.$$

In particular, μ is continuous.

(2) Let $s \in (9, 1127/125]$ be a rational number, and write $s = k_2/k_1$, where k_1, k_2 are coprime positive integers. Then

$$\mu(s) = \frac{T_{\mathcal{O}_{\mathbb{P}^2}(1)}(k_1 v_s)}{k_1} = \frac{\sqrt{k_1 k_2}}{k_1} = \sqrt{s}.$$

Since μ is continuous, $\mu(s) = \sqrt{s}$ for any real number $s \in (9, 1127/125]$, and $\mu(9) = \lim_{s \to 9+} \sqrt{s} = 3$. Moreover, $\mu(8) = 17/6$ by the choice of x, y.

(3) This follows from the second paragraph of this proof, since x, y satisfy the conditions from [Tev26] for $s \in \{1127/125, 4507/500\}$.

(4) This follows from $\gamma_2 \neq 0$. □

Set-up 4.2. Let p, x, y, v_s, μ, π, Z, E, H, o, z, w_t, J_ξ, W_h, E_h, H_h, W_l, E_l, H_l, s_h, s_l, t_h, t_l, β_h, β_l, q, ℓ_1, ℓ_2, c_0, r, and α_0 be as follows.

(1) $p \in \mathbb{P}^2$ is a closed point, and x, y are algebraic local parameters of $\mathbb{P}^2$ at p satisfying (1)–(4) of Lemma 4.1.

(2) For any real number $s \geq 1$, v_s is the quasi-monomial valuation over $\mathbb{P}^2$ such that

$$v_s\Big(\sum_{i,j} c_{ij} x^i y^j\Big) = \min\{i + sj \mid c_{ij} \neq 0\}$$

for any non-zero element $\sum_{i,j} c_{ij}x^iy^j$ of $\widehat{\mathcal{O}}_{\mathbb{P}^2,p} = \mathbb{C}[[x,y]]$, and $\mu(s) := T_{\mathcal{O}_{\mathbb{P}^2}(1)}(v_s)$.

(3) $\pi : Z \to \mathbb{P}^2$ is the blow-up of p, E is the exceptional curve, and $H := \pi^*\mathcal{O}_{\mathbb{P}^2}(1)$.
(4) $o \in E$ is the point corresponding to the tangent direction of $\{y = 0\}$ at p, and $z := y/x$, so that x, z are local parameters of Z at o and $E = \{x = 0\}$ near o.
(5) For any real number $t \geq 1$, w_t is the quasi-monomial valuation over Z such that

$$w_t\Big(\sum_{i,j} c_{ij}x^iz^j\Big) = \min\{i + tj \mid c_{ij} \neq 0\}$$

for any non-zero element $\sum_{i,j} c_{ij}x^iz^j$ of $\widehat{\mathcal{O}}_{Z,o} = \mathbb{C}[[x,z]]$, so that $w_t = v_{t+1}$.
(6) For any closed point $\xi \in E$, $J_\xi \subset \mathbb{P}^2$ is the line passing through p whose tangent direction at p corresponds to ξ.
(7) $W_h \to \mathbb{P}^2$ and $W_l \to \mathbb{P}^2$ are the normalized blow-ups of the ideal sheaves which are equal to (x^{1127}, y^{125}) and (x^{4507}, y^{500}), respectively, near p and trivial away from p.
(8) E_h and E_l are the exceptional divisors of $W_h \to \mathbb{P}^2$ and $W_l \to \mathbb{P}^2$, respectively.
(9) H_h and H_l are the pullbacks of $\mathcal{O}_{\mathbb{P}^2}(1)$ to W_h and W_l, respectively.
(10) $s_h := 1127/125$, $s_l := 4507/500$, $t_h := s_h - 1$, $t_l := s_l - 1$, $\beta_h := 15/16$, $\beta_l := 611/10000$, $q := \beta_h + \beta_l$, $\ell_1 := 17646/10^8$, and $\ell_2 := 5435/10^8$.
(11) $c_0 := \beta_h - \beta_l t_l$,

$$r := \frac{\ell_2}{\ell_1} + \sqrt{\frac{\ell_2^2}{\ell_1^2} + 16c_0}, \quad \text{and} \quad \alpha_0 := \frac{r}{8\ell_1}.$$

4.2. **Curves on weighted blow-ups.** In this subsection, we construct plane curves all of whose branches at p have first Puiseux pair (k_1, k_2). They are the images of general members of very ample linear systems on the weighted blow-ups of Lemma 4.1(3), and their strict transforms on Z are the components of the boundary in Construction-Proposition 4.4 that pass through o.

Lemma 4.3. *Notation as in Set-up 4.2. Let (W, E_W, H_W, k_1, k_2) be $(W_h, E_h, H_h, 125, 1127)$ or $(W_l, E_l, H_l, 500, 4507)$. Let d, k be positive integers such that $k_1k_2 \mid k$ and $k^2 < k_1k_2d^2$. Then there exists a positive integer n_1 satisfying the following. For any integer $n \geq n_1$, the divisor $ndH_W - nkE_W$ is very ample, and a general member D of $|ndH_W - nkE_W|$ is a smooth irreducible curve which does not pass through the singular points of W and intersects E_W transversally at exactly $nk/(k_1k_2)$ points. The image G of D in $\mathbb{P}^2$ is an irreducible curve of degree nd which is smooth away from p, and G has exactly $nk/(k_1k_2)$ analytic branches at p. Each branch has a parametrization $x = \tau^{k_1}\varphi_1(\tau)$, $y = \tau^{k_2}\varphi_2(\tau)$ with $\varphi_1, \varphi_2 \in \mathbb{C}[[\tau]]$ and $\varphi_1(0)\varphi_2(0) \neq 0$. In particular, $\operatorname{mult}_p G = nk/k_2$, and $\pi_*^{-1}G$ intersects E only at o.*

Proof. Let $f : W \to \mathbb{P}^2$ be the induced morphism. We first describe W near E_W. Let X_1, X_2 be the coordinates of $\mathbb{A}^2$. Since dx and dy are linearly independent at p, there exists an open neighborhood $\mathcal{U}$ of p such that (x, y) defines an étale morphism $\mathcal{U} \to \mathbb{A}^2$, p is the only point of $\mathcal{U}$ over the origin, and the ideal (x^{k_2}, y^{k_1}) on $\mathcal{U}$ is the pullback of $(X_1^{k_2}, X_2^{k_1})$. Since blowing up commutes with flat base change and normalization commutes with étale base change, $f^{-1}(\mathcal{U}) \cong \mathcal{U} \times_{\mathbb{A}^2} W_0$, where W_0 is the normalized blow-up of $(X_1^{k_2}, X_2^{k_1})$ on $\mathbb{A}^2$, and E_W is isomorphic to the exceptional curve E_0 of W_0. Since the normalized blow-up of a monomial ideal $\mathfrak{a}$ is the toric variety of the normal fan of the Newton polyhedron of $\mathfrak{a}$ [CLS11, Corollary 11.3.2(a)] (which treats the ideal $\mathfrak{a}'$ spanned by the monomials with exponents in the Newton polyhedron of $\mathfrak{a}$, and the normalized blow-ups of $\mathfrak{a}$ and $\mathfrak{a}'$ coincide, since the Rees algebras of $\mathfrak{a}$ and $\mathfrak{a}'$ have equal normalizations by [CLS11, Proposition 1.3.8]), and the Newton polyhedron of $(X_1^{k_2}, X_2^{k_1})$ is $\operatorname{conv}\{(k_2, 0), (0, k_1)\} + \mathbb{R}^2_{\geq 0}$, the surface W_0 is the toric surface whose fan consists of the cones generated by $(1, 0), (k_1, k_2)$ and by $(k_1, k_2), (0, 1)$ together with their faces, and $E_0 \cong \mathbb{P}^1$ is the curve corresponding to the ray spanned by (k_1, k_2) (see also [Tev26, Section 5]). Since $\gcd(k_1, k_2) = 1$, the cones generated by $(1, 0), (k_1, k_2)$ and by $(k_1, k_2), (0, 1)$ have multiplicities k_2 and k_1, and $k_2 > k_1 > 1$. Thus the singular locus of W_0 consists of two points of E_0, which are

cyclic quotient singularities of orders k_2 and k_1, and $k_1k_2E_0$ is Cartier, because the class group of a cyclic quotient singularity of order m is $\mathbb{Z}/m\mathbb{Z}$. Since $W\setminus E_W \cong \mathbb{P}^2\setminus\{p\}$ and being Cartier is an étale local property, the singular locus of W consists of two points of E_W, which are cyclic quotient singularities of orders k_2 and k_1, and $k_1k_2E_W$ is Cartier. On the open torus orbit of E_0, the divisors of X_1 and X_2 are k_1E_0 and k_2E_0, because the strict transforms of $\{X_1=0\}$ and $\{X_2=0\}$ intersect E_0 only at the singular points of W_0. Hence for any point $\eta\in E_W$ which is not a singular point of W, there exist local parameters ζ, u of W at η such that $E_W=\{\zeta=0\}$ near η, $x=\zeta^{k_1}\epsilon_1$, and $y=\zeta^{k_2}\epsilon_2$ for some units $\epsilon_1,\epsilon_2\in\mathcal{O}_{W,\eta}$.

Since $k^2<k_1k_2d^2$, we have $0<k/d<\sqrt{k_1k_2}$. Thus $H_W-(k/d)E_W$ belongs to the interior of

$$\operatorname{Nef}(W)=\mathbb{R}_{\geq 0}H_W+\mathbb{R}_{\geq 0}(H_W-\sqrt{k_1k_2}E_W)$$

(Lemma 4.1(3)), so $H_W-(k/d)E_W$ is ample by [Laz04, Theorem 1.4.23(ii)]. Since $k_1k_2\mid k$ and $k_1k_2E_W$ is Cartier, dH_W-kE_W is an ample Cartier divisor. By [Laz04, Theorem 1.2.6(iv)], there exists a positive integer n_1 such that $n(dH_W-kE_W)$ is very ample for any integer $n\geq n_1$.

Fix an integer $n\geq n_1$, and let D be a general member of $|ndH_W-nkE_W|$. The linear system $|ndH_W-nkE_W|$ embeds W into a projective space, so D does not pass through the singular points of W and does not contain E_W. The restrictions of $|ndH_W-nkE_W|$ to W_{reg} and to the smooth curve $E_W\cap W_{\mathrm{reg}}$ are base point free. By [Har77, Corollary III.10.9 and Remark III.10.9.2], D is smooth and $D\cap E_W$ is reduced. Since W is a normal projective surface and $D\geq 0$ is an ample divisor, D is connected by [Har77, Corollary III.7.9]. Thus D is a smooth irreducible curve, and D intersects E_W transversally at exactly

$$(ndH_W-nkE_W)\cdot E_W=-nkE_W^2=\frac{nk}{k_1k_2}$$

points, none of which is a singular point of W.

Since f is an isomorphism over $\mathbb{P}^2\setminus\{p\}$ and $f^{-1}(p)=E_W$, the image $G=f(D)$ is an irreducible curve which is smooth away from p, $D\to G$ is the normalization of G, and $f^*G=D+cE_W$ for some integer c. Since $H_W^2=1$ and $H_W\cdot E_W=0$,

$$\deg G=f^*G\cdot H_W=D\cdot H_W=nd.$$

The analytic branches of G at p correspond to the points of $D\cap E_W$ (see e.g. [CA00, Sections 2.5–2.6]), so G has exactly $nk/(k_1k_2)$ branches at p. Let $\eta\in D\cap E_W$, and let ζ, u be local parameters of W at η and $\epsilon_1,\epsilon_2\in\mathcal{O}_{W,\eta}$ units such that $E_W=\{\zeta=0\}$ near η, $x=\zeta^{k_1}\epsilon_1$, and $y=\zeta^{k_2}\epsilon_2$, which exist by the first paragraph of this proof. Since D is smooth at η and transverse to $E_W=\{\zeta=0\}$, the restriction τ of ζ to D is a local parameter of D at η. Thus the branch of G at p corresponding to η has the parametrization $x=\tau^{k_1}\varphi_1(\tau)$, $y=\tau^{k_2}\varphi_2(\tau)$, where $\varphi_1,\varphi_2\in\mathbb{C}[[\tau]]$ are the restrictions of ϵ_1,ϵ_2 to D, so $\varphi_1(0)\varphi_2(0)\neq 0$. Since $k_1<k_2$, each branch has multiplicity k_1 at p, and

$$\operatorname{mult}_p G=\frac{nk}{k_1k_2}\cdot k_1=\frac{nk}{k_2}.$$

Finally, x, z are local parameters of Z at o and $E=\{x=0\}$ near o (Set-up 4.2(4)). On each branch of G at p, we have

$$z=\frac{y}{x}=\tau^{k_2-k_1}\frac{\varphi_2(\tau)}{\varphi_1(\tau)}$$

with $k_2>k_1$, so the strict transform of the branch passes through the point $x=z=0$, which is o. Since $\pi_*^{-1}G\cap E$ consists of the points of E through which the strict transforms of the branches of G at p pass, $\pi_*^{-1}G$ intersects E only at o. □

4.3. **The pair.** In this subsection, we construct the pair (Z,B). The boundary has the form C/N with C a reduced divisor, because Section 5 uses a boundary all of whose components have coefficient $1/N$. Besides the curves of Lemma 4.3, C contains the strict transforms of general lines through p, and their number makes Lemma 4.6(2) and Set-up 5.1(3) hold.

Construction-Proposition 4.4. Notation as in Set-up 4.2. There exists a positive integer n_0 satisfying the following. Let $n \geq n_0$ be an integer and $N := 10^8 n$. Let $D_{h,1}, \dots, D_{h,9375}$ be general members of $|30027nH_h - 11270000nE_h|$, $D_{l,1}, \dots, D_{l,2444}$ general members of $|7506nH_l - 11267500nE_l|$, and $\xi_1, \dots, \xi_{134565n}$ general closed points of E. Let C be the sum of the strict transforms on Z of $J_{\xi_1}, \dots, J_{\xi_{134565n}}$ and of the images in $\mathbb{P}^2$ of $D_{h,1}, \dots, D_{h,9375}, D_{l,1}, \dots, D_{l,2444}$, and let $B := C/N$ and $L := \ell_1 H - \ell_2 E$. Then the following hold.

(1) C is a reduced divisor, $E \not\subset \operatorname{Supp} C$, $o \notin \pi_*^{-1} J_{\xi_i}$ for any i, and C is simple normal crossing on $Z \backslash \{o\}$.
(2) $C \sim 299982354nH - 99994565nE$, $-(K_Z + B) \sim_{\mathbb{Q}} L$, and L is ample.
(3) For any normalized quasi-monomial valuation w over Z with center o and $m(w) = 1$, we have $w(B) = \beta_h \theta + \beta_l \min\{\theta, t_l\}$, where $\theta := s(w \wedge w_{t_h})$.
(4) (Z, B) is klt.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we define n_0 and prove (1).

Let n_h and n_l be the integers n_1 of Lemma 4.3 for

$$(W, E_W, H_W, k_1, k_2; d, k) = (W_h, E_h, H_h, 125, 1127; 30027, 11270000) \quad \text{and}$$
$$(W, E_W, H_W, k_1, k_2; d, k) = (W_l, E_l, H_l, 500, 4507; 7506, 11267500),$$

respectively. Lemma 4.3 can be applied in both cases, because $k^2 < k_1 k_2 d^2$ is equivalent to $(k/(k_1 k_2))^2 k_1 k_2 < d^2$, and

$$11270000 = 80 \cdot 125 \cdot 1127, \qquad 80^2 \cdot 125 \cdot 1127 = 901600000 < 901620729 = 30027^2,$$
$$11267500 = 5 \cdot 500 \cdot 4507, \qquad 5^2 \cdot 500 \cdot 4507 = 56337500 < 56340036 = 7506^2.$$

Let $n_0 := \max\{n_h, n_l\}$, and fix an integer $n \geq n_0$.

For $1 \leq i \leq 9375$ and $1 \leq j \leq 2444$, let $G_{h,i}$ and $G_{l,j}$ be the images of $D_{h,i}$ and $D_{l,j}$ in $\mathbb{P}^2$. By Lemma 4.3, $G_{h,i}$ is an irreducible curve of degree $30027n$ which is smooth away from p, $G_{h,i}$ has exactly $80n$ branches at p, each with a parametrization $x = \tau^{125}\varphi_1(\tau)$, $y = \tau^{1127}\varphi_2(\tau)$ such that $\varphi_1(0)\varphi_2(0) \neq 0$, $\operatorname{mult}_p G_{h,i} = 10000n$, and $\pi_*^{-1} G_{h,i}$ intersects E only at o. Similarly, $G_{l,j}$ is an irreducible curve of degree $7506n$ which is smooth away from p, $G_{l,j}$ has exactly $5n$ branches at p, each with a parametrization $x = \tau^{500}\varphi_1(\tau)$, $y = \tau^{4507}\varphi_2(\tau)$ such that $\varphi_1(0)\varphi_2(0) \neq 0$, $\operatorname{mult}_p G_{l,j} = 2500n$, and $\pi_*^{-1} G_{l,j}$ intersects E only at o. In particular, $\pi_*^{-1} G_{h,i}$ and $\pi_*^{-1} G_{l,j}$ are smooth on $Z \backslash \{o\}$.

We choose the members and the points one at a time. At each stage, the members or points satisfying the conditions below form a dense open subset, so the tuples satisfying all the conditions contain a dense open subset of the product of the linear systems and of $E^{134565n}$, and the conditions hold for general members and general points. The linear systems $|30027nH_h - 11270000nE_h|$ and $|7506nH_l - 11267500nE_l|$ are very ample, hence base point free on $W_h \backslash E_h$ and on $W_l \backslash E_l$, both of which are isomorphic to $\mathbb{P}^2 \backslash \{p\}$. Assume that some of the curves $G_{h,i}$ and $G_{l,j}$ are already fixed, and let Σ be the finite set of points other than p at which two of them intersect. A general member D of either linear system is different from the fixed members, so the image G of D is different from the fixed curves $G_{h,i}$ and $G_{l,j}$, which have degree $30027n$ or $7506n$. Moreover, $G \cap \Sigma = \emptyset$, and by [Har77, Corollary III.10.9 and Remark III.10.9.2] applied to the restriction of the linear system to the smooth curve $G' \backslash \{p\}$ for any fixed curve G', G intersects G' transversally away from p. Thus the curves $G_{h,i}$ and $G_{l,j}$ are distinct, pairwise transverse away from p, and no three of them pass through a common point other than p.

For any closed point $\xi \in E$, $\pi_*^{-1} J_\xi$ is the member of the base point free pencil $|H - E|$ passing through ξ, and $\pi_*^{-1} J_\xi$ intersects E transversally at ξ only. Since $(H - E)^2 = 0$, distinct members of $|H - E|$ are disjoint. Assume that some of the points ξ_i are already fixed. A general closed point $\xi \in E$ is different from o and from the fixed points, and $\pi_*^{-1} J_\xi$ does not pass through the finite set of points other than o at which two of the curves $\pi_*^{-1} G_{h,i}$ and $\pi_*^{-1} G_{l,j}$ intersect. By

[Har77, Corollary III.10.9 and Remark III.10.9.2] applied to the restriction of $|H-E|$ to the smooth curves $\pi_*^{-1}G_{h,i}\backslash\{o\}$ and $\pi_*^{-1}G_{l,j}\backslash\{o\}$, $\pi_*^{-1}J_\xi$ intersects them transversally. Since $\xi \neq o$ and $\pi_*^{-1}J_\xi \cap E = \{\xi\}$, we have $o \notin \pi_*^{-1}J_\xi$.

Thus the components of C are distinct, $E \not\subset \operatorname{Supp} C$, and $o \notin \pi_*^{-1}J_{\xi_i}$ for any i. Moreover, on $Z\backslash\{o\}$, the components of C are smooth, pairwise transverse, and no three of them pass through a common point. This proves (1).

Step 2. In this step, we prove (2).

For any plane curve G, Lemma 2.10(2) implies $\pi_*^{-1}G \sim (\deg G)H - (\operatorname{mult}_p G)E$. By **Step 1** and $\deg J_{\xi_i} = \operatorname{mult}_p J_{\xi_i} = 1$, the image π_*C satisfies

$$\deg \pi_*C = 9375\cdot 30027n + 2444\cdot 7506n + 134565n = 299982354n,$$

$$\operatorname{mult}_p \pi_*C = 9375\cdot 10000n + 2444\cdot 2500n + 134565n = 99994565n.$$

Thus $C \sim 299982354nH - 99994565nE$. Since $-K_Z \sim 3H - E$ (Lemma 2.10(1)) and $N = 10^8 n$,

$$-(K_Z+B) \sim_{\mathbb{Q}} \Big(3 - \frac{299982354}{10^8}\Big)H - \Big(1 - \frac{99994565}{10^8}\Big)E = \ell_1 H - \ell_2 E = L.$$

Since $\ell_1 = 17646/10^8 > \ell_2 = 5435/10^8 > 0$, L is ample by Lemma 2.10(1). This proves (2).

Step 3. In this step, we prove (3).

In this step, skewness, tree multiplicity, and $\wedge$ are taken at o, and $\mathfrak{m}_o$ denotes the maximal ideal of $\widehat{\mathcal{O}}_{Z,o} = \mathbb{C}[[x,z]]$. For any real numbers $1 \le t \le t'$ and integers $i,j \ge 0$, we have $i+tj \le i+t'j$ and $i+tj \le t(i+j)$. Hence $w_t \le w_{t'}$, $w_t(\phi) \le t \operatorname{mult}_o \phi$ for any non-zero $\phi \in \mathfrak{m}_o$, and $w_t(z) = t$, so $s(w_t) = t$. By Lemma 2.9(2) applied to $w_{t'}$, any normalized valuation $\omega \le w_{t'}$ at o with $s(\omega) = t$ is equal to w_t.

Let w be a normalized quasi-monomial valuation over Z with center o and $m(w) = 1$, and let $\theta := s(w \wedge w_{t_h})$. Since $w \wedge w_{t_h} \le w_{t_h}$, we have $w \wedge w_{t_h} = w_\theta$ and $1 \le \theta \le t_h$.

Let $g \in \mathfrak{m}_o$ be an irreducible element defining a branch of $\pi_*^{-1}G_{h,i}$ at o for some i. By **Step 1** and $z = y/x$, the branch has a parametrization $x = \tau^{125}\varphi_1(\tau)$, $z = \tau^{1002}\varphi_2(\tau)/\varphi_1(\tau)$ such that $\varphi_1(0)\varphi_2(0) \neq 0$. Thus $\operatorname{mult}_o g = 125$ and $(g\cdot x^i z^j)_o = 125i + 1002j$ (see e.g. [CA00, Sections 2.5–2.6]), so

$$\nu_g(x^i z^j) = i + t_h j$$

for any integers $i,j \ge 0$. Hence $\nu_g \ge w_{t_h}$ and $\nu_g(z) = t_h$. Let $u := w \wedge \nu_g$ and $\sigma := s(u)$. By Lemma 2.9(2), $u = w^{(\sigma)}$ is a normalized quasi-monomial valuation, and by Lemma 2.9(4), $m(u) \le m(w) = 1$.

We show that $\sigma \le t_h$. Suppose that $\sigma > t_h$. Since $u \le \nu_g$ and $w_{t_h} \le \nu_g$, the valuations u and w_{t_h} are comparable by [FJ04, Theorem 3.14]. If $u \le w_{t_h}$, then $\sigma \le s(w_{t_h}) = t_h$ by the definition of skewness, which is impossible. Thus $w_{t_h} \le u \le \nu_g$, and

$$t_h = w_{t_h}(z) \le u(z) \le \nu_g(z) = t_h.$$

On the other hand, since $m(u) = 1$, Lemma 2.9(3) implies that there exist generators x', y' of $\mathfrak{m}_o$ such that

$$u\Big(\sum_{i,j} c_{ij} x'^i y'^j\Big) = \min\{i + \sigma j \mid c_{ij} \neq 0\}.$$

Hence $u(z) = i + \sigma j$ for some integers $i,j \ge 0$. Since $u(z) = t_h < \sigma$, we have $j = 0$, so $t_h = u(z)$ is an integer, which contradicts $t_h = 1002/125$. Thus $\sigma \le t_h$.

By [FJ04, Theorem 3.14] again, the valuations u and w_{t_h} are comparable. If $w_{t_h} \le u$, then $t_h \le \sigma$, so $\sigma = t_h$, and Lemma 2.9(2) applied to u implies $w_{t_h} = u^{(t_h)} = u$. Thus $u \le w_{t_h}$ in both cases. Since $u \le w$, we have $u \le w \wedge w_{t_h}$. Conversely, $w \wedge w_{t_h} \le w$ and $w \wedge w_{t_h} \le w_{t_h} \le \nu_g$, so $w \wedge w_{t_h} \le w \wedge \nu_g = u$. Therefore $w \wedge \nu_g = w \wedge w_{t_h} = w_\theta$, and Lemma 2.9(1) implies

$$w(g) = \operatorname{mult}_o g \cdot s(w \wedge \nu_g) = 125\theta.$$

Let $\psi \in \mathfrak{m}_o$ be an irreducible element defining a branch of $\pi_*^{-1}G_{l,j}$ at o for some j. By **Step 1**, the branch has a parametrization $x = \tau^{500}\varphi_1(\tau)$, $z = \tau^{4007}\varphi_2(\tau)/\varphi_1(\tau)$ such that $\varphi_1(0)\varphi_2(0) \neq 0$. Thus $\operatorname{mult}_o \psi = 500$, $\nu_\psi \geq w_{t_l}$, and $\nu_\psi(z) = t_l$. Since $t_l = 4007/500$ is not an integer, replacing g and t_h with ψ and t_l, the argument of the previous two paragraphs implies $w \wedge \nu_\psi = w \wedge w_{t_l}$. Let $\theta' := \min\{\theta, t_l\}$. Since $w_{\theta'} \leq w_\theta \leq w$ and $w_{\theta'} \leq w_{t_l}$, we have $w_{\theta'} \leq w \wedge w_{t_l}$. Conversely, $w \wedge w_{t_l} \leq w$ and $w \wedge w_{t_l} \leq w_{t_l} \leq w_{t_h}$, so $w \wedge w_{t_l} \leq w \wedge w_{t_h} = w_\theta$. Thus

$$\theta' \leq s(w \wedge w_{t_l}) \leq \min\{s(w_\theta), s(w_{t_l})\} = \theta'$$

by the definition of skewness. Hence Lemma 2.9(1) implies

$$w(\psi) = \operatorname{mult}_o \psi \cdot s(w \wedge \nu_\psi) = 500 \min\{\theta, t_l\}.$$

By **Step 1**, the components of C passing through o are the curves $\pi_*^{-1}G_{h,i}$ and $\pi_*^{-1}G_{l,j}$. Each $\pi_*^{-1}G_{h,i}$ has exactly $80n$ branches at o, and each $\pi_*^{-1}G_{l,j}$ has exactly $5n$ branches at o. Since $N = 10^8 n$,

$$\begin{aligned} w(B) &= \frac{9375 \cdot 80n \cdot 125\theta + 2444 \cdot 5n \cdot 500 \min\{\theta, t_l\}}{10^8 n} \\ &= \frac{15}{16}\theta + \frac{611}{10000}\min\{\theta, t_l\} = \beta_h \theta + \beta_l \min\{\theta, t_l\}. \end{aligned}$$

This proves (3).

Step 4. In this step, we prove (4).

By (1), any prime divisor on Z has coefficient at most $1/N < 1$ in B. Let $\xi \in Z$ be a closed point, and let w be a normalized quasi-monomial valuation over Z with center ξ, where the skewness $s(w)$ is taken at ξ. By the definition of skewness, $w(B) \leq s(w) \operatorname{mult}_\xi B$, and by Lemma 2.9(5), $a(w, Z, 0) \geq 1 + s(w)$. Thus by Lemma 2.3,

$$a(w, Z, B) = a(w, Z, 0) - w(B) \geq 1 + s(w)(1 - \operatorname{mult}_\xi B).$$

If $\xi \neq o$, then C is simple normal crossing at ξ by (1), so $\operatorname{mult}_\xi C \leq 2$ and $\operatorname{mult}_\xi B \leq 2/N < 1$. If $\xi = o$, then the curves $\pi_*^{-1}J_{\xi_i}$ do not pass through o, and by **Step 1**,

$$\operatorname{mult}_o \pi_*^{-1}G_{h,i} = 80n \cdot 125 = 10000n \quad \text{and} \quad \operatorname{mult}_o \pi_*^{-1}G_{l,j} = 5n \cdot 500 = 2500n.$$

Hence

$$\operatorname{mult}_o B = \frac{9375 \cdot 10000n + 2444 \cdot 2500n}{10^8 n} = \frac{15}{16} + \frac{611}{10000} = \frac{4993}{5000} = q < 1.$$

Thus $a(w, Z, B) > 0$ in both cases. For any prime divisor F over Z whose center on Z is a closed point ξ, the valuation defined by F is cw for some real number $c > 0$ and some normalized quasi-monomial valuation w over Z with center ξ, and $a(F, Z, B) = c \cdot a(w, Z, B) > 0$. Therefore (Z, B) is klt. This proves (4). □

4.4. **Maximal vanishing orders on the blow-up.** In this subsection, we bound the maximal vanishing orders $T_L(w)$ for valuations w over Z, and we prove the arithmetic properties of the constants of Set-up 4.2. For any positive integer j divisible by 10^8, the sections of $\mathcal{O}_Z(jL)$ are the plane forms of degree $j\ell_1$ with multiplicity at least $j\ell_2$ at p (Lemma 2.10(2)). Thus (1), (3), (4), and (5) of the next lemma are elementary bounds by degrees of plane forms and intersection numbers on Z, while the exact value of $T_L(w_t)$ in (2) uses Lemma 4.1(2).

Lemma 4.5. *Notation as in Construction-Proposition 4.4. Then the following hold.*

(1) For any valuation w over Z,

$$T_L(w) \leq \ell_1 T_{\mathcal{O}_{\mathbb{P}^2}(1)}(\pi_* w) - \ell_2 w(E).$$

(2) $T_L(w_t) = \ell_1\sqrt{t+1} - \ell_2$ for any real number $t \in (8, t_h]$.

(3) $T_L(w) \leq \ell_1 s(w)$ for any normalized quasi-monomial valuation w over Z whose center is a closed point of $Z \backslash E$.

(4) *Let $\xi \in E$ be a closed point and w a normalized quasi-monomial valuation over Z with center ξ and $m(w) = 1$. Then $T_L(w) \le \ell_1(s(w)+1) - \ell_2$. If moreover $w(E) = w(\pi_*^{-1}J_\xi) = 1$, then $T_L(w) \le \ell_1 \max\{s(w), 2\} - \ell_2$.*

(5) *$T_L(E) = \ell_1 - \ell_2$, and $T_L(F) \le \ell_1$ for any prime divisor $F \ne E$ on Z.*

Proof. For a valuation w over Z, $w(\pi^*D_0) = (\pi_*w)(D_0)$ for any $\mathbb{R}$-divisor D_0 on $\mathbb{P}^2$ (Definition 2.2).

(1) Let $D \ge 0$ be a $\mathbb{Q}$-divisor on Z such that $D \sim_{\mathbb{Q}} L$. Let j be a positive integer divisible by 10^8 such that jD is a divisor and $jD \sim jL = j\ell_1 H - j\ell_2 E$. By Lemma 2.10(2), there exists a homogeneous polynomial g of degree $j\ell_1$ such that $\operatorname{mult}_p g \ge j\ell_2$ and

$$jD = \pi^* \operatorname{div}(g) - j\ell_2 E. \tag{4.1}$$

Since $G := \operatorname{div}(g)/(j\ell_1)$ is a $\mathbb{Q}$-divisor such that $G \ge 0$ and $G \sim_{\mathbb{Q}} \mathcal{O}_{\mathbb{P}^2}(1)$, we have

$$(\pi_*w)(\operatorname{div}(g)) = j\ell_1(\pi_*w)(G) \le j\ell_1 T_{\mathcal{O}_{\mathbb{P}^2}(1)}(\pi_*w).$$

Thus

$$w(D) = \frac{1}{j}\Big((\pi_*w)(\operatorname{div}(g)) - j\ell_2 w(E)\Big) \le \ell_1 T_{\mathcal{O}_{\mathbb{P}^2}(1)}(\pi_*w) - \ell_2 w(E).$$

Taking the supremum over D, we obtain (1).

(2) Let $t \in (8, t_h]$ and $s := t+1 \in (9, s_h]$. By Set-up 4.2(4)(5), $\pi_*w_t = v_s$ and $w_t(E) = w_t(x) = 1$. By (1) and Lemma 4.1(2),

$$T_L(w_t) \le \ell_1\mu(s) - \ell_2 = \ell_1\sqrt{s} - \ell_2.$$

We prove the reverse inequality. Any $\mathbb{Q}$-divisor $D_0 \ge 0$ on $\mathbb{P}^2$ such that $D_0 \sim_{\mathbb{Q}} \mathcal{O}_{\mathbb{P}^2}(1)$ is equal to $\operatorname{div}(P)/d$ for a non-zero homogeneous polynomial P of some degree $d \ge 1$. Since $\mu(s) = \sqrt{s}$, there exist non-zero homogeneous polynomials P_k of degrees $d_k \ge 1$, $k = 1, 2, \dots$, such that $\lim_{k\to+\infty} v_s(P_k)/d_k = \sqrt{s}$. For any non-zero $f \in \mathbb{C}[[x, y]]$, a monomial x^iy^j of f with $i + j = \operatorname{mult}_p f$ has value $i + sj \le s \operatorname{mult}_p f$, so $v_s(f) \le s \operatorname{mult}_p f$. Hence

$$\liminf_{k\to+\infty} \frac{\operatorname{mult}_p P_k}{d_k} \ge \lim_{k\to+\infty} \frac{v_s(P_k)}{sd_k} = \frac{1}{\sqrt{s}}.$$

Since $\ell_2/\ell_1 = 5435/17646 < 31/100$ and $\sqrt{s} \le \sqrt{s_h} < 16/5$, we have $(\ell_2/\ell_1)\sqrt{s} < 124/125 < 1$. Thus $\operatorname{mult}_p P_k > (\ell_2/\ell_1)d_k$ for any sufficiently large k. For such k, let $m_k := 10^8 d_k$. Then P_k^{17646} is a homogeneous polynomial of degree $17646d_k = m_k\ell_1$, and

$$\operatorname{mult}_p P_k^{17646} = 17646 \operatorname{mult}_p P_k > 5435 d_k = m_k\ell_2.$$

By Lemma 2.10(2),

$$D_k := \frac{1}{m_k}\Big(\pi^* \operatorname{div}(P_k^{17646}) - m_k\ell_2 E\Big)$$

is a $\mathbb{Q}$-divisor such that $D_k \ge 0$ and $m_kD_k \sim m_kL$. Since $w_t(E) = 1$,

$$w_t(D_k) = \frac{17646 v_s(P_k) - m_k\ell_2}{m_k} = \ell_1 \frac{v_s(P_k)}{d_k} - \ell_2,$$

which tends to $\ell_1\sqrt{s} - \ell_2$ as $k \to +\infty$. Hence $T_L(w_t) \ge \ell_1\sqrt{s} - \ell_2$, and we obtain (2).

(3) Let ζ be the center of w. Since $\zeta \notin E$, $w(E) = 0$. Let g be a non-zero homogeneous polynomial of degree d. Since π is an isomorphism near ζ, a local equation of $\pi^* \operatorname{div}(g)$ at ζ has multiplicity $\operatorname{mult}_{\pi(\zeta)} g$ at ζ, and $\operatorname{mult}_{\pi(\zeta)} g \le d$ because the dehomogenization of g at $\pi(\zeta)$ is a non-zero polynomial of degree at most d. By the definition of $s(w)$ (Definition 2.7),

$$w(\pi^* \operatorname{div}(g)) \le s(w) \operatorname{mult}_{\pi(\zeta)} g \le s(w)d.$$

Thus $T_{\mathcal{O}_{\mathbb{P}^2}(1)}(\pi_*w) \le s(w)$, and (3) follows from (1).

(4) Let $J' := \pi_*^{-1}J_\xi$. For any prime divisor D on $\mathbb{P}^2$, let $d_D := \deg D$, $m_D := \operatorname{mult}_p D$, and $D' := \pi_*^{-1}D$. Then $\pi^*D = D' + m_DE$ and $D' \sim d_DH - m_DE$ [Har77, Proposition V.3.6]. In particular, $J' \sim H - E$, and $(E \cdot J') = 1$ by Lemma 2.10(1). Since $\xi \in J'$, [Har77, Proposition V.1.4]

implies that $E \cap J' = \{\xi\}$ and $(E \cdot J')_\xi = 1$. Since E and $J' \cong J_\xi$ are smooth curves, $E + J'$ is a simple normal crossing divisor. Hence $(Z, E + J')$ is lc [KM98, Corollary 2.31(3)]. Since local equations of E and J' at ξ belong to the maximal ideal of $\widehat{\mathcal{O}}_{Z,\xi}$ and have multiplicity 1, we have $1 \le w(E) \le s(w)$ and $1 \le w(J') \le s(w)$ (Definition 2.7).

We show that

$$w(E) + w(J') \le 1 + s(w). \tag{4.2}$$

By Lemma 2.3, $a(w, Z, E + J') = a(w, Z, 0) - w(E) - w(J')$. Since $(Z, E + J')$ is lc and $a(w, Z, E + J')$ is a combination with non-negative coefficients of log discrepancies of prime divisors (Definition 2.2), $a(w, Z, E + J') \ge 0$. Since $m(w) = 1$, $a(w, Z, 0) = 1 + s(w)$ by Lemma 2.9(5). Thus (4.2) holds.

Let $D \ne J_\xi$ be a prime divisor on $\mathbb{P}^2$. We show that

$$m_D + \operatorname{mult}_\xi D' \le d_D. \tag{4.3}$$

By Lemma 2.10(1), $(D' \cdot J') = ((d_D H - m_D E) \cdot (H - E)) = d_D - m_D$. Let η_1, η_2 be local parameters of Z at ξ such that $J' = \{\eta_1 = 0\}$ near ξ, and let g_D be a local equation of D' at ξ. Since $D' \ne J'$, $g_D(0, \eta_2) \ne 0$ in $\mathbb{C}[[\eta_2]]$, and [Har77, Proposition V.1.4] implies that

$$d_D - m_D = (D' \cdot J') \ge \dim_{\mathbb{C}} \mathcal{O}_{Z,\xi}/(\eta_1, g_D) = \dim_{\mathbb{C}} \mathbb{C}[[\eta_2]]/(g_D(0, \eta_2)) \ge \operatorname{mult}_\xi D'.$$

Thus (4.3) holds. Since $w(E) \le s(w)$ and $w(D') \le s(w) \operatorname{mult}_\xi D'$, (4.3) implies that

$$w(\pi^* D) = m_D w(E) + w(D') \le s(w)(m_D + \operatorname{mult}_\xi D') \le s(w) d_D.$$

Moreover, $w(\pi^* J_\xi) = w(E) + w(J') \le 1 + s(w)$ by (4.2).

Let g be a non-zero homogeneous polynomial of degree d, and write $\operatorname{div}(g) = cJ_\xi + \sum_D c_D D$, where $c, c_D \ge 0$ and D runs over the prime divisors on $\mathbb{P}^2$ different from J_ξ. Then $d = c + \sum_D c_D d_D$, and

$$w(\pi^* \operatorname{div}(g)) \le c(1 + s(w)) + s(w) \sum_D c_D d_D \le (s(w) + 1) d.$$

Thus $T_{\mathcal{O}_{\mathbb{P}^2}(1)}(\pi_* w) \le s(w) + 1$. Since $w(E) \ge 1$, (1) implies that $T_L(w) \le \ell_1(s(w) + 1) - \ell_2$. If moreover $w(E) = w(J') = 1$, then $w(\pi^* J_\xi) = 2$, so

$$w(\pi^* \operatorname{div}(g)) \le 2c + s(w) \sum_D c_D d_D \le \max\{s(w), 2\} d.$$

Thus $T_{\mathcal{O}_{\mathbb{P}^2}(1)}(\pi_* w) \le \max\{s(w), 2\}$, and (1) implies that $T_L(w) \le \ell_1 \max\{s(w), 2\} - \ell_2$.

(5) Let $D \ge 0$ be a $\mathbb{Q}$-divisor on Z such that $D \sim_{\mathbb{Q}} L$, and write jD as in (4.1). By [Har77, Proposition V.3.6], $\operatorname{mult}_E \pi^* \operatorname{div}(g) = \operatorname{mult}_p g \le j\ell_1$. Hence

$$\operatorname{mult}_E(jD) = \operatorname{mult}_p g - j\ell_2 \le j\ell_1 - j\ell_2,$$

so $T_L(E) \le \ell_1 - \ell_2$. Since $\ell_1 > \ell_2$, we have $\ell_1 \pi_*^{-1} J_o + (\ell_1 - \ell_2)E \ge 0$, and

$$\ell_1 \pi_*^{-1} J_o + (\ell_1 - \ell_2)E \sim_{\mathbb{Q}} \ell_1(H - E) + (\ell_1 - \ell_2)E = L.$$

The coefficient of E in $\ell_1 \pi_*^{-1} J_o + (\ell_1 - \ell_2)E$ is $\ell_1 - \ell_2$, so $T_L(E) = \ell_1 - \ell_2$. Let $F \ne E$ be a prime divisor on Z and $F_0 := \pi(F)$. Then F_0 is a prime divisor on $\mathbb{P}^2$ of degree at least 1, and

$$\operatorname{mult}_F(jD) = \operatorname{mult}_{F_0} \operatorname{div}(g) \le \deg g = j\ell_1.$$

Thus $T_L(F) \le \ell_1$. □

The next lemma collects the arithmetic of the constants c_0, r, and α_0 which is used in Subsection 4.5. For the constants of Set-up 4.2(10), the inequalities in (2) and (4) of the next lemma hold, and so do the estimates by $\mu(8)$ and $\mu(9)$ in **Step 1** of the proof of Proposition 4.7.

Lemma 4.6. *Notation as in Set-up 4.2. Then the following hold.*

(1) $c_0 = 2239223/5000000$, *and* $r^2 = 2(\ell_2/\ell_1)r + 16c_0$.

(2) $3753/1250 < r < 15013/5000$. *In particular,* $s_l < r^2 < s_h$.

(3) The real numbers r, r^2, and α_0 are irrational.
(4) $\alpha_0 > 1$, $\ell_1\alpha_0 < 31/80$, and $\ell_1\alpha_0 < (3-\ell_1)/4$.
(5) $c_0 + s/16 - \alpha_0(\ell_1\sqrt{s} - \ell_2) = (\sqrt{s}-r)^2/16$ for any real number $s \geq 0$.

Proof. (1) Since $t_l = 4007/500$,

$$c_0 = \frac{15}{16} - \frac{611}{10000}\cdot\frac{4007}{500} = \frac{4687500 - 2448277}{5000000} = \frac{2239223}{5000000}.$$

By the definition of r, $(r-\ell_2/\ell_1)^2 = (\ell_2/\ell_1)^2 + 16c_0$, which is equivalent to $r^2 = 2(\ell_2/\ell_1)r + 16c_0$.

(2) Let

$$\chi(\lambda) := \lambda^2 - 2\frac{\ell_2}{\ell_1}\lambda - 16c_0 = \lambda^2 - \frac{5435}{8823}\lambda - \frac{2239223}{312500}.$$

By (1), $\chi(r) = 0$, and $r > 0$ by the definition of r. Since $\chi(0) < 0$, r is the unique positive root of χ, $\chi(\lambda) < 0$ for any $\lambda \in [0, r)$, and $\chi(\lambda) > 0$ for any $\lambda > r$. A direct calculation shows that

$$1562500 \cdot 8823 \cdot \chi\Big(\frac{3753}{1250}\Big) = -8231988 \quad \text{and} \quad 25000000 \cdot 8823 \cdot \chi\Big(\frac{15013}{5000}\Big) = 106023767.$$

Thus $3753/1250 < r < 15013/5000$. Since

$$\Big(\frac{3753}{1250}\Big)^2 = 9.01440576 > 9.014 = s_l \quad \text{and} \quad \Big(\frac{15013}{5000}\Big)^2 = 9.01560676 < 9.016 = s_h,$$

we have $s_l < r^2 < s_h$.

(3) Since $5 \nmid 17646$, the 5-adic valuation of $(\ell_2/\ell_1)^2 = (5435/17646)^2$ is non-negative. Since $312500 = 2^2 \cdot 5^7$ and $5 \nmid 2239223$, the 5-adic valuation of $16c_0 = 2239223/312500$ is -7. Thus the 5-adic valuation of $(\ell_2/\ell_1)^2 + 16c_0$ is -7, which is odd, so $(\ell_2/\ell_1)^2 + 16c_0$ is not the square of a rational number. Hence $\sqrt{(\ell_2/\ell_1)^2 + 16c_0}$ is irrational, and so is r. Since ℓ_2/ℓ_1 is a non-zero rational number, (1) implies that r^2 is irrational. Since ℓ_1 is rational, $\alpha_0 = r/(8\ell_1)$ is irrational.

(4) By (2), $3 < r < 31/10$. Since $\ell_1 < 3/8$, $\alpha_0 = r/(8\ell_1) > 3/(8\ell_1) > 1$. Moreover, $\ell_1\alpha_0 = r/8 < 31/80$. Since $31/80 < (3-\ell_1)/4$ if and only if $\ell_1 < 29/20$, and $\ell_1 = 17646/10^8 < 29/20$, we have $\ell_1\alpha_0 < (3-\ell_1)/4$.

(5) By (1), $r^2/16 = c_0 + r\ell_2/(8\ell_1) = c_0 + \alpha_0\ell_2$. Since $\alpha_0\ell_1 = r/8$,

$$\frac{(\sqrt{s}-r)^2}{16} = \frac{s}{16} - \frac{r\sqrt{s}}{8} + \frac{r^2}{16} = c_0 + \frac{s}{16} - \alpha_0\ell_1\sqrt{s} + \alpha_0\ell_2.$$

□

4.5. **The alpha invariant of the pair.** In this subsection, we compute the alpha invariant of the pair (Z, B) of Construction-Proposition 4.4. We compare $a(w, Z, B)$ with $\alpha_0 T_L(w)$ first along the valuations w_t, where Lemma 4.5(2) computes $T_L(w_t)$, and then for the other quasi-monomial valuations, which are compared with the valuations w_t by Lemma 3.1, estimated by the upper bounds of Lemma 4.5, or reduced to the case $m(w) = 1$ by Lemma 3.2. Only the inequality along the valuations w_t is tight: equality holds at $t = r^2 - 1$ by Lemma 4.6(5), and among the estimates that use upper bounds of μ, the one with the smallest margin is at $t = 144/17 - 1$, where only $\mu(8)$ and $\mu(9)$ are available.

Proposition 4.7. *Notation as in Construction-Proposition 4.4. Then the following hold.*

(1) $a(F, Z, B) > \alpha_0 T_L(F)$ for any prime divisor F over Z.
(2) w_{r^2-1} is a quasi-monomial valuation of rational rank 2, and

$$a(w_{r^2-1}, Z, B) = \alpha_0 T_L(w_{r^2-1}).$$

(3) $\alpha(Z, B; L) = \alpha_0$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we compute $a(w, Z, B)$ for normalized quasi-monomial valuations w over Z with center o and $m(w) = 1$, and we prove that

$$a(w_t, Z, B) \geq \alpha_0 T_L(w_t) \text{ for any } t \in [1, t_h], \text{ with equality if and only if } t = r^2 - 1. \tag{4.4}$$

As in **Step 3** of the proof of Construction-Proposition 4.4, $w_t \leq w_{t'}$ for any $1 \leq t \leq t'$ and $s(w_t) = t$. Moreover, w_t is normalized since $w_t(x) = 1$, and $m(w_t) = 1$ by Lemma 2.9(3). In particular, $w_t \wedge w_{t_h} = w_t$ for any $t \in [1, t_h]$.

Let w be a normalized quasi-monomial valuation over Z with center o and $m(w) = 1$, and let $\theta := s(w \wedge w_{t_h})$. As in **Step 3** of the proof of Construction-Proposition 4.4, $w \wedge w_{t_h} = w_\theta$ and $1 \leq \theta \leq t_h$. Since $w_\theta \leq w$, we also have $\theta \leq s(w)$. By Lemma 2.3, $a(w, Z, B) = a(w, Z, 0) - w(B)$. Hence Lemma 2.9(5) and Construction-Proposition 4.4(3) imply that

$$a(w, Z, B) = 1 + s(w) - \beta_h \theta - \beta_l \min\{\theta, t_l\}. \tag{4.5}$$

For $w = w_t$ with $t \in [1, t_h]$, we have $\theta = t$. Let $s := t+1 \in [2, s_h]$. Since $q = \beta_h + \beta_l$, $1 - \beta_h = 1/16$, and $c_0 = \beta_h - \beta_l t_l$, (4.5) implies that

$$a(w_{s-1}, Z, B) = \begin{cases} q + (1-q)s & \text{if } 2 \leq s \leq s_l, \\ c_0 + s/16 & \text{if } s_l \leq s \leq s_h. \end{cases} \tag{4.6}$$

Since $\pi_* w_{s-1} = v_s$ and $w_{s-1}(E) = w_{s-1}(x) = 1$, Lemma 4.5(1)(2) implies that

$$\begin{aligned} T_L(w_{s-1}) &\leq \ell_1 \mu(s) - \ell_2 \quad \text{for any } s \in [2, s_h], \\ T_L(w_{s-1}) &= \ell_1 \sqrt{s} - \ell_2 \quad \text{for any } s \in (9, s_h]. \end{aligned} \tag{4.7}$$

For any $s \in [2, s_l]$, let

$$U(s) := \frac{1}{\ell_1}\Big(\frac{a(w_{s-1}, Z, B)}{\alpha_0} + \ell_2\Big).$$

The function U is used only in this subsection. Since $\alpha_0 \ell_1 = r/8$, Lemma 4.6(1) implies that $\ell_2/\ell_1 = (r^2 - 16c_0)/(2r)$. Since $16(q - c_0) = 16\beta_l(1 + t_l) = (611/625)s_l$ and $16(1-q) = 14/625$, (4.6) implies that

$$U(s) = \frac{8(q + (1-q)s)}{r} + \frac{\ell_2}{\ell_1} = \frac{r^2 + (611/625)s_l + (14/625)s}{2r}. \tag{4.8}$$

In particular, U is strictly increasing. By (4.7), for any $s \in [2, 9]$ such that $\mu(s) < U(s)$, we have

$$\alpha_0 T_L(w_{s-1}) \leq \alpha_0(\ell_1 \mu(s) - \ell_2) < \alpha_0(\ell_1 U(s) - \ell_2) = a(w_{s-1}, Z, B).$$

We consider four ranges of s.

The range $2 \leq s \leq 8$. By Lemma 4.6(2), $r < 31/10$. Since $\ell_2/\ell_1 = 5435/17646 > 3/10$ and $q + 2(1-q) = 2 - q > 1$, (4.8) implies that

$$U(2) > \frac{80}{31} + \frac{3}{10} = \frac{893}{310} > \frac{17}{6}.$$

By Lemma 4.1(1)(2), $\mu(s) \leq \mu(8) = 17/6 < U(2) \leq U(s)$.

The range $8 \leq s \leq 9$. By Lemma 4.1(1)(2), $\mu(s) \leq (s/8)\mu(8) = 17s/48$ and $\mu(s) \leq \mu(9) = 3$. Let $s_c := 144/17 \in (8, 9)$, so that $17s_c/48 = 3$. By (4.8), a direct calculation shows that

$$2r\big(U(s_c) - 3\big) = (r-3)^2 + \frac{611}{625}s_l + \frac{14}{625} \cdot \frac{144}{17} - 9 = (r-3)^2 + \frac{9709}{5312500} > 0.$$

On $[8, s_c]$, the function $U(s) - 17s/48$ is affine with slope $7/(625r) - 17/48 < 0$, and $U(s_c) - 17s_c/48 > 0$. Hence $\mu(s) \leq 17s/48 < U(s)$ for any $s \in [8, s_c]$. For any $s \in [s_c, 9]$, $\mu(s) \leq 3 < U(s_c) \leq U(s)$.

The range $9 < s \le s_l$. Since $q - c_0 = \beta_l s_l$ and $1 - q = 1/16 - \beta_l$, (4.6) implies that $a(w_{s-1}, Z, B) = c_0 + s/16 + \beta_l(s_l - s)$. By (4.7) and Lemma 4.6(5),

$$a(w_{s-1}, Z, B) - \alpha_0 T_L(w_{s-1}) = \frac{(\sqrt{s} - r)^2}{16} + \beta_l(s_l - s).$$

If $s < s_l$, then $\beta_l(s_l - s) > 0$. If $s = s_l$, then $(\sqrt{s} - r)^2 > 0$ since $r^2 > s_l$ (Lemma 4.6(2)). Thus $a(w_{s-1}, Z, B) > \alpha_0 T_L(w_{s-1})$.

The range $s_l \le s \le s_h$. By (4.6), (4.7), and Lemma 4.6(5),

$$a(w_{s-1}, Z, B) - \alpha_0 T_L(w_{s-1}) = \frac{(\sqrt{s} - r)^2}{16} \ge 0,$$

with equality if and only if $s = r^2$. By Lemma 4.6(2), $s_l < r^2 < s_h$.

The four ranges cover $s \in [2, s_h]$, that is, $t \in [1, t_h]$. Thus (4.4) holds. Moreover, since $a(w_1, Z, B) = 2 - q$ by (4.6) and $U(2) > 17/6 > 2$,

$$2 - q = \alpha_0(\ell_1 U(2) - \ell_2) > \alpha_0(2\ell_1 - \ell_2). \tag{4.9}$$

Step 2. In this step, we prove that $a(w, Z, B) \ge \alpha_0 T_L(w)$ for any normalized quasi-monomial valuation w over Z with center o and $m(w) = 1$, with equality only if $w = w_{r^2-1}$. Let $\theta := s(w \wedge w_{t_h})$, so that $w \wedge w_{t_h} = w_\theta$ and (4.5) holds (**Step 1**). By Lemma 4.5(4) with $\xi = o$, $T_L(w) \le \ell_1(s(w) + 1) - \ell_2$.

The case $\theta = 1$. By (4.5), $a(w, Z, B) = 1 + s(w) - q$. Thus

$$a(w, Z, B) - \alpha_0 T_L(w) \ge (1 - \alpha_0\ell_1)(s(w) - 1) + 2 - q - \alpha_0(2\ell_1 - \ell_2) > 0,$$

where the last inequality follows from $\alpha_0\ell_1 = r/8 < 1$, $s(w) \ge 1$, and (4.9).

The case $1 < \theta \le 1/q$. Since $w_\theta \le w$, we have $w(z) \ge w_\theta(z) = \theta > 1$. Since x, z generate the maximal ideal of $\mathbb{C}[[x, z]]$ and w is normalized, $\min\{w(x), w(z)\} = 1$, so $w(E) = w(x) = 1$. By Set-up 4.2(4)(6), J_o is the tangent line of $\{y = 0\}$ at p, so $J_o = \{y = \phi(x)\}$ near p for a power series $\phi \in \mathbb{C}[[x]]$, and the order of ϕ is $(J_o \cdot \{y = 0\})_p = 2$ by Lemma 4.1(4). Since $y = xz$ near o, we have $\pi_*^{-1}J_o = \{z = \phi(x)/x\}$ near o, and $\phi(x)/x$ is x times a unit of $\mathbb{C}[[x]]$. Thus $w(\phi(x)/x) = 1 < w(z)$, so $w(\pi_*^{-1}J_o) = w(z - \phi(x)/x) = 1$. By Lemma 4.5(4),

$$T_L(w) \le \ell_1 \max\{s(w), 2\} - \ell_2.$$

Since $\theta \le 1/q < t_l$, (4.5) implies that $a(w, Z, B) = 1 + s(w) - q\theta$, where $q\theta \le 1$. If $s(w) \le 2$, then, since $\theta \le s(w)$ and $s(w) \ge 1$, (4.9) implies that

$$a(w, Z, B) \ge 1 + (1 - q)s(w) \ge 2 - q > \alpha_0(2\ell_1 - \ell_2) \ge \alpha_0 T_L(w).$$

If $s(w) > 2$, then

$$a(w, Z, B) \ge s(w) > \frac{r}{8} s(w) = \alpha_0\ell_1 s(w) > \alpha_0(\ell_1 s(w) - \ell_2) \ge \alpha_0 T_L(w).$$

The case $\theta > 1/q$. By Construction-Proposition 4.4(3), $w(B) = \beta_h\theta + \beta_l \min\{\theta, t_l\}$. If $\theta \le t_l$, then $w(B) = q\theta > 1$. If $\theta > t_l$, then $w(B) = \beta_h\theta + \beta_l t_l > qt_l > 1$. Since $w_\theta \wedge w_{t_h} = w_\theta$, (4.5) applied to w_θ and the formula for $w(B)$ above imply that $a(w_\theta, Z, B) = 1 + \theta - w(B)$. By Lemma 3.1 applied to $w_\theta \le w$, $\theta T_L(w) \le s(w)T_L(w_\theta)$. Since $\theta \in (1/q, t_h]$, (4.4) implies that $\alpha_0 T_L(w_\theta) \le a(w_\theta, Z, B)$. Hence, by (4.5),

$$\begin{aligned}\theta\big(a(w, Z, B) - \alpha_0 T_L(w)\big) &\ge \theta a(w, Z, B) - s(w)\alpha_0 T_L(w_\theta) \\ &= (s(w) - \theta)(w(B) - 1) + s(w)\big(a(w_\theta, Z, B) - \alpha_0 T_L(w_\theta)\big) \ge 0.\end{aligned}$$

If equality holds, then $s(w) = \theta$ since $w(B) > 1$, and $a(w_\theta, Z, B) = \alpha_0 T_L(w_\theta)$, so $\theta = r^2 - 1$ by (4.4). Since $w_\theta \le w$ and $s(w_\theta) = s(w)$, Lemma 2.9(2) implies that $w = w_\theta = w_{r^2-1}$.

Step 3. In this step, we prove that $a(w, Z, B) > \alpha_0 T_L(w)$ for any normalized quasi-monomial valuation w over Z whose center is a closed point $\zeta \ne o$ and $m(w) = 1$, and that $a(F, Z, B) > \alpha_0 T_L(F)$ for any prime divisor F on Z.

By Construction-Proposition 4.4(1), C is a reduced divisor which is simple normal crossing near ζ, so (Z, C) is lc near ζ [KM98, Corollary 2.31(3)]. Let $h : Y \to Z$, $y \in Y$, $y_1, \dots, y_k$, and $t_1, \dots, t_k$ represent w as in Definition 2.2, where h is a log resolution of (Z, C). Let E_i be the prime divisor on Y defined by y_i near y. Since the center of E_i on Z contains ζ, $a(E_i, Z, C) \geq 0$ for any i. Thus $a(w, Z, C) = \sum_i t_i a(E_i, Z, C) \geq 0$. Since $a(w, Z, C) = a(w, Z, 0) - w(C)$ and $a(w, Z, B) = a(w, Z, 0) - w(C)/N$ (Lemma 2.3), Lemma 2.9(5) implies that

$$a(w, Z, B) \geq \Big(1 - \frac{1}{N}\Big) a(w, Z, 0) \geq \Big(1 - \frac{1}{N}\Big)(1 + s(w)).$$

On the other hand, $T_L(w) < \ell_1(s(w) + 1)$ by Lemma 4.5(3) if $\zeta \notin E$, and by Lemma 4.5(4) if $\zeta \in E \backslash \{o\}$. Since $\ell_1 \alpha_0 < 31/80$ (Lemma 4.6(4)) and $N \geq 10^8$,

$$\alpha_0 T_L(w) < \frac{31}{80}(1 + s(w)) < \Big(1 - \frac{1}{N}\Big)(1 + s(w)) \leq a(w, Z, B).$$

Let F be a prime divisor on Z. Since C is reduced, $a(F, Z, B) = 1 - \operatorname{mult}_F B \geq 1 - 1/N$. By Lemma 4.5(5), $T_L(F) \leq \ell_1$. Thus

$$\alpha_0 T_L(F) \leq \alpha_0 \ell_1 < \frac{31}{80} < 1 - \frac{1}{N} \leq a(F, Z, B).$$

Step 4. In this step, we prove that $a(w, Z, B) > \alpha_0 T_L(w)$ for any normalized quasi-monomial valuation w over Z whose center is a closed point ζ and $m(w) \geq 2$. By Construction-Proposition 4.4(2)(4), B is a $\mathbb{Q}$-divisor, (Z, B) is klt, and L is an ample $\mathbb{Q}$-divisor. By Lemma 3.2, there exists a normalized divisorial valuation $u \leq w$ with $m(u) = 1$ such that

$$a(u, Z, B) T_L(w) < a(w, Z, B) T_L(u).$$

The center of u is ζ, so $\alpha_0 T_L(u) \leq a(u, Z, B)$ by **Steps 2 and 3**. Since (Z, B) is klt, $a(u, Z, B) > 0$ and $a(w, Z, B) > 0$ (Definition 2.2). Hence

$$\alpha_0 a(u, Z, B) T_L(w) < a(w, Z, B) \alpha_0 T_L(u) \leq a(w, Z, B) a(u, Z, B),$$

so $\alpha_0 T_L(w) < a(w, Z, B)$.

Step 5. In this step, we conclude the proof.

(1) Let F be a prime divisor over Z. If F is a divisor on Z, then (1) follows from **Step 3**. Otherwise, the center of F on Z is a closed point ζ. Let c be the positive real number such that c times the valuation defined by F is normalized at ζ, and let w be c times the valuation defined by F. Then w is quasi-monomial [JM12, Remark 3.9], $a(w, Z, B) = c a(F, Z, B)$, and $T_L(w) = c T_L(F)$ (Definitions 2.1 and 2.4). The value group of w is $c\mathbb{Z}$, while $w_{r^2-1}(x) = 1$ and $w_{r^2-1}(z) = r^2 - 1$ are linearly independent over $\mathbb{Q}$ by Lemma 4.6(3). Thus $w \neq w_{r^2-1}$. By **Steps 2, 3, and 4**, $a(w, Z, B) > \alpha_0 T_L(w)$, so $a(F, Z, B) > \alpha_0 T_L(F)$.

(2) By Lemma 4.6(2), $r^2 - 1 > t_l > 1$, so w_{r^2-1} is defined. By Definition 2.2 with $Y = Z$, $y = o$, the local parameters x, z, and the real numbers $1, r^2 - 1$, w_{r^2-1} is a quasi-monomial valuation of rational rank $\dim_{\mathbb{Q}}(\mathbb{Q} + \mathbb{Q}(r^2 - 1)) = 2$, since r^2 is irrational (Lemma 4.6(3)). Since $r^2 - 1 \in [1, t_h]$, (4.4) implies that $a(w_{r^2-1}, Z, B) = \alpha_0 T_L(w_{r^2-1})$.

(3) By Construction-Proposition 4.4(2)(4), (Z, B) is a projective klt pair and L is an ample $\mathbb{Q}$-divisor. By Lemma 2.5(1) and Proposition 4.7(1), $\alpha(Z, B; L) = \inf_F a(F, Z, B)/T_L(F) \geq \alpha_0$. By Lemma 2.5(2) and Proposition 4.7(2),

$$\alpha(Z, B; L) T_L(w_{r^2-1}) \leq a(w_{r^2-1}, Z, B) = \alpha_0 T_L(w_{r^2-1}).$$

Since $r^2 - 1 \in (8, t_h]$, Lemma 4.5(2) implies that $T_L(w_{r^2-1}) = \ell_1 r - \ell_2 > 0$. Thus $\alpha(Z, B; L) \leq \alpha_0$, and so $\alpha(Z, B; L) = \alpha_0$. □

5. Construction of the Fano variety

The goal of this section is to construct, from a pair (Z, B) as in Set-up 5.1, a klt Fano variety X of dimension N without boundary such that $-K_X$ corresponds to the character L (Construction-Theorem 5.2), and to lift any non-trivial quasi-monomial valuation v over Z to a quasi-monomial valuation w over X whose rational rank is equal to that of v, with $a(w, X, 0) = a(v, Z, B)$ and $T_{-K_X}(w) = T_L(v)$ (Proposition 5.3). This section and Section 6 work under the general conditions of Set-up 5.1, and only Section 7 substitutes the pair (Z, B) of Section 4.

5.1. **Construction of X.** In this subsection, we construct X and prove the properties of X used in the rest of the paper.

Set-up 5.1. Let Z, H, E, $\mathcal{S}$, N, C, F_C, B, L, ℓ_1, ℓ_2, δ_N, and γ be as follows.

(1) Z is the blow-up of $\mathbb{P}^2$ at a closed point, E is the exceptional curve, H is the pullback of $\mathcal{O}_{\mathbb{P}^2}(1)$, and $\mathcal{S} = \mathbb{C}[x_0, x_1, x_2, e]$ is the Cox ring of Z (Lemma 2.10(3)).
(2) $N \geq 3$ is an integer, $C \neq 0$ is a reduced divisor on Z, $F_C \in \mathcal{S}_{[C]}$ is an element whose divisor is C, and $B := C/N$.
(3) (Z, B) is klt, and $L := -(K_Z + B) \sim_{\mathbb{Q}} \ell_1 H - \ell_2 E$ for rational numbers $\ell_1 > 3\ell_2 > 0$.
(4) $\delta_N := (N-2)/N$ and $\gamma := \ell_1/(3 - \ell_1)$.

Since $C \neq 0$, and since $\ell_1 > 3\ell_2$ excludes $C = E$ (**Step 3** of the proof of Construction-Theorem 5.2), we have $3 - \ell_1 = (H \cdot B) > 0$, so γ is well defined and positive. The condition $\ell_1 > 3\ell_2$ also implies that L is not proportional to $[C]$ or to the degree of any of x_0, x_1, x_2, e, so the quotient in Construction-Theorem 5.2 is geometric (**Step 3** of the proof).

In Construction-Theorem 5.2 below, the sections of $-jK_X$ are the elements of degree jL of a ring containing $\mathcal{S}$, so that T_{-K_X} on X can be compared directly with T_L on Z. Since $[C]/N$ need not belong to $\mathrm{Pic}(Z)$, the variables U_i, whose N-th powers add up to $-F_C$, are graded by the group $\mathrm{Pic}(Z) + \mathbb{Z}[C]/N$, and X is a quotient by the corresponding torus. The sketch of the proof in Section 1 explains why we use $N-1$ roots and one equation instead of a cyclic cover.

Construction-Theorem 5.2. Notation and conditions as in Set-up 5.1. Let

$$\tau := [C]/N \in \mathrm{Pic}(Z)_{\mathbb{Q}} \quad \text{and} \quad \Lambda := \mathrm{Pic}(Z) + \mathbb{Z}\tau \subset \mathrm{Pic}(Z)_{\mathbb{Q}},$$

which contains the class $3H - E - \tau$ of L. Let

$$R := \mathcal{S}[U_1, \ldots, U_{N-1}]/(F_C + U_1^N + \cdots + U_{N-1}^N),$$

graded by Λ with $\deg U_i = \tau$ for any i, let $\overline{X} := \operatorname{Spec} R$ with the induced action of the torus $\mathbb{T} := \mathrm{Hom}(\Lambda, \mathbb{C}^*)$, let $\widehat{X} \subset \overline{X}$ be the union of the open subsets $\{g \neq 0\}$ for all $g \in R_{jL}$ and all integers $j \geq 1$, and let

$$X := \operatorname{Proj} \bigoplus_{j \geq 0} R_{jL}.$$

Then the following hold.

(1) R is a normal domain, and $\overline{X} \setminus \widehat{X}$ is of codimension at least 2 in $\overline{X}$.
(2) The natural morphism $h : \widehat{X} \to X$ is a geometric quotient by $\mathbb{T}$, any $\mathbb{T}$-orbit in $\widehat{X}$ is of dimension 2 and has finite stabilizer, and the points of $\widehat{X}$ with non-trivial stabilizer form a closed subset of codimension at least 2 in $\widehat{X}$.
(3) X is a klt Fano variety of dimension N.
(4) There exists a positive integer j_0 such that for any positive integer j divisible by j_0, $-jK_X$ is Cartier, $jL \in \mathrm{Pic}(Z)$, and there exists an isomorphism $R_{jL} \to H^0(X, \mathcal{O}_X(-jK_X))$ such that for any $0 \neq g \in R_{jL}$, the divisor D_g of the section corresponding to g satisfies $h^* D_g = \mathrm{div}(g|_{\widehat{X}})$.
(5) The inclusion $\mathcal{S} \subset R$ induces a dominant rational map $\rho : X \dashrightarrow Z$.
(6) For any $g \in R$, the following hold.

(i) There exist unique elements $a_I \in \mathcal{S}$, where $I = (I_1, \dots, I_{N-1})$ runs over $\mathbb{Z}_{\geq 0}^{N-1}$, with the following properties.
  (a) $a_I = 0$ for all but finitely many I.
  (b) $a_I = 0$ whenever $I_{N-1} \geq N$.
  (c) $g = \sum_I a_I U^I$, where $U^I := U_1^{I_1} \cdots U_{N-1}^{I_{N-1}}$.
  We call $\sum_I a_I U^I$ the *normal form* of g.
(ii) If $g \neq 0$, then we let $\deg_U(g) := \max\{|I| \mid a_I \neq 0\}$, where $|I| := I_1 + \cdots + I_{N-1}$.
(iii) If $g \in R_Q$ for some $Q \in \Lambda$, then $a_I \in \mathcal{S}_{Q-|I|\tau}$ for any I, where $\mathcal{S}_{Q'} := 0$ for any $Q' \in \Lambda \setminus \operatorname{Pic}(Z)$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove that R is a normal domain and prove (6).

Since $K_Z \sim -3H + E$ (Lemma 2.10(1)) and $[C] = N\tau$, the class of $L = -(K_Z + B)$ is $3H - E - \tau \in \Lambda$. Since $F_C \in \mathcal{S}_{N\tau}$ and $\deg U_i^N = N\tau$, the polynomial $F_C + \sum_i U_i^N$ is homogeneous of degree $N\tau$, so R is graded by Λ.

Let $\mathbb{K}$ be the fraction field of $\mathcal{S}$ and $\Phi := F_C + \sum_{i=1}^{N-2} U_i^N$. Since $N \geq 3$, the variable U_1 appears in Φ. If the square of an irreducible polynomial $P \in \mathbb{K}[U_1, \dots, U_{N-2}]$ divides Φ, then P divides $\partial\Phi/\partial U_1 = NU_1^{N-1}$, so P is a scalar multiple of U_1. This is impossible because

$$\Phi|_{U_1=0} = F_C + \sum_{i=2}^{N-2} U_i^N \neq 0.$$

Thus any irreducible factor P of Φ in $\mathbb{K}[U_1, \dots, U_{N-2}]$ has multiplicity one, and $U_{N-1}^N + \Phi$ is an Eisenstein polynomial with respect to P. Hence $U_{N-1}^N + \Phi$ is irreducible over the fraction field of $\mathcal{S}[U_1, \dots, U_{N-2}]$ [Lan02, Theorem IV.3.1]. Since $U_{N-1}^N + \Phi$ is monic in U_{N-1}, it is irreducible, hence prime, in the factorial ring $\mathcal{S}[U_1, \dots, U_{N-1}]$ by [Lan02, Theorem IV.2.3]. Therefore R is a domain.

Since $F_C + \sum_i U_i^N$ is monic in U_{N-1} of degree N, R is a free $\mathcal{S}[U_1, \dots, U_{N-2}]$-module with basis $1, U_{N-1}, \dots, U_{N-1}^{N-1}$. Writing the coefficients in the basis as polynomials in $U_1, \dots, U_{N-2}$, we obtain the existence and the uniqueness of the elements a_I in (6). Let $g \in R_Q$ with $g = \sum_I a_I U^I$ as in (6), and write $a_I = \sum_{Q' \in \operatorname{Pic}(Z)} a_{I,Q'}$ with $a_{I,Q'} \in \mathcal{S}_{Q'}$, where we let $a_{I,Q'} := 0$ for $Q' \in \Lambda \setminus \operatorname{Pic}(Z)$. Since the quotient map $\mathcal{S}[U_1, \dots, U_{N-1}] \to R$ preserves the Λ-grading, g is also the image of the degree Q part

$$\sum_I a_{I,Q-|I|\tau} U^I$$

of $\sum_I a_I U^I$. The uniqueness in (6) implies that $a_I = a_{I,Q-|I|\tau} \in \mathcal{S}_{Q-|I|\tau}$ for any I.

For any $y \in \mathbb{A}^4 = \operatorname{Spec} \mathcal{S}$, the fiber of the projection $\overline{X} \to \mathbb{A}^4$ over y is the hypersurface $\{F_C(y) + \sum_i U_i^N = 0\}$ in $\mathbb{A}^{N-1}$, which is of dimension $N-2$. Thus $\dim \overline{X} = N + 2$, and the preimage in $\overline{X}$ of any closed subset $W \subset \mathbb{A}^4$ is of dimension at most $\dim W + N - 2$. Since $\partial(F_C + \sum_i U_i^N)/\partial U_i = NU_i^{N-1}$, the singular locus of the hypersurface $\overline{X} \subset \mathbb{A}^{N+3}$ is contained in $\{U_1 = \cdots = U_{N-1} = 0,\ F_C = 0\}$, which is of dimension 3, hence of codimension $N - 1 \geq 2$ in $\overline{X}$. By [Har77, Proposition II.8.23(b)], $\overline{X}$ is normal.

Step 2. In this step, we prove that $\overline{X} \setminus \widehat{X}$ is of codimension at least 2 in $\overline{X}$, which finishes the proof of (1).

Let

$$\mathbb{A}^4_\circ := \{(x_1, x_2) \neq (0,0),\ (x_0, e) \neq (0,0)\} \subset \mathbb{A}^4$$

be the open subset of Lemma 2.10(3), and let $\overline{X}_Z \subset \overline{X}$ be its preimage. Let j be a positive integer such that $j\ell_1, j\ell_2 \in \mathbb{Z}$. Since $\ell_1 - \ell_2 > 0$, $\ell_2 > 0$, and

$$j(\ell_1 - \ell_2)H + j\ell_2(H - E) = j(\ell_1 - \ell_2)E + j\ell_1(H - E) = jL,$$

we have

$$x_0^{j(\ell_1-\ell_2)}x_i^{j\ell_2} \in R_{jL} \quad \text{and} \quad e^{j(\ell_1-\ell_2)}x_i^{j\ell_1} \in R_{jL} \quad \text{for } i = 1, 2.$$

At any point of $\overline{X}_Z$, one of the monomials above does not vanish. Thus $\overline{X}_Z \subset \widehat{X}$. The complement $\overline{X}\backslash\overline{X}_Z$ is the preimage of $\{x_1 = x_2 = 0\} \cup \{x_0 = e = 0\}$, which is of dimension 2 in $\mathbb{A}^4$. By **Step 1**, $\dim(\overline{X}\backslash\overline{X}_Z) \leq N$. Thus $\overline{X}\backslash\widehat{X}$ is of codimension at least 2 in $\overline{X}$.

Step 3. In this step, we prove that X is a normal projective variety and prove (2).

We use the basis $(E, H - E)$ of $\mathrm{Pic}(Z)_{\mathbb{Q}}$ and write (c_1, c_2) for $c_1E + c_2(H - E)$. By **Step 1**,

$$H - E = (0, 1), \quad H = (1, 1), \quad E = (1, 0), \quad \tau = (2 - \ell_1 + \ell_2, 3 - \ell_1), \quad L = (\ell_1 - \ell_2, \ell_1).$$

Since $F_C \neq 0$ is a linear combination of monomials in x_0, x_1, x_2, e of degree $N\tau$, we have $2 - \ell_1 + \ell_2 \geq 0$. We have $(H \cdot C) = N(3 - \ell_1)$. If $C = E$, then $\ell_1 = 3$ and $\ell_2 = 1 + 1/N$, which contradicts $\ell_1 > 3\ell_2$. Thus C has a component other than E, and $3 - \ell_1 > 0$. Moreover,

$$(3 - \ell_1)(\ell_1 - \ell_2) - \ell_1(2 - \ell_1 + \ell_2) = \ell_1 - 3\ell_2 > 0 \quad \text{and} \quad \ell_1 - (\ell_1 - \ell_2) = \ell_2 > 0,$$

so L belongs to the interior of the cone generated by H and τ. Thus L is not proportional to any of H, $H - E$, E, and τ. Hence any cone generated by some of H, $H - E$, E, τ that contains L is of dimension 2, and L belongs to the interior of the cone. By the coordinates of L, τ, and E above,

$$L = \gamma\tau + \frac{\ell_1 - 3\ell_2}{3 - \ell_1}E.$$

We first prove three facts on $\mathcal{R} := \bigoplus_{j\geq 0} R_{jL}$ and X.

(i) $\mathcal{R}$ is a finitely generated $\mathbb{C}$-algebra. Let $\deg : \mathbb{R}^{N+3} \to \Lambda \otimes \mathbb{R}$ be the linear map sending the standard basis vectors to the degrees of $x_0, x_1, x_2, e, U_1, \ldots, U_{N-1}$. For any $j \geq 0$, R_{jL} is spanned by the images of the monomials in $x_0, x_1, x_2, e, U_1, \ldots, U_{N-1}$ of degree jL. The exponent vectors u of the monomials with $\deg(u) \in \mathbb{Z}_{\geq 0}L$ form the intersection of the rational polyhedral cone

$$\{u \in \mathbb{R}^{N+3}_{\geq 0} \mid \deg(u) \in \mathbb{R}_{\geq 0}L\}$$

with the lattice $\{u \in \mathbb{Z}^{N+3} \mid \deg(u) \in \mathbb{Z}L\}$. By Gordan's lemma [CLS11, Proposition 1.2.17], the intersection is a finitely generated monoid, so $\mathcal{R}$ is finitely generated.

(ii) $\mathcal{R}_0 = R_0 = \mathbb{C}$. The linear function $c_1 + c_2$ is positive on H, $H - E$, E, and τ, because $2 - \ell_1 + \ell_2 \geq 0$ and $3 - \ell_1 > 0$. Thus the degree of any non-constant monomial is non-zero. By (i) and (ii), X is projective.

(iii) X is a normal variety. For any homogeneous $g \in R_{jL}$ with $j \geq 1$, the degree zero part of $\mathcal{R}_g$ is equal to the Λ-degree zero part of R_g, which is the ring $(R_g)^{\mathbb{T}}$ of $\mathbb{T}$-invariants. Thus $D_+(g) = \mathrm{Spec}(R_g)^{\mathbb{T}}$. If an element of the fraction field of $(R_g)^{\mathbb{T}}$ is integral over $(R_g)^{\mathbb{T}}$, then it belongs to R_g because R_g is a normal domain (**Step 1**), and it is $\mathbb{T}$-invariant, so it belongs to $(R_g)^{\mathbb{T}}$. Thus $D_+(g)$ is normal. Since $\mathcal{R} \subset R$ is a domain, X is a normal variety.

For homogeneous $g \in R_{jL}$ with $j \geq 1$, the inclusion $(R_g)^{\mathbb{T}} \subset R_g$ defines the quotient $\{g \neq 0\} \to D_+(g)$ of the affine variety $\{g \neq 0\} \subset \overline{X}$ by the reductive group $\mathbb{T}$, which is a good quotient [MFK94, Chapter 1, §2]. Gluing over g, we obtain a good quotient $h : \widehat{X} \to X$ with $h^{-1}(D_+(g)) = \{g \neq 0\}$ for any such g. Note that $\widehat{X}$ is the set of semistable points of $\overline{X}$ with respect to the character L.

For $x \in \overline{X}$, let $\omega(x) \subset \Lambda \otimes \mathbb{R}$ be the cone generated by the degrees of the coordinates among $x_0, x_1, x_2, e, U_1, \ldots, U_{N-1}$ that do not vanish at x. Then $x \in \widehat{X}$ if and only if $L \in \omega(x)$ [ADHL15, Proposition 3.1.1.10 and Lemma 3.1.2.7]: if $g(x) \neq 0$ for some $g \in R_{jL}$ with $j \geq 1$, then a monomial of degree jL in a lift of g to $\mathcal{S}[U_1, \ldots, U_{N-1}]$ does not vanish at x, so $L \in \omega(x)$. Conversely, if $L \in \omega(x)$, then, since $\omega(x)$ is a rational cone, some monomial in the coordinates that do not vanish at x is of degree jL for some $j \geq 1$, so $x \in \widehat{X}$.

Let $x \in \widehat{X}$. Then $\omega(x)$ is generated by some of H, $H - E$, E, τ and contains L, so $\omega(x)$ is of dimension 2. Hence the degrees of the coordinates that do not vanish at x span $\Lambda \otimes \mathbb{Q}$. Thus the $\mathbb{T}$-orbit of x is of dimension 2, and the stabilizer of x, which is the intersection of the kernels

of the characters of $\mathbb{T}$ given by the degrees of the coordinates that do not vanish at x, is finite. Since the closure of a $\mathbb{T}$-orbit is the union of the orbit and orbits of smaller dimension [Bor91, Proposition I.1.8], any $\mathbb{T}$-orbit in $\widehat{X}$ is closed in $\widehat{X}$. Thus any point of $\widehat{X}$ is stable, and h is a geometric quotient [MFK94, Theorem 1.10]. In particular, $\dim X = \dim \widehat{X} - 2 = N$. If $\ell_1 = 3\ell_2$, then $L = \gamma\tau$, and the points of $\overline{X}$ with $x_0 = x_1 = x_2 = e = 0$ and $U \neq 0$ would be semistable with 1-dimensional orbits. The condition $\ell_1 > 3\ell_2$ excludes the wall $L \in \mathbb{R}\tau$.

The stabilizer of $x \in \overline{X}$ depends only on which coordinates vanish at x, so only finitely many subgroups of $\mathbb{T}$ are stabilizers. Thus the set of points of $\widehat{X}$ with non-trivial stabilizer is the union of the fixed point sets of the finitely many non-trivial stabilizers, which is closed. If $x \in \overline{X}_Z$ and $U_i(x) \neq 0$ for some i, then the degrees of the coordinates that do not vanish at x contain $H - E$, one of H and E, and τ, which generate Λ by Lemma 2.10(1). Thus the stabilizer of x is trivial. Hence the points of $\widehat{X}$ with non-trivial stabilizer belong to

$$(\widehat{X} \backslash \overline{X}_Z) \cup \{U_1 = \cdots = U_{N-1} = 0\},$$

which is of codimension at least 2 in $\widehat{X}$ by **Step 2** and **Step 1**. This finishes the proof of (2).

Finally, we describe h over $\overline{X}_Z$. Let Γ be the kernel of the restriction $\mathbb{T} \to \mathbb{T}_0$, where $\mathbb{T}_0$ is the torus of Lemma 2.10(3). Then $\Gamma = \operatorname{Hom}(\Lambda/\operatorname{Pic}(Z), \mathbb{C}^*)$ is a finite cyclic group, which acts trivially on x_0, x_1, x_2, e and multiplies each U_i by the character of Γ given by the class of τ in $\Lambda/\operatorname{Pic}(Z)$, which is faithful because τ generates $\Lambda/\operatorname{Pic}(Z)$. Let $a \in \{x_0, e\}$ and $b \in \{x_1, x_2\}$. Let $\overline{X}_{ab} := \{ab \neq 0\} \subset \overline{X}$ and $V_{ab} := \overline{X} \cap \{a = b = 1\}$. Let $Z_{ab} \subset Z$ be the image of $\{ab \neq 0\} \subset \mathbb{A}^4$ under the quotient in Lemma 2.10(3), which is isomorphic to $\mathbb{A}^2$ with coordinates the other two Cox coordinates because $\deg a$ and $\deg b$ form a basis of $\operatorname{Pic}(Z)$. Let ϕ_{ab} be the restriction of F_C to $\{a = b = 1\}$, which is an equation of $C \cap Z_{ab}$. Then

$$V_{ab} = \left\{\phi_{ab} + \sum_{i=1}^{N-1} U_i^N = 0\right\} \subset Z_{ab} \times \mathbb{A}^{N-1}.$$

The polynomial $\phi_{ab} + \sum_i U_i^N$ is irreducible by the argument of **Step 1**, with the polynomial ring $\mathbb{C}[Z_{ab}]$ in place of $\mathcal{S}$ and $\phi_{ab} \neq 0$ in place of F_C. Thus V_{ab} is irreducible. Since $\deg a$ and $\deg b$ form a basis of $\operatorname{Pic}(Z)$, (a, b) defines a morphism $\overline{X}_{ab} \to \mathbb{T}_0$ with fiber V_{ab} over 1, and the morphism $\mathbb{T} \times V_{ab} \to \overline{X}_{ab}$, $(t, v) \to t \cdot v$, is the base change of the isogeny $\mathbb{T} \to \mathbb{T}_0$ along it. Thus $\mathbb{T} \times V_{ab} \to \overline{X}_{ab}$ is finite étale, and it is the quotient by the free action of Γ given by $\zeta \cdot (t, v) = (t\zeta^{-1}, \zeta \cdot v)$. By **Step 2**, $\overline{X}_{ab} = \{g \neq 0\}$ for one of the monomials $g \in R_{jL}$ of **Step 2**, so $\overline{X}_{ab} = h^{-1}(D_+(g))$ and

$$h(\overline{X}_{ab}) = D_+(g) \cong V_{ab}/\Gamma.$$

Step 4. In this step, we prove (4) and that $-K_X$ is ample.

Let $g_1, \ldots, g_r$ be homogeneous generators of $\mathcal{R}$ with $g_s \in R_{k_s L}$ and $k_s \geq 1$, which exist by **Step 3**(i)(ii). Let $k := \operatorname{lcm}(k_1, \ldots, k_r)$ and $j_1 := rk$. Writing a monomial of degree mj_1 in $g_1, \ldots, g_r$ as a product of the elements g_s^{k/k_s} and a monomial of degree less than j_1, and grouping the factors into products of degree j_1, we see that $\bigoplus_{m \geq 0} R_{mj_1 L}$ is generated by $R_{j_1 L}$. For any integer j, let $\mathcal{O}_X(j)$ be the sheaf on X associated to the graded $\mathcal{R}$-module $\bigoplus_{m \geq 0} R_{(j+m)L}$, whose sections over $D_+(g)$ are the elements of $(R_g)_{jL}$. For any homogeneous $g \in \mathcal{R}$ of positive degree, $D_+(g) = D_+(g^{j_1})$ and the degree zero parts of $\mathcal{R}_g$ and $\mathcal{R}_{g^{j_1}}$ coincide, so $X = \operatorname{Proj} \bigoplus_{m \geq 0} R_{mj_1 L}$. Moreover, for $g \in R_{j_1 L}$ and any j divisible by j_1, the sections over $D_+(g)$ of $\mathcal{O}_X(j)$ and of the twist $\mathcal{O}(j/j_1)$ defined by $\bigoplus_{m \geq 0} R_{mj_1 L}$ are both $(R_g)_{jL}$. Thus a basis of $R_{j_1 L}$ defines a closed embedding of X into a projective space under which $\mathcal{O}(1)$ restricts to $\mathcal{O}_X(j_1)$ [Har77, Exercise II.3.12(a) and Proposition II.5.12(c)]. Thus for any positive integer j divisible by j_1, the open subsets $D_+(g)$ for $g \in R_{jL}$ cover X, $\mathcal{O}_X(j)|_{D_+(g)} = \mathcal{O}_{D_+(g)} \cdot g$, and $\mathcal{O}_X(j) \cong \mathcal{O}_X(j_1)^{\otimes j/j_1}$ is invertible [Har77, Proposition II.5.12(a)(b)] and ample [Har77, Theorem II.7.6]. Let j_0 be a positive integer divisible by j_1 such that $j_0\ell_1, j_0\ell_2 \in \mathbb{Z}$. Let j be a positive integer divisible by j_0. Then $jL = j\ell_1 H - j\ell_2 E \in \operatorname{Pic}(Z)$.

We compute the sections of $\mathcal{O}_X(j)$. A section of $\mathcal{O}_X(j)$ over X is a family of elements of $(R_g)_{jL}$ for $g \in R_{jL}$ that are compatible on the overlaps, that is, a regular function on $\widehat{X}$ that is semi-invariant of weight jL. By **Step 1**, **Step 2**, and [Har77, Proposition II.6.3A], any regular function on $\widehat{X}$ extends to $\overline{X}$. Thus

$$H^0(X, \mathcal{O}_X(j)) = R_{jL}.$$

For $0 \neq g \in R_{jL}$, let D_g be the divisor of the corresponding section of $\mathcal{O}_X(j)$. On $D_+(g')$ with $g' \in R_{jL}$, D_g is defined by $g/g' \in (R_{g'})^{\mathbb{T}}$. Since g' is invertible on $\{g' \neq 0\} = h^{-1}(D_+(g'))$, we have $h^*D_g = \mathrm{div}(g|_{\widehat{X}})$ over $D_+(g')$. Thus $h^*D_g = \mathrm{div}(g|_{\widehat{X}})$.

We compute the canonical class. Let

$$\widehat{X}^\circ := \overline{X}_Z \backslash \{U_1 = \cdots = U_{N-1} = 0\}$$

and $X^\circ := h(\widehat{X}^\circ)$. By **Step 1** and **Step 3**, $\widehat{X}^\circ$ is smooth and $\mathbb{T}$ acts freely on $\widehat{X}^\circ$. Since h is a geometric quotient, X° is open and $h^{-1}(X^\circ) = \widehat{X}^\circ$. By **Step 1** and **Step 2**, $\widehat{X} \backslash \widehat{X}^\circ$ is of dimension at most N. Since any $\mathbb{T}$-orbit in $\widehat{X}$ is of dimension 2, $X \backslash X^\circ$ is of dimension at most $N-2$. Let

$$\Omega := \frac{dx_0 \wedge dx_1 \wedge dx_2 \wedge de \wedge dU_1 \wedge \cdots \wedge dU_{N-2}}{U_{N-1}^{N-1}}$$

on $\overline{X} \cap \{U_{N-1} \neq 0\}$. Since Ω is N times the Poincaré residue of

$$\frac{dx_0 \wedge dx_1 \wedge dx_2 \wedge de \wedge dU_1 \wedge \cdots \wedge dU_{N-1}}{F_C + \sum_i U_i^N}$$

along the hypersurface $\overline{X} \subset \mathbb{A}^{N+3}$, Ω extends to a nowhere vanishing section of $\omega_{\overline{X}}$ by adjunction. Since $\mathbb{T}$ acts on any coordinate by the character given by the degree of the coordinate, the formula for Ω shows that Ω has weight

$$\deg x_0 + \deg x_1 + \deg x_2 + \deg e + (N-2)\tau - (N-1)\tau = (3H - E) - \tau = L,$$

that is, $t^*\Omega = t(L)\Omega$ for any $t \in \mathbb{T} = \mathrm{Hom}(\Lambda, \mathbb{C}^*)$. Let ξ_1, ξ_2 be the vector fields on $\overline{X}$ induced by a basis of the Lie algebra of $\mathbb{T}$, which are $\mathbb{T}$-invariant since $\mathbb{T}$ is commutative. Let $\Theta := \iota_{\xi_1}\iota_{\xi_2}\Omega$. Since $\mathbb{T}$ acts freely on $\widehat{X}^\circ$, Θ is a nowhere vanishing N-form on $\widehat{X}^\circ$, $\iota_{\xi_1}\Theta = \iota_{\xi_2}\Theta = 0$, and $t^*\Theta = t(L)\Theta$ for any $t \in \mathbb{T}$. For $0 \neq g \in R_{jL}$, the form $g^{-1}\Theta^{\otimes j}$ is $\mathbb{T}$-invariant. By **Step 3**, over $\overline{X}_{ab} \backslash \{U_1 = \cdots = U_{N-1} = 0\}$, the morphism h is étale locally the projection $\mathbb{T} \times V_{ab}^\circ \to V_{ab}^\circ$, where $V_{ab}^\circ := V_{ab} \backslash \{U_1 = \cdots = U_{N-1} = 0\}$ and Γ acts freely on V_{ab}°. Thus $g^{-1}\Theta^{\otimes j}$ descends to a nowhere vanishing section σ_g of $\omega_{X^\circ}^{\otimes j}$ over $D_+(g) \cap X^\circ$, and $\sigma_g = (g'/g)\sigma_{g'}$ for $g, g' \in R_{jL}$. Comparing with $\mathcal{O}_X(j)|_{D_+(g)} = \mathcal{O}_{D_+(g)} \cdot g$, we obtain an isomorphism

$$\mathcal{O}_X(j)|_{X^\circ} \otimes \omega_{X^\circ}^{\otimes j} \to \mathcal{O}_{X^\circ}, \qquad g \otimes \sigma_g \to 1.$$

Since X is normal and $X \backslash X^\circ$ is of codimension at least 2, $\mathcal{O}_X(jK_X) \cong \mathcal{O}_X(j)^{-1}$. Thus $-jK_X$ is Cartier, and $\mathcal{O}_X(-jK_X) \cong \mathcal{O}_X(j)$ is ample. Composing $R_{jL} = H^0(X, \mathcal{O}_X(j))$ with the map on global sections induced by $\mathcal{O}_X(j) \cong \mathcal{O}_X(-jK_X)$, we obtain (4).

Step 5. In this step, we prove (3).

By **Step 1**, $\widehat{X}$ is smooth at any point where $U_i \neq 0$ for some i. Let $x \in \widehat{X}$ with $U_1(x) = \cdots = U_{N-1}(x) = 0$. If $x_1(x) = x_2(x) = 0$, then $\omega(x)$ is contained in the cone $\{c_1 \geq c_2 \geq 0\}$ generated by H and E, which does not contain $L = (\ell_1 - \ell_2, \ell_1)$. If $x_0(x) = e(x) = 0$, then $\omega(x) \subset \mathbb{R}_{\geq 0}(H - E)$, which does not contain L. Thus $x \in \overline{X}_{ab}$ for some $a \in \{x_0, e\}$ and $b \in \{x_1, x_2\}$. Since $\mathbb{T} \times V_{ab} \to \overline{X}_{ab}$ is finite étale (**Step 3**), V_{ab} is normal, and by [KM98, Proposition 5.20], $\overline{X}_{ab}$ is klt if $\mathbb{T} \times V_{ab}$ is klt. Since the product of a log resolution of V_{ab} with $\mathbb{T}$ is a log resolution of $\mathbb{T} \times V_{ab}$, and $\mathbb{T} \times D$ has log discrepancy $a(D, V_{ab}, 0)$ with respect to $\mathbb{T} \times V_{ab}$ for any prime divisor D on the log resolution of V_{ab}, it suffices to prove that the normal variety $V := V_{ab}$, which is irreducible by **Step 3**, is klt. Let $\phi := \phi_{ab}$. Since V is a hypersurface in $Z_{ab} \times \mathbb{A}^{N-1}$, K_V is Cartier.

Let $\psi : Z' \to Z$ be a log resolution of (Z, C) that is an isomorphism over $Z\backslash C$. For any prime divisor D on Z', let $A_D := a(D, Z, 0)$ and $m_D := \mathrm{mult}_D\, \psi^* C$. Since (Z, B) is klt,

$$A_D - \frac{m_D}{N} = a(D, Z, B) > 0.$$

Let $V_{Z'} := V \times_{Z_{ab}} \psi^{-1}(Z_{ab})$, which is the hypersurface $\{\psi^*\phi + \sum_i U_i^N = 0\}$ in $\psi^{-1}(Z_{ab}) \times \mathbb{A}^{N-1}$. The singular locus of $V_{Z'}$ is contained in $\{U_1 = \cdots = U_{N-1} = 0,\ \psi^*\phi = 0\}$, which is of dimension 1, so $V_{Z'}$ is normal by [Har77, Proposition II.8.23(b)]. The part of $V_{Z'}$ over $\psi^{-1}(C)$ is of dimension at most $N-1$, and the part over $Z'\backslash\psi^{-1}(C)$ is isomorphic to the part of V over $Z\backslash C$, and any irreducible component of $V_{Z'}$ is of dimension N, so, since V is irreducible (**Step 3**), $V_{Z'}$ is irreducible and $V_{Z'} \to V$ is proper and birational. Let

$$B_{V_{Z'}} := \sum_D (1 - A_D)(D \times \mathbb{A}^{N-1})|_{V_{Z'}}.$$

Since $K_{Z'} = \psi^* K_Z + \sum_D (A_D - 1)D$ and $V_{Z'}$ is the pullback of the Cartier divisor V, adjunction implies that $K_{V_{Z'}} + B_{V_{Z'}}$ is the pullback of K_V. Thus V is klt if and only if $(V_{Z'}, B_{V_{Z'}})$ is sub-klt. Over $Z'\backslash\psi^{-1}(C)$, $\psi^*\phi$ is invertible, so $V_{Z'}$ is smooth, and $B_{V_{Z'}} = 0$ because $\mathrm{Exc}(\psi) \subset \psi^{-1}(C)$.

Let $z' \in \psi^{-1}(C)$. Choose local coordinates z_1, z_2 of Z' at z' such that $\psi^{-1}(C) \cup \mathrm{Exc}(\psi)$ is contained in $\{z_1 z_2 = 0\}$ near z', and let $m_j := m_{D_j}$ and $A_j := A_{D_j}$, where $D_j := \{z_j = 0\}$ near z'. Then $\psi^*\phi = \epsilon z_1^{m_1} z_2^{m_2}$ for a function ϵ invertible near z', and $(m_1, m_2) \neq (0, 0)$. After an étale base change of a neighborhood of z' that adds an N-th root $\epsilon^{1/N}$ of ϵ, the morphism

$$(z', U) \to (z_1, z_2, \epsilon^{-1/N} U_1, \ldots, \epsilon^{-1/N} U_{N-1})$$

is étale, $V_{Z'}$ is the preimage of

$$V' := \left\{ z_1^{m_1} z_2^{m_2} + \sum_{i=1}^{N-1} U_i^N = 0 \right\} \subset \mathbb{A}^{N+1},$$

and $B_{V_{Z'}}$ is the pullback of $B_{V'} := \sum_j (1 - A_j)\{z_j = 0\}|_{V'}$. By the argument of **Step 1**, with $\mathbb{C}[z_1, z_2]$ in place of $\mathcal{S}$ and $z_1^{m_1} z_2^{m_2}$ in place of F_C, V' is a normal variety. Since log discrepancies do not change under étale base change [Kol13, 2.14(2)], it suffices to prove that $(V', B_{V'})$ is sub-klt.

The polynomial $\Phi' := z_1^{m_1} z_2^{m_2} + \sum_i U_i^N$ is Newton non-degenerate at the origin in the sense of [AGM24, Definition 1.2]: $0 \in V'$ since $(m_1, m_2) \neq (0, 0)$, and for any $v \in \mathbb{R}^{N+1}_{\geq 0}$, the initial form $\mathrm{In}_v(\Phi')$ either contains some U_i^N, and then its derivative $N U_i^{N-1}$ with respect to U_i does not vanish on the torus $(\mathbb{C}^*)^{N+1}$, or is equal to $z_1^{m_1} z_2^{m_2}$, which has no zero on the torus. Let Σ'' be a smooth refinement of the dual fan of the Newton polyhedron [CLS11, Theorem 11.1.9], let $Y \to \mathbb{A}^{N+1}$ be the induced proper birational toric morphism, let V'' be the strict transform of V', and let $\pi'' : V'' \to V'$ be the induced morphism.

Let κ be a cone of Σ'' and v a vector in the relative interior of κ. Since κ is contained in a cone of the dual fan of the Newton polyhedron, there is a monomial μ'' of Φ' whose exponent attains the minimum of $\langle v', \cdot \rangle$ over the exponents of Φ' for every $v' \in \kappa$. Then Φ'/μ'' is regular on the affine chart of κ, and the restriction of Φ'/μ'' to the torus orbit $O(\kappa)$ of Y is $\mathrm{In}_v(\Phi')/\mu'' \neq 0$, since the restriction to $O(\kappa)$ of a monomial regular on the chart is the corresponding character of $O(\kappa)$ if the exponent of the monomial is orthogonal to κ, and is 0 otherwise [CLS11, Lemma 3.2.5]. Since every toric divisor of the affine chart of κ contains $O(\kappa)$ [CLS11, Theorem 3.2.6(c)(d)], the zero locus of Φ'/μ'' on the chart contains no toric divisor and is equal to V'' there, and $V'' \cap O(\kappa)$ is defined in $O(\kappa)$ by $\mathrm{In}_v(\Phi')/\mu''$. The exponents of $\mathrm{In}_v(\Phi')/\mu''$ are orthogonal to κ, so the zero locus of $\mathrm{In}_v(\Phi')$ in $(\mathbb{C}^*)^{N+1}$ is the preimage of $V'' \cap O(\kappa)$ under the smooth surjective homomorphism $(\mathbb{C}^*)^{N+1} \to O(\kappa)$ (since $\kappa^\perp \cap \mathbb{Z}^{N+1}$ is a saturated sublattice of $\mathbb{Z}^{N+1}$, the torus $(\mathbb{C}^*)^{N+1}$ is the product of $O(\kappa)$ and a torus, and the homomorphism is the projection [CLS11, Lemma 3.2.5]), and by the non-degeneracy above, $V'' \cap O(\kappa)$ is empty or a smooth hypersurface of $O(\kappa)$. Hence V'' is smooth along $O(\kappa)$ and transverse to every toric divisor of Y containing $O(\kappa)$

(cf. [AGM24, Theorem 1.3]). As Y is the union of the orbits $O(\kappa)$, V'' is smooth and transverse to the toric boundary of Y. Thus π'' is a log resolution of $(V', B_{V'})$, and any π''-exceptional prime divisor and any component of the strict transform of $\operatorname{Supp} B_{V'}$ is a component of $P_\beta \cap V''$ for some toric prime divisor P_β of Y.

Let P_β be the toric prime divisor of Y corresponding to a ray of Σ'' with primitive generator

$$(\alpha_1, \alpha_2, \beta_1, \dots, \beta_{N-1}) \in \mathbb{Z}_{\geq 0}^{N+1},$$

and let

$$t := \min\{m_1\alpha_1 + m_2\alpha_2, N\beta_1, \dots, N\beta_{N-1}\}.$$

The coefficient of P_β in K_Y minus the pullback of $K_{\mathbb{A}^{N+1}}$ is $\sum_j \alpha_j + \sum_i \beta_i - 1$ [CLS11, Lemma 11.4.10]. The pullback of z_j vanishes to order α_j along P_β, and a direct calculation shows that the pullback of V' contains P_β with multiplicity t: by [CLS11, Proposition 4.1.2], the pullback of a monomial vanishes along P_β to the order obtained by pairing its exponent vector with $(\alpha_1, \alpha_2, \beta_1, \dots, \beta_{N-1})$, the monomials of $z_1^{m_1} z_2^{m_2} + \sum_i U_i^N$ give the orders $m_1\alpha_1 + m_2\alpha_2, N\beta_1, \dots, N\beta_{N-1}$, and the terms of order t do not cancel along P_β, because they restrict to distinct characters of the torus orbit of P_β. The boundary $B_{V'}$ contributes $\sum_j (A_j - 1)\alpha_j$. By adjunction, any component of $P_\beta \cap V''$ has log discrepancy

$$\sum_j A_j\alpha_j + \sum_i \beta_i - t \geq \sum_j \Big(A_j - \frac{m_j}{N}\Big)\alpha_j + \frac{m_1\alpha_1 + m_2\alpha_2 - t}{N}$$

with respect to $(V', B_{V'})$, where the inequality follows from $\beta_i \geq t/N$ for any i. Since $A_j - m_j/N = a(D_j, Z, B) > 0$ and $t \leq m_1\alpha_1 + m_2\alpha_2$, the right-hand side is positive if $\alpha_1 + \alpha_2 > 0$. If $\alpha_1 = \alpha_2 = 0$, then $t = 0$ and the left-hand side is $\sum_i \beta_i > 0$. Any other prime divisor on V'' has log discrepancy 1.

Let $K_{V''} + \Delta'' = \pi''^*(K_{V'} + B_{V'})$. Since V'' is smooth and transverse to the toric boundary of Y, Δ'' is supported on a simple normal crossing divisor, and by the computation above every coefficient of Δ'' is less than 1. Hence (V'', Δ'') is sub-klt by [KM98, Corollary 2.31(3)], and $(V', B_{V'})$ is sub-klt by [KM98, Lemma 2.30]. Thus V is klt, and $\widehat{X}$ is klt.

Since $\widehat{X}$ is klt and h is a good quotient by the torus $\mathbb{T}$ onto the projective variety X, there exists an $\mathbb{R}$-divisor $\Delta_X \geq 0$ on X such that (X, Δ_X) is klt [BGLM24, Theorem 2]. By **Step 4**, K_X is $\mathbb{Q}$-Cartier, so Δ_X is $\mathbb{R}$-Cartier, and

$$a(P, X, 0) \geq a(P, X, \Delta_X) > 0$$

for any prime divisor P over X. Thus X is klt. By **Step 4**, $-K_X$ is ample, and by **Step 3**, $\dim X = N$. Thus X is a klt Fano variety of dimension N.

Step 6. In this step, we prove (5).

Let $a \in \{x_0, e\}$, $b \in \{x_1, x_2\}$, and let $g \in R_{jL}$ be the monomial of **Step 2** with $\overline{X}_{ab} = \{g \neq 0\}$. By Lemma 2.10(3), Z_{ab} is the spectrum of the degree zero part of $\mathcal{S}_g$, which is contained in the degree zero part $(R_g)^{\mathbb{T}}$ of R_g. Thus the inclusion $\mathcal{S} \subset R$ defines a morphism $D_+(g) \to Z_{ab}$, whose composite with h is the $\mathbb{T}$-invariant morphism $\overline{X}_{ab} \to Z_{ab}$. Under $D_+(g) \cong V_{ab}/\Gamma$ (**Step 3**), the morphism $D_+(g) \to Z_{ab}$ is induced by the projection $V_{ab} \to Z_{ab}$, which is surjective because the hypersurface $\{c + \sum_i U_i^N = 0\} \subset \mathbb{A}^{N-1}$ is non-empty for any $c \in \mathbb{C}$. Gluing over a and b, we obtain a surjective morphism $h(\overline{X}_Z) \to Z$. Since $h(\overline{X}_Z)$ is a non-empty open subset of X, we obtain the dominant rational map $\rho : X \dashrightarrow Z$ induced by $\mathcal{S} \subset R$. □

5.2. **Lifting quasi-monomial valuations.** In this subsection, we lift a quasi-monomial valuation v over Z to a quasi-monomial valuation w over X such that the rational rank of w is equal to that of v, $a(w, X, 0) = a(v, Z, B)$, and $T_{-K_X}(w) = T_L(v)$. The weight c/N that the lift gives to each U_i, where c is the value of a local equation of C, is explained in part (c) of the sketch of the proof in Section 1.

In the proof, the weights c/N of the U_i define a valuation because the initial form $\mathrm{in}_v(F_C) + \sum_i U_i^N$ of the equation $F_C + \sum_i U_i^N$ with respect to v and the weights c/N is still irreducible (**Step 3**). The lift does not increase T because, for each term $a_I U^I$ of an element of R_{jL}, a power of a_I times the corresponding power of F_C is a section of a multiple of jL on Z whose value bounds the value of the term (**Step 7**). The log discrepancy is computed on a toric resolution when $c > 0$ (**Step 5**) and on the log resolution Z' of Z when $c = 0$ (**Step 6**).

Proposition 5.3. *Notation as in Construction-Theorem 5.2. Let v be a non-trivial quasi-monomial valuation over Z. Then there exists a quasi-monomial valuation w over X such that $\rho_* w = v$, the rational rank of w is equal to that of v, $a(w, X, 0) = a(v, Z, B)$, and $T_{-K_X}(w) = T_L(v)$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we cover an open subset of X by quotients V_W/Γ of hypersurfaces V_W by a finite cyclic group Γ.

Let

$$\mathbb{A}^4_\circ := \{(x_1, x_2) \neq (0,0),\ (x_0, e) \neq (0,0)\} \subset \operatorname{Spec} \mathcal{S} = \mathbb{A}^4,$$

and let $\overline{X}_Z \subset \overline{X}$ be the inverse image of $\mathbb{A}^4_\circ$ under the morphism $\overline{X} \to \mathbb{A}^4$ induced by $\mathcal{S} \subset R$. Let j be a positive integer such that $j\ell_1, j\ell_2 \in \mathbb{Z}$. For $i \in \{1, 2\}$, the elements

$$x_0^{j(\ell_1-\ell_2)} x_i^{j\ell_2} \quad \text{and} \quad e^{j(\ell_1-\ell_2)} x_i^{j\ell_1}$$

of $\mathcal{S}$ have degree $j\ell_1 H - j\ell_2 E = jL$ (Lemma 2.10(3)), so they belong to R_{jL}. Since $\ell_1 > \ell_2 > 0$, the open subsets of $\overline{X}$ where the elements above do not vanish cover $\overline{X}_Z$, hence $\overline{X}_Z \subset \widehat{X}$. Since $\overline{X}_Z$ is $\mathbb{T}$-invariant and h is a geometric quotient (Construction-Theorem 5.2(2)), $X_Z := h(\overline{X}_Z)$ is an open subset of X and $h^{-1}(X_Z) = \overline{X}_Z$. The composition $\overline{X}_Z \to \mathbb{A}^4_\circ \to Z$ is $\mathbb{T}$-invariant, so it descends to a morphism $X_Z \to Z$, which is the restriction of ρ.

The open subsets $\{x_0 x_1 \neq 0\}$, $\{x_0 x_2 \neq 0\}$, $\{e x_1 \neq 0\}$, and $\{e x_2 \neq 0\}$ cover Z. Let W be one of them. We describe the case $W = \{x_0 x_1 \neq 0\}$. The other cases follow by renaming the coordinates. Since the degrees H and $H - E$ of x_0 and x_1 form a basis of $\operatorname{Pic}(Z)$ (Lemma 2.10(1)) and $\mathbb{T}_0$ acts freely on $\mathbb{A}^4_\circ$ (Lemma 2.10(3)), the slice $\{x_0 = x_1 = 1\} \cong \mathbb{A}^2$ with coordinates x_2, e maps isomorphically onto W. Let $f_W := F_C(1, 1, x_2, e)$, which is a local equation of C on W, and let

$$V_W := \Big\{ f_W + \sum_{i=1}^{N-1} U_i^N = 0 \Big\} \subset W \times \mathbb{A}^{N-1}$$

be the inverse image of the slice in $\overline{X}$. Let

$$\Gamma := \operatorname{Ker}(\mathbb{T} \to \mathbb{T}_0) = \operatorname{Hom}(\Lambda/\operatorname{Pic}(Z), \mathbb{C}^*).$$

Since $\Lambda/\operatorname{Pic}(Z)$ is generated by τ, Γ is a finite cyclic group, any $\zeta \in \Gamma$ acts trivially on $\mathcal{S}$ and multiplies each U_i by $\zeta(\tau)$, and the character $\zeta \to \zeta(\tau)$ is injective. Moreover, $|\Gamma|$ divides N, as $N\tau = [C] \in \operatorname{Pic}(Z)$. Let $X_W := X_Z \cap \rho^{-1}(W)$. Since $\mathbb{T} \to \mathbb{T}_0$ is surjective and $\mathbb{T}_0$ acts freely on $\mathbb{A}^4_\circ$, the morphism $\mathbb{T} \times V_W \to h^{-1}(X_W)$, $(t, x) \to t \cdot x$, is the quotient by the free action $\zeta \cdot (t, x) = (t\zeta^{-1}, \zeta x)$ of Γ. Hence $X_W \cong V_W/\Gamma$, and under the isomorphism, $q_W := h|_{V_W} : V_W \to X_W$ is the quotient morphism and $\rho \circ q_W$ is the projection $V_W \to W$. Since $\overline{X}$ is normal (Construction-Theorem 5.2(1)) and $\mathbb{T} \times V_W \to h^{-1}(X_W)$ is étale, V_W is normal. Moreover, V_W is irreducible: the polynomial $f_W + \sum_{i<N-1} U_i^N$ is non-constant, as $N - 1 \geq 2$, and it is not divisible by the square of any irreducible polynomial, because the partial derivative of $f_W + \sum_{i<N-1} U_i^N$ with respect to U_1 is NU_1^{N-1} and U_1 does not divide $f_W + \sum_{i<N-1} U_i^N$. Hence $U_{N-1}^N + f_W + \sum_{i<N-1} U_i^N$ is irreducible by Eisenstein's criterion.

If $\zeta \in \Gamma$ is not the identity, then any fixed point of ζ on V_W satisfies $U_1 = \cdots = U_{N-1} = 0$, and $\{U_1 = \cdots = U_{N-1} = 0\} \cap V_W = \{f_W = 0\} \times \{0\}$ is of codimension $N - 1 \geq 2$ in V_W. Hence q_W is quasi-étale, that is, finite and étale in codimension one, and $K_{V_W} = q_W^* K_{X_W}$ by the Hurwitz

formula [Kol13, (2.41.3)]. Finally, by Construction-Theorem 5.2(4), for any positive integer j divisible by j_0 and any $0 \neq g \in R_{jL}$, we have

$$q_W^* D_g = \mathrm{div}(g|_{V_W}).$$

Step 2. In this step, we choose coordinates in which v and C are monomial.

By [JM12, Lemma 3.6(ii)], there exists a good log smooth pair (Z_1, D_1) over Z adapted to v. Let $\psi : Z' \to Z$ be a log resolution of (Z, C) which factors through Z_1 and on which the pullbacks of D_1 and C have simple normal crossing support. By [JM12, Lemma 3.6(i)], there exist a closed point $y \in Z'$, a regular system of parameters y_1, y_2 of $\mathcal{O}_{Z',y}$, and real numbers $t_1, t_2 \geq 0$, such that v is the monomial valuation in y_1, y_2 with weights t_1, t_2 (Definition 2.2) and $\mathrm{Supp}(\psi^* C) \cup \mathrm{Exc}(\psi) \subset \{y_1 y_2 = 0\}$ near y. When the center of v on Z' is a curve, y is a general closed point of it and the second coordinate has weight 0 [JM12, Proposition 3.1]. Let P_1, P_2 be the prime divisors on Z' defined by y_1, y_2 near y. By **Step 1**, we may take W such that $\psi(y) \in W$. Replacing Z' with $\psi^{-1}(W)$, we may assume that $\psi(Z') \subset W$, and $\psi : Z' \to W$ is still a log resolution of $(W, C|_W)$. Let $f := f_W$. Since f is a local equation of C, there exist a unit $\epsilon \in \mathcal{O}_{Z',y}$ and integers $m_1, m_2 \geq 0$ such that $\psi^* f = \epsilon y_1^{m_1} y_2^{m_2}$. Let $c := v(f)$. Then

$$c = m_1 t_1 + m_2 t_2 \geq 0 \quad \text{and} \quad a(v, Z, 0) = t_1 a(P_1, Z, 0) + t_2 a(P_2, Z, 0).$$

Since $\mathrm{mult}_{P_j} \psi^* B = m_j/N$, we have $a(P_j, Z, B) = a(P_j, Z, 0) - m_j/N$ for $j \in \{1, 2\}$, hence

$$a(v, Z, B) = a(v, Z, 0) - \frac{c}{N}. \tag{5.1}$$

Step 3. In this step, we construct w and show that $\rho_* w = v$ and that the rational rank of w is equal to that of v.

Since $f + \sum_i U_i^N$ is monic of degree N in U_{N-1}, any regular function Ψ on V_W can be written uniquely as

$$\Psi = \sum_J \varphi_J U^J,$$

where J runs over the elements of $\mathbb{Z}_{\geq 0}^{N-1}$ with $J_{N-1} < N$ and the φ_J are regular functions on W, all but finitely many equal to 0. As for the elements of R in Construction-Theorem 5.2(6), we call $\sum_J \varphi_J U^J$ the *normal form* of Ψ. For $\Psi \neq 0$, we let $|J| := J_1 + \cdots + J_{N-1}$ and

$$\widetilde{w}(\Psi) := \min\left\{ v(\varphi_J) + \frac{|J|c}{N} \middle| \varphi_J \neq 0 \right\}. \tag{5.2}$$

Formula (5.2), with $\bar{v}$ in place of v and the elements a_I of Construction-Theorem 5.2(6) in place of φ_J, defines the number $b(g)$ in Lemma 6.6(2).

We show that $\widetilde{w}$ extends to a valuation over V_W. Let $G := \bigoplus_{s \in \mathbb{R}} \{v \geq s\}/\{v > s\}$ be the graded ring of v on the rational functions on Z. Then G is a domain whose non-zero homogeneous elements are units. For $\varphi \neq 0$, let $\mathrm{in}_v(\varphi) \in G$ be the class of φ in degree $v(\varphi)$. Let

$$A := G[\overline{U}_1, \ldots, \overline{U}_{N-1}] \Big/ \Big(\mathrm{in}_v(f) + \sum_{i=1}^{N-1} \overline{U}_i^N \Big), \qquad \deg \overline{U}_i = \frac{c}{N}.$$

The monomials $\overline{U}^J$ with $J_{N-1} < N$ form a basis of A over G, so A embeds into $A \otimes_G \mathrm{Frac}(G)$, which is a domain by the argument of **Step 1** over $\mathrm{Frac}(G)$, as $\mathrm{in}_v(f) \neq 0$. Hence A is a domain.

For $\Psi' = \sum_J \varphi'_J U^J$ with the φ'_J regular functions on W, not necessarily in normal form, and a real number s such that $v(\varphi'_J) + |J|c/N \geq s$ for any J, let $[\Psi']_s \in A$ be the sum over J of the class of φ'_J in $\{v \geq s - |J|c/N\}/\{v > s - |J|c/N\}$ times $\overline{U}^J$. Then $[\Psi']_s$ is additive in Ψ'. Since $v(f) = c$, replacing a term $\varphi' U^J U_{N-1}^N$ of Ψ' with $-\varphi' f U^J - \sum_{i<N-1} \varphi' U^J U_i^N$ keeps the inequality and does not change $[\Psi']_s$. Let Ψ_1, Ψ_2 be non-zero normal forms and $s_i := \widetilde{w}(\Psi_i)$. Then $[\Psi_i]_{s_i} \neq 0$, since the monomials $\overline{U}^J$ with $J_{N-1} < N$ form a basis of A over G. Finitely many

replacements of terms $\varphi' U^J U_{N-1}^N$ turn the product of Ψ_1 and Ψ_2 as polynomials in $U_1, \dots, U_{N-1}$ into the normal form Ψ_3 of $\Psi_1\Psi_2$, so

$$[\Psi_3]_{s_1+s_2} = [\Psi_1]_{s_1}[\Psi_2]_{s_2} \neq 0,$$

as A is a domain. Thus $\widetilde{w}(\Psi_1\Psi_2) = s_1 + s_2$. Since the normal form of $\Psi_1 + \Psi_2$ is the sum of the normal forms, $\widetilde{w}(\Psi_1 + \Psi_2) \geq \min\{s_1, s_2\}$. Thus $\widetilde{w}$ extends uniquely to a valuation on the function field of V_W, and $\widetilde{w}(\varphi) = v(\varphi)$ for any non-zero rational function φ on Z. Let ξ be the generic point of $\operatorname{center}_Z v$. Then $\xi \in W$, and normal forms of elements of $\mathcal{O}_{W,\xi}[U_1, \dots, U_{N-1}]$ have coefficients in $\mathcal{O}_{W,\xi}$, so $\widetilde{w}$ is non-negative on them. Hence $\widetilde{w}$ is a valuation over V_W.

Any $\zeta \in \Gamma$ multiplies U^J by $\zeta(\tau)^{|J|}$, so $\widetilde{w}$ is Γ-invariant. Let w be the valuation over X defined by $w := (q_W)_*\widetilde{w}$, where X_W is an open subset of X. By **Step 1**,

$$\rho_* w = (\rho \circ q_W)_* \widetilde{w} = v.$$

The value group of $\widetilde{w}$ contains the value group of v and is contained in the value group of v plus $\mathbb{Z}\frac{c}{N}$, and c belongs to the value group of v. Hence the rational rank of $\widetilde{w}$ is equal to that of v. Let $\phi : Y_1 \to Y_2$ be a dominant morphism between normal varieties with $\dim Y_1 = \dim Y_2$, for example q_W, or an étale morphism between irreducible normal varieties, and let μ be a valuation over Y_1. Since the function field of Y_1 is a finite extension of that of Y_2, the rational rank of μ is equal to that of $\phi_*\mu$ [ZS60, Chapter VI, §11, Lemma 1], and the transcendence degree of the residue field of μ is equal to that of $\phi_*\mu$ [ZS60, Chapter VI, §6, Lemma 2, Corollary 2]. Since a valuation is quasi-monomial if and only if the sum of the rational rank and the transcendence degree of the residue field is equal to the dimension [JM12, Proposition 3.7], applied on resolutions of Y_1 and Y_2, μ is quasi-monomial if and only if $\phi_*\mu$ is quasi-monomial. For $\phi = q_W$, the rational rank of w is equal to that of v, and w is quasi-monomial if $\widetilde{w}$ is quasi-monomial. It remains to show that w is quasi-monomial, $a(w, X, 0) = a(v, Z, B)$, and $T_{-K_X}(w) = T_L(v)$.

Step 4. In this step, we compare canonical divisors after the base change by ψ.

Let

$$V' := V_W \times_W Z' = \left\{\psi^* f + \sum_{i=1}^{N-1} U_i^N = 0\right\} \subset Z' \times \mathbb{A}^{N-1},$$

with the induced action of Γ. Since V' is flat over Z' and the generic fiber of $V' \to Z'$ is the generic fiber of $V_W \to W$, V' is irreducible. The singular points of the hypersurface V' satisfy $U_1 = \cdots = U_{N-1} = 0$, hence are contained in $\{\psi^* f = 0\} \times \{0\}$, which is of codimension $N - 1 \geq 2$ in V'. Thus V' is normal by [Har77, Proposition II.8.23(b)], and hence so is V'/Γ. Since Γ acts trivially on W and taking Γ-invariants commutes with base change in characteristic zero, $V'/\Gamma = X_W \times_W Z'$. Let $\psi_V : V' \to V_W$ and $\psi_X : V'/\Gamma \to X_W$ be the projections, which are proper and birational, let $p' : V'/\Gamma \to Z'$ be the projection, and let $q' : V' \to V'/\Gamma$ be the quotient morphism. By adjunction for the hypersurfaces $V_W \subset W \times \mathbb{A}^{N-1}$ and $V' \subset Z' \times \mathbb{A}^{N-1}$, $K_{V'} = \psi_V^* K_{V_W} + (p' \circ q')^* K_{Z'/W}$. As in **Step 1**, the fixed points of non-identity elements of Γ on V' are contained in $\{\psi^* f = 0\} \times \{0\}$, so q' is quasi-étale and $K_{V'} = q'^* K_{V'/\Gamma}$. Together with $K_{V_W} = q_W^* K_{X_W}$, we obtain

$$K_{V'/\Gamma} = \psi_X^* K_{X_W} + p'^* K_{Z'/W}.$$

Since $K_{Z'/W} = (a(P_1, Z, 0) - 1)P_1 + (a(P_2, Z, 0) - 1)P_2$ near y and $w(p'^* y_j) = v(y_j) = t_j$ by $\rho_* w = v$, if w is quasi-monomial, then

$$a(w, X, 0) = a(w, V'/\Gamma, 0) + \sum_{j=1}^{2} (a(P_j, Z, 0) - 1)t_j. \tag{5.3}$$

Step 5. In this step, we assume that $c > 0$ and show that w is quasi-monomial and $a(w, X, 0) = a(v, Z, B)$.

We compare w with a toric valuation through étale morphisms, which preserve quasi-monomiality by **Step 3** and log discrepancies by [Kol13, 2.14(1)], as applied in the last paragraph

of this step. Possibly replacing Z' with an étale neighborhood of y on which $\epsilon = \epsilon_1^N$ for a unit ϵ_1, and U_i with $\epsilon_1^{-1}U_i$ for any i, which commutes with the action of Γ and multiplies U^J by a unit of value 0, we may assume that $\epsilon = 1$. Let

$$\Phi := y_1^{m_1}y_2^{m_2} + \sum_{i=1}^{N-1} U_i^N \quad \text{and} \quad V_0 := \{\Phi = 0\} \subset \mathbb{A}^{N+1},$$

where $y_1, y_2, U_1, \dots, U_{N-1}$ are the coordinates of $\mathbb{A}^{N+1}$ and Γ acts on $\mathbb{A}^{N+1}$ through the action on the U_i. By the argument of **Step 4**, V_0 is normal, and hence so is V_0/Γ. Since $\psi^* f = y_1^{m_1}y_2^{m_2}$, we have $V' = Z' \times_{\mathbb{A}^2} V_0$, where $Z' \to \mathbb{A}^2$ is the morphism (y_1, y_2), which is étale near y because y_1, y_2 is a regular system of parameters at y. Since Γ acts trivially on Z' and on $\mathbb{A}^2$ and taking Γ-invariants commutes with base change in characteristic zero (as in **Step 4**), $V'/\Gamma = Z' \times_{\mathbb{A}^2} (V_0/\Gamma)$. Hence, over a neighborhood of y in Z', the morphisms $V' \to V_0$ and $V'/\Gamma \to V_0/\Gamma$ are base changes of an étale morphism, so they are étale, and the first one is Γ-equivariant.

Let $\mathcal{N}_0 := \mathbb{Z}^{N+1}$, whose dual lattice $\mathcal{M}_0$ is the lattice of exponents of monomials in $y_1, y_2, U_1, \dots, U_{N-1}$, and let

$$\mathcal{N} := \mathcal{N}_0 + \mathbb{Z}\frac{1}{|\Gamma|}(0, 0, 1, \dots, 1).$$

The dual lattice $\mathcal{M} \subset \mathcal{M}_0$ of $\mathcal{N}$ is the lattice of exponents of Γ-invariant monomials, so $\mathbb{A}^{N+1}/\Gamma$ is the affine toric variety of the positive orthant σ_+ with respect to $\mathcal{N}$. Since $N - 1 \geq 2$, a vector of $\mathcal{N}$ which is a multiple of a coordinate vector belongs to $\mathcal{N}_0$, so the primitive generators in $\mathcal{N}$ of the rays of σ_+ are the coordinate vectors. Hence the support function of $K_{\mathbb{A}^{N+1}/\Gamma}$, the negative of the sum of the toric prime divisors, is $u \to |u|$, the sum of the coordinates of u. For $u \in \sigma_+$, let

$$\lambda(u) := \min\{m_1u_1 + m_2u_2, Nu_3, \dots, Nu_{N+1}\},$$

the minimum of $\langle u, \alpha\rangle$ over the exponents α of Φ. Let Σ_Φ be the dual fan of the Newton polyhedron of Φ, that is, the fan with support σ_+ whose cones are the closures of the subsets on which the set of exponents attaining λ is constant. For $u \in \sigma_+$, let $\mathrm{In}_u(\Phi)$ be the sum of the terms of Φ whose exponents attain $\lambda(u)$. The polynomial Φ is Newton non-degenerate in the sense of [AGM24, Definition 1.2]: $0 \in V_0$ since $(m_1, m_2) \neq (0, 0)$, and for any $u \in \sigma_+$, $\mathrm{In}_u(\Phi)$ either contains some U_i^N, whose derivative NU_i^{N-1} with respect to U_i does not vanish on the torus, or is equal to $y_1^{m_1}y_2^{m_2}$, which has no zero on the torus. Since the exponents of Φ belong to $\mathcal{M}$ and the torus of $\mathcal{N}$ is the quotient of the torus of $\mathcal{N}_0$ by Γ, Φ is also Newton non-degenerate in the sense of [AGM24, Definition 1.2] for the affine toric variety $\mathbb{A}^{N+1}/\Gamma$ of the cone σ_+ with respect to $\mathcal{N}$, whose closed orbit, the image of 0, is contained in V_0/Γ.

Let $\beta := (t_1, t_2, c/N, \dots, c/N)$. Then β belongs to the cone

$$\sigma_\Phi := \{u \in \sigma_+ \mid m_1u_1 + m_2u_2 = Nu_3 = \cdots = Nu_{N+1}\}$$

of Σ_Φ, on which all exponents of Φ attain λ, and $\lambda(\beta) = c$. If v has rational rank 1, then β spans a rational ray, and we replace Σ_Φ with the star subdivision at the ray $\mathbb{R}_{\geq 0}\beta$. By [CLS11, Theorem 11.1.9], there exists a smooth refinement Σ of Σ_Φ with respect to $\mathcal{N}$ which contains the ray $\mathbb{R}_{\geq 0}\beta$ in the case of rational rank 1. Let $\sigma \in \Sigma$ be the smallest cone containing β, let $u_1, \dots, u_d$ be the primitive generators of the rays of σ, and write $\beta = \sum_k s_k u_k$ with $s_k > 0$. Since σ_Φ is contained in a rational plane and β does not belong to a rational line when t_1, t_2 are linearly independent over $\mathbb{Q}$, d is the rational rank of v and $s_1, \dots, s_d$ are linearly independent over $\mathbb{Q}$. Let $\pi_\Sigma : X_\Sigma \to \mathbb{A}^{N+1}/\Gamma$ be the toric morphism, D_u the toric prime divisor of a ray with primitive generator u, and V_Σ the strict transform of V_0/Γ. Let σ' be a cone of Σ, u a vector in the relative interior of σ', $\sigma'' \supset \sigma'$ a maximal cone of Σ, and α'' an exponent of Φ attaining λ on σ''. Then $\chi^{-\alpha''}\Phi$ is regular on the affine toric open subset of X_Σ of the cone σ'', and the restriction of $\chi^{-\alpha''}\Phi$ to the orbit $O(\sigma')$ is $\chi^{-\alpha''}\,\mathrm{In}_u(\Phi) \neq 0$, since the restriction to $O(\sigma')$ of a monomial regular on the affine toric open subset of the cone σ'' is the corresponding

character of $O(\sigma')$ if the exponent of the monomial is orthogonal to σ', and is 0 otherwise [CLS11, Lemma 3.2.5]. Hence the zero locus of $\chi^{-\alpha''}\Phi$ contains no toric divisor and is equal to V_Σ there, and $V_\Sigma \cap O(\sigma')$ is defined in $O(\sigma')$ by $\chi^{-\alpha''}\operatorname{In}_u(\Phi)$, whose exponents are orthogonal to σ'. As the torus of $\mathcal{N}$ maps onto $O(\sigma')$ by a smooth surjective homomorphism (since $\sigma'^{\perp}\cap\mathcal{M}$ is a saturated sublattice of $\mathcal{M}$, the torus of $\mathcal{N}$ is the product of $O(\sigma')$ and a torus, and the homomorphism is the projection [CLS11, Lemma 3.2.5]), the zero locus of $\operatorname{In}_u(\Phi)$ in the torus of $\mathcal{N}$ is the preimage of $V_\Sigma \cap O(\sigma')$, and by the non-degeneracy above, $V_\Sigma \cap O(\sigma')$ is empty or a smooth hypersurface of $O(\sigma')$. Thus V_Σ is smooth along $O(\sigma')$ and transverse to every toric divisor containing $O(\sigma')$ (cf. [AGM24, Theorem 1.3] for an affine normal toric variety). As X_Σ is the union of the orbits $O(\sigma')$, V_Σ is smooth and transverse to the toric boundary. Since $\sigma \subset \sigma_\Phi$, we have $\Phi = \chi^{\alpha_\Phi}\widehat{\Phi}$, where $\alpha_\Phi := (m_1, m_2, 0, \dots, 0)$ and $\widehat{\Phi} \in \mathbb{C}[\sigma^\perp \cap \mathcal{M}]$, and $V_\Sigma \cap O(\sigma) = \{\widehat{\Phi} = 0\}$ in the orbit $O(\sigma) = \operatorname{Spec}\mathbb{C}[\sigma^\perp \cap \mathcal{M}]$. The polynomial Φ is irreducible by the argument of **Step 1** and is not divisible by any monomial, so Φ is a prime element of the Laurent polynomial ring $\mathbb{C}[\mathcal{M}_0]$. Since Φ is Γ-invariant and $\mathbb{C}[\mathcal{M}] = \mathbb{C}[\mathcal{M}_0]^\Gamma$, we have

$$\Phi\mathbb{C}[\mathcal{M}_0] \cap \mathbb{C}[\mathcal{M}] = \Phi\mathbb{C}[\mathcal{M}],$$

so Φ, and hence $\widehat{\Phi} = \chi^{-\alpha_\Phi}\Phi$, is a prime element of $\mathbb{C}[\mathcal{M}]$. Since $\sigma^\perp \cap \mathcal{M}$ is a saturated sublattice of $\mathcal{M}$, $\mathbb{C}[\mathcal{M}]$ is a Laurent polynomial ring over $\mathbb{C}[\sigma^\perp \cap \mathcal{M}]$, so $\widehat{\Phi}$ is a prime element of $\mathbb{C}[\sigma^\perp \cap \mathcal{M}]$, and $V_\Sigma \cap O(\sigma)$ is irreducible. Let w_β be the monomial valuation, at the generic point of $V_\Sigma \cap O(\sigma)$, in local equations of $D_{u_1}|_{V_\Sigma}, \dots, D_{u_d}|_{V_\Sigma}$ with weights $s_1, \dots, s_d$ [JM12, Proposition 3.1]. It is a quasi-monomial valuation over V_0/Γ of rational rank d.

We show that w_β corresponds to w under the étale morphism above. Let $\Psi = \sum_J \varphi_J U^J$ be a Γ-invariant normal form whose coefficients φ_J are power series in y_1, y_2, and let $\operatorname{val}_\beta(\Psi)$ be the minimum of $\langle\beta, \alpha\rangle$ over the exponents α of the monomials of Ψ. Then $\operatorname{val}_\beta(\Psi) = \widetilde{w}(\Psi)$ by (5.2). By [CLS11, Proposition 4.1.2], χ^α is a unit near the generic point of $O(\sigma)$ times $\prod_k z_k^{\langle u_k, \alpha\rangle}$, where z_k is a local equation of D_{u_k}. Hence $w_\beta(\Psi) \geq \operatorname{val}_\beta(\Psi)$, with equality if the sum $\operatorname{In}_\beta(\Psi)$ of the terms of Ψ attaining $\operatorname{val}_\beta(\Psi)$ does not vanish on $V_\Sigma \cap O(\sigma)$. Since $s_1, \dots, s_d$ are linearly independent over $\mathbb{Q}$,

$$\operatorname{In}_\beta(\Psi) = \chi^{\alpha_\Psi}\widehat{\Psi}$$

for some $\alpha_\Psi \in \mathcal{M}$ and $\widehat{\Psi} \in \mathbb{C}[\sigma^\perp \cap \mathcal{M}]$. The exponents of U_{N-1} in $\operatorname{In}_\beta(\Psi)$ are in $[0, N-1]$, while any non-zero multiple of Φ has two exponents of U_{N-1} differing by at least N. Thus $\widehat{\Phi}$ does not divide $\widehat{\Psi}$, and $w_\beta(\Psi) = \widetilde{w}(\Psi) = w(\Psi)$. If $t_1, t_2 > 0$, then, since the normal form of a Γ-invariant element is Γ-invariant and any rational function on V'/Γ is a quotient of Γ-invariant elements, w is equal to w_β under the étale morphism above, without renormalization.

Suppose that $t_j = 0$ for some j, and let j' be the index with $\{j, j'\} = \{1, 2\}$. By **Step 2**, the center of v on Z' is the curve $P_{j'}$ and y is a general closed point of $P_{j'}$, so $m_j = 0$, y_j does not appear in Φ, and the j-th coordinate of β is 0. In this case, we take Σ to be the product of the ray $\mathbb{R}_{\geq 0}e_j$, where e_j is the j-th coordinate vector, and a smooth refinement, with respect to the projection of $\mathcal{N}$ to the other N coordinates, of the dual fan of the Newton polyhedron of Φ as a polynomial in the variables other than y_j, which contains the projection of the ray $\mathbb{R}_{\geq 0}\beta$. Then V_Σ is the product of $\mathbb{A}^1_{y_j}$ and a variety of dimension $N-1$, and $w_\beta(y_j) = 0$. Since the center of w_β on V_0/Γ is not the image of the origin, we take Ψ above only among the normal forms of elements of $\mathbb{C}[y_1, y_2, U_1, \dots, U_{N-1}]^\Gamma$, which are regular functions on V_0/Γ generating its function field. The coefficients φ_J of such a normal form Ψ are polynomials in y_1, y_2, so Ψ is a finite sum of monomials, $\operatorname{In}_\beta(\Psi)$ is a finite sum, and the argument above shows

$$w_\beta(\Psi) = \operatorname{val}_\beta(\Psi) = \widetilde{w}(\Psi) = w(\Psi).$$

Thus the pushforward of w to V_0/Γ is w_β. Moreover, v has rational rank 1, so $d = 1$ and w_β is s_1 times the valuation defined by a component of $D_{u_1}|_{V_\Sigma}$. Since the ramification index of the étale base change along any prime divisor is 1, w is s_1 times the valuation defined by a component of

the preimage of that component, which is the description of w used in the last paragraph of this step.

Since w_β is quasi-monomial and w corresponds to w_β under the étale morphisms above, w is quasi-monomial by **Step 3**.

By [CLS11, Lemma 11.4.10], the coefficient of D_u in $K_{X_\Sigma} - \pi_\Sigma^* K_{\mathbb{A}^{N+1}/\Gamma}$ is $|u| - 1$, and since the valuation defined by D_u is the monomial valuation of weight u on $\mathbb{C}[\mathcal{M}]$ (cf. [CLS11, Proposition 4.1.2]) and V_0/Γ is reduced,

$$\pi_\Sigma^* \operatorname{div}(\Phi) = V_\Sigma + \sum_u \lambda(u) D_u,$$

where u runs over the primitive generators of the rays of Σ. The non-identity elements of Γ fix only $\{U_1 = \cdots = U_{N-1} = 0\}$, and

$$V_0 \cap \{U_1 = \cdots = U_{N-1} = 0\} = \{y_1^{m_1} y_2^{m_2} = 0\} \times \{0\}$$

is of codimension $N - 1 \geq 2$ in V_0. Thus $\mathbb{A}^{N+1} \to \mathbb{A}^{N+1}/\Gamma$ is étale over every codimension one point of V_0/Γ, so $\mathbb{A}^{N+1}/\Gamma$ is smooth there and adjunction holds there. Since V_0/Γ is normal,

$$(K_{\mathbb{A}^{N+1}/\Gamma} + V_0/\Gamma)|_{V_0/\Gamma} = K_{V_0/\Gamma},$$

and $K_{V_0/\Gamma}$ is $\mathbb{Q}$-Cartier. By adjunction on the smooth V_Σ,

$$K_{V_\Sigma} = (\pi_\Sigma|_{V_\Sigma})^* K_{V_0/\Gamma} + \sum_u (|u| - 1 - \lambda(u)) D_u|_{V_\Sigma}.$$

Since V_Σ is transverse to the toric boundary, any component of $D_u|_{V_\Sigma}$ has log discrepancy $|u| - \lambda(u)$ with respect to V_0/Γ. Since $u \to |u|$ and λ are linear on $\sigma \subset \sigma_\Phi$,

$$a(w_\beta, V_0/\Gamma, 0) = \sum_{k=1}^{d} s_k\big(|u_k| - \lambda(u_k)\big) = |\beta| - \lambda(\beta) = t_1 + t_2 + \frac{(N-1)c}{N} - c = t_1 + t_2 - \frac{c}{N}.$$

The base change of $\pi_\Sigma|_{V_\Sigma} : V_\Sigma \to V_0/\Gamma$ along the étale morphism above is a log resolution of the étale neighborhood in V'/Γ, and on it w is the monomial valuation in the pullbacks of local equations of $D_{u_1}|_{V_\Sigma}, \ldots, D_{u_d}|_{V_\Sigma}$ with weights $s_1, \ldots, s_d$, as w is equal to w_β under the étale morphism when $t_1, t_2 > 0$, and by the description of w above when $t_j = 0$. Since log discrepancies do not change under étale base change [Kol13, 2.14(1)], $a(w, V'/\Gamma, 0) = a(w_\beta, V_0/\Gamma, 0)$. Here [Kol13, 2.14(1)] is applied twice: to the étale morphism to V_0/Γ above, and to the étale morphism from V'/Γ to V'/Γ before the replacement of Z' with an étale neighborhood of y at the beginning of this step, along which a log resolution of V'/Γ before the replacement near the center of w pulls back to a log resolution of V'/Γ after it. By (5.3) and (5.1),

$$a(w, X, 0) = t_1 + t_2 - \frac{c}{N} + \sum_{j=1}^{2} (a(P_j, Z, 0) - 1) t_j = a(v, Z, 0) - \frac{c}{N} = a(v, Z, B).$$

Step 6. In this step, we assume that $c = 0$ and show that w is quasi-monomial and $a(w, X, 0) = a(v, Z, B)$.

Since $c = m_1 t_1 + m_2 t_2 = 0$, we have $m_j = 0$ whenever $t_j > 0$. Let ξ' be the generic point of the intersection of the P_j with $t_j > 0$, that is, of the center of v on Z'. Then $\psi^* f$ is a unit at ξ', and v is the monomial valuation at ξ' in the y_j with $t_j > 0$ [JM12, Proposition 3.1]. Since the partial derivatives NU_i^{N-1} of $\psi^* f + \sum_i U_i^N$ vanish simultaneously only on $\{U_1 = \cdots = U_{N-1} = 0\}$, which is disjoint from $V' \cap \{\psi^* f \neq 0\}$, the projection $V' \to Z'$ is smooth over $\{\psi^* f \neq 0\}$. The fiber $V'_{\xi'} = \{\overline{f} + \sum_i U_i^N = 0\}$, where $0 \neq \overline{f} \in \kappa(\xi')$ is the value of $\psi^* f$, is irreducible by the argument of **Step 1**, as $N - 1 \geq 2$. Let η' be the generic point of $V'_{\xi'}$, and let $\widetilde{w}'$ be the monomial valuation at η' in the pullbacks of the y_j with $t_j > 0$, with weights t_j. Then $\widetilde{w}'$ is quasi-monomial, and the rational rank of $\widetilde{w}'$ is equal to that of v. The monomials U^J with $J_{N-1} < N$ form a basis of $\kappa(\xi')[U_1, \ldots, U_{N-1}]/(\overline{f} + \sum_i U_i^N)$ over $\kappa(\xi')$, so for any normal form $\Psi = \sum_J \varphi_J U^J$ with

$\varphi_J \in \mathcal{O}_{Z',\xi'}$, expanding the φ_J in the y_j shows $\widetilde{w}'(\Psi) = \min_J v(\varphi_J)$. Thus $\widetilde{w}'$ satisfies (5.2) with $c = 0$, and $\widetilde{w}'$ corresponds to $\widetilde{w}$ under the birational morphism ψ_V. Hence $\widetilde{w}$ is quasi-monomial, and so is w by **Step 3**.

Any non-identity element of Γ fixes only points with $U_1 = \cdots = U_{N-1} = 0$, which are not on $V' \cap \{\psi^* f \neq 0\}$. Hence q' is étale near η', V'/Γ is smooth over Z' near $q'(\eta')$, and since the y_j are Γ-invariant, w is the monomial valuation at $q'(\eta')$ in the pullbacks of the y_j with $t_j > 0$, with weights t_j. The pullback of P_j to V'/Γ is smooth near $q'(\eta')$, so $a(w, V'/\Gamma, 0) = t_1 + t_2$. By (5.3) and (5.1) with $c = 0$,

$$a(w, X, 0) = t_1 a(P_1, Z, 0) + t_2 a(P_2, Z, 0) = a(v, Z, 0) = a(v, Z, B).$$

Step 7. In this step, we show that $T_{-K_X}(w) = T_L(v)$.

For $Q \in \mathrm{Pic}(Z)$ and $0 \neq a \in \mathcal{S}_Q$, let $\mathrm{div}_Z(a) \geq 0$ be the divisor of a regarded as a section of $\mathcal{O}_Z(Q)$ (Lemma 2.10(3)). Since the slice of **Step 1** maps isomorphically onto W, $a|_W := a(1, 1, x_2, e)$ is a local equation of $\mathrm{div}_Z(a)$ on W. In particular, $a|_W \neq 0$ and $v(\mathrm{div}_Z(a)) = v(a|_W)$. Let j be a positive integer divisible by j_0, so that $jL \in \mathrm{Pic}(Z)$ and $-jK_X$ is Cartier (Construction-Theorem 5.2(4)), and let $0 \neq g \in R_{jL}$. Since $q_W^* D_g = \mathrm{div}(g|_{V_W})$ (**Step 1**) and the center of $\widetilde{w}$ maps to the center of w, we have

$$w(D_g) = \widetilde{w}(g|_{V_W}).$$

We show $T_{-K_X}(w) \leq T_L(v)$. Write $g = \sum_I a_I U^I$ in the normal form of Construction-Theorem 5.2(6). By Construction-Theorem 5.2(6), $a_I \in \mathcal{S}_{jL-|I|\tau}$ for any I. In particular, $a_I = 0$ if $jL - |I|\tau \notin \mathrm{Pic}(Z)$. Since $I_{N-1} < N$ whenever $a_I \neq 0$, $g|_{V_W} = \sum_I a_I|_W U^I$ is the normal form of **Step 3**. Let I be a multi-index such that $a_I \neq 0$, and let k be a positive integer such that $k|I|/N \in \mathbb{Z}$. Since $\tau = [C]/N$,

$$0 \neq a_I^k F_C^{k|I|/N} \in \mathcal{S}_{kjL},$$

and the divisor of the corresponding section of $\mathcal{O}_Z(kjL)$ is $\mathbb{Q}$-linearly equivalent to kjL. Since $F_C|_W = f$ and $v(f) = c$, Definition 2.4 implies

$$kv(a_I|_W) + \frac{k|I|c}{N} = v\big(\mathrm{div}_Z(a_I^k F_C^{k|I|/N})\big) \leq kjT_L(v),$$

that is, $v(a_I|_W) + |I|c/N \leq jT_L(v)$. By (5.2), $w(D_g) = \widetilde{w}(g|_{V_W}) \leq jT_L(v)$. Let $D' \geq 0$ be an $\mathbb{R}$-divisor on X such that $D' \sim_{\mathbb{Q}} -K_X$. Then $jD' \sim -jK_X$ for some positive integer j divisible by j_0, so $jD' = D_g$ for some $0 \neq g \in R_{jL}$ by Construction-Theorem 5.2(4). Hence $w(D') \leq T_L(v)$, and $T_{-K_X}(w) \leq T_L(v)$.

We show $T_{-K_X}(w) \geq T_L(v)$. Let $D \geq 0$ be an $\mathbb{R}$-divisor on Z such that $D \sim_{\mathbb{Q}} L$. Then there exist a positive integer j divisible by j_0 and $0 \neq a \in \mathcal{S}_{jL}$ such that $jD = \mathrm{div}_Z(a)$ (Lemma 2.10(3)). Let $g := a \in \mathcal{S}_{jL} \subset R_{jL}$. Since $g|_{V_W} = a|_W$ is in normal form, (5.2) implies

$$w(D_g) = \widetilde{w}(a|_W) = v(a|_W) = jv(D).$$

Since $D_g/j \sim_{\mathbb{Q}} -K_X$, we have $T_{-K_X}(w) \geq v(D)$, hence $T_{-K_X}(w) \geq T_L(v)$.

By **Steps 3, 5, 6, and 7**, w satisfies the conditions of the proposition. □

6. Prime divisors over the Fano variety

In this section, we prove Proposition 6.7. For any prime divisor P over X, Proposition 6.7 controls $a(P, X, 0)$ and $T_{-K_X}(P)$ by the image of P under ρ.

6.1. **Invariant extensions.**

In this subsection, we compare prime divisors over X with torus-invariant valuations over $\overline{X}$ and $\mathbb{A}^4$.

Lemma 6.1. *Let $f : Y \dashrightarrow T$ be a dominant rational map between normal varieties such that f is a morphism or T is projective. Let P be a prime divisor over Y and v_P the valuation defined by P. Then either the image of P under f is dense in T and $f_* v_P$ is trivial, or there exist a projective birational morphism $T' \to T$ from a smooth variety T' and a prime divisor F on T'*

such that the induced rational map $Y \dashrightarrow T'$ maps P onto F and f_*v_P is k times the valuation defined by F, where k is the multiplicity of the pullback of F along P.

Proof. Let $Y' \to Y$ be a proper birational morphism from a normal variety Y' such that P is a prime divisor on Y'. Let $f' : Y' \dashrightarrow T$ be the induced rational map. If f is a morphism, then so is f'. If T is projective, then f' is defined at the generic point of P by the valuative criterion of properness, since $\mathcal{O}_{Y',P}$ is a discrete valuation ring. Let $\xi \in T$ be the image of the generic point of P under f'. If ξ is the generic point of T, then the image of P under f is dense in T and f_*v_P is trivial, since f'^* maps the non-zero rational functions on T, which are units of $\mathcal{O}_{T,\xi}$, to units of $\mathcal{O}_{Y',P}$. Thus we may assume that ξ is not the generic point of T.

Since f'^* maps $\mathcal{O}_{T,\xi}$ into $\mathcal{O}_{Y',P}$ and maps the maximal ideal of $\mathcal{O}_{T,\xi}$ into the maximal ideal of $\mathcal{O}_{Y',P}$, the center of f_*v_P on T is the closure of ξ. In particular, f_*v_P is non-trivial. The value group of f_*v_P is a non-zero subgroup of the value group $\mathbb{Z}$ of v_P, hence is isomorphic to $\mathbb{Z}$.

Since f is dominant, f^* embeds the function field of T into the function field of Y, and the function field of Y has transcendence degree $\dim Y - \dim T$ over the function field of T. Let κ be the residue field of f_*v_P. The residue field of v_P is the function field of P, which is of transcendence degree $\dim Y - 1$ over $\mathbb{C}$. Applying the relative Abhyankar inequality [Bou72, Chapter VI, §10, no. 3, Corollary 1] to v_P over f_*v_P, and to f_*v_P, whose value group is of rational rank 1, over the trivial valuation of $\mathbb{C}$, we obtain

$$(\dim Y - 1) - \operatorname{trdeg}_{\mathbb{C}} \kappa \leq \dim Y - \dim T \quad \text{and} \quad \operatorname{trdeg}_{\mathbb{C}} \kappa + 1 \leq \dim T.$$

Thus $\operatorname{trdeg}_{\mathbb{C}} \kappa = \dim T - 1$.

By [KM98, Lemma 2.45], there exist a variety T_1, a morphism $T_1 \to T$ which is a composite of finitely many blow-ups, and a prime divisor F_1 on T_1 such that $\mathcal{O}_{T_1,F_1}$ is the valuation ring of f_*v_P. In particular, T_1 is smooth at the generic point of F_1. Let $T' \to T_1$ be a projective birational morphism from a smooth variety T' which is an isomorphism over the smooth locus of T_1 [KM98, Theorem 0.2(1)], and let F be the strict transform of F_1 on T'. Then $\mathcal{O}_{T',F}$ is the valuation ring of f_*v_P. Let φ be a local equation of F at the generic point of F. Since $\mathcal{O}_{T',F}$ is a discrete valuation ring with uniformizer φ, f_*v_P is $(f_*v_P)(\varphi)$ times the valuation defined by F, and $\operatorname{center}_{T'}(f_*v_P) = F$. Since $T' \to T$ is proper, the induced rational map $f'' : Y' \dashrightarrow T'$ is defined at the generic point of P by the valuative criterion of properness. As in the second paragraph, f'' maps the generic point of P to the generic point of $\operatorname{center}_{T'}(f_*v_P) = F$. Thus the induced rational map $Y \dashrightarrow T'$ maps P onto F, and $(f_*v_P)(\varphi) = v_P(f''^*\varphi)$ is the multiplicity k of the pullback of F along P. □

Definition 6.2. Let Y be a normal variety with an action of a torus $\mathbb{G}$. For any $t \in \mathbb{G}$ and any non-zero rational function ϕ on Y, let $t^*\phi$ be the rational function on Y defined by $(t^*\phi)(y) := \phi(t \cdot y)$. A valuation w over Y is $\mathbb{G}$-*invariant* if $w(t^*\phi) = w(\phi)$ for any $t \in \mathbb{G}$ and any non-zero rational function ϕ on Y.

Lemma 6.3. *Let $f : Y_1 \to Y_2$ be a finite surjective morphism between normal quasi-projective varieties and Δ_2 a $\mathbb{Q}$-divisor on Y_2 such that $K_{Y_2} + \Delta_2$ is $\mathbb{Q}$-Cartier. Let Δ_1 be the $\mathbb{Q}$-divisor on Y_1 such that $K_{Y_1} + \Delta_1 = f^*(K_{Y_2} + \Delta_2)$. Let w be a positive real multiple of the valuation defined by a prime divisor over Y_1. Then f_*w is a positive real multiple of the valuation defined by a prime divisor over Y_2, and $a(f_*w, Y_2, \Delta_2) = a(w, Y_1, \Delta_1)$.*

Proof. Write $w = cv_{G_1}$, where $c > 0$ and v_{G_1} is the valuation defined by a prime divisor G_1 over Y_1. Since f is finite, the image of G_1 under f is not dense in Y_2, so by Lemma 6.1, $f_*v_{G_1} = kv_{G_2}$, where v_{G_2} is the valuation defined by a prime divisor G_2 over Y_2 and k is the multiplicity of the pullback of G_2 along G_1, that is, the ramification index of f along G_1. In characteristic zero the ramification is tame, so, with $K_{Y_1} + \Delta_1 = f^*(K_{Y_2} + \Delta_2)$ as in [Kol13, 2.41], [Kol13, (2.42.2)] gives $a(G_1, Y_1, \Delta_1) = k \cdot a(G_2, Y_2, \Delta_2)$ (in [Kol13], a denotes the discrepancy, which is the log discrepancy minus 1). Here [Kol13, 2.42] is applied on the normalization, in the function field of Y_1, of a smooth variety which is projective and birational over Y_2 and on which G_2 is a prime

divisor (Lemma 6.1). Since the valuation ring of v_{G_1} contains the local ring of that smooth variety at G_2, G_1 is a prime divisor on the normalization over G_2 with ramification index k. Hence

$$a(w, Y_1, \Delta_1) = ck \cdot a(G_2, Y_2, \Delta_2) = a(f_* w, Y_2, \Delta_2).$$

□

For any positive integer j divisible by j_0, the sections of $\mathcal{O}_X(-jK_X)$ are the elements of R_{jL} (Construction-Theorem 5.2(4)). To estimate the orders of the elements of R_{jL} along a prime divisor P over X through their normal forms (Construction-Theorem 5.2(6)), we replace the valuation defined by P with a $\mathbb{T}$-invariant valuation over $\overline{X}$.

Lemma 6.4. *Notation as in Construction-Theorem 5.2. Let P be a prime divisor over X. Then there exists a $\mathbb{T}$-invariant valuation $\widehat{w}$ over $\overline{X}$ satisfying the following.*

(1) $\widehat{w}$ is a divisorial valuation with center on $\widehat{X}$.
(2) $h_*\widehat{w}$ is the valuation defined by P, and $a(\widehat{w}, \overline{X}, 0) = a(P, X, 0)$.
(3) $\widehat{w}(g) = \operatorname{mult}_P D_g$ for any positive integer j divisible by j_0 and any $0 \neq g \in R_{jL}$, where D_g is as in Construction-Theorem 5.2(4).

Proof. For any $\chi \in \Lambda$ and $t \in \mathbb{T} = \operatorname{Hom}(\Lambda, \mathbb{C}^*)$, we write $\chi(t) := t(\chi)$. Thus $g(t \cdot x) = \chi(t)g(x)$ for any $g \in R_\chi$ and any $x \in \overline{X}$. Let v_P be the valuation defined by P. Since $\overline{X}$ is a hypersurface in $\mathbb{A}^{N+3}$, $K_{\overline{X}}$ is Cartier. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct a quasi-étale morphism $q : V \to X^\circ$ onto an open subset $X^\circ \subset X$ which contains the generic point of $\operatorname{center}_X P$, together with a finite étale morphism $\mathbb{T} \times V \to h^{-1}(X^\circ)$.

Pick a closed point $\bar{x} \in \widehat{X}$ such that $h(\bar{x}) \in \operatorname{center}_X P$, and let O be the closure of $\mathbb{T} \cdot \bar{x}$ in $\overline{X}$ with the reduced structure. The ideal of O is $\mathbb{T}$-stable, so the coordinate ring of O is graded by Λ. Let $\Lambda_O \subset \Lambda$ be the subgroup generated by the degrees of the non-zero homogeneous elements of the coordinate ring of O. If $\operatorname{rank} \Lambda_O \leq 1$, then the subgroup

$$\{t \in \mathbb{T} \mid \chi(t) = 1 \text{ for any } \chi \in \Lambda_O\}$$

of $\mathbb{T}$ is of dimension at least 1 and acts trivially on O, which contradicts the finiteness of the stabilizer of $\bar{x}$ (Construction-Theorem 5.2(2)). Thus there exist linearly independent $\chi_1, \chi_2 \in \Lambda$ and elements $y_1 \in R_{\chi_1}$, $y_2 \in R_{\chi_2}$ whose restrictions to O are non-zero. Since $\mathbb{T} \cdot \bar{x}$ is dense in O and $y_i(t \cdot \bar{x}) = \chi_i(t) y_i(\bar{x})$, we have $y_1(\bar{x}) y_2(\bar{x}) \neq 0$.

Let $\overline{X}_{y_1 y_2} := \{y_1 y_2 \neq 0\} \subset \overline{X}$, $V_0 := \{y_1 = y_2 = 1\} \subset \overline{X}$, and

$$\Gamma_0 := \{t \in \mathbb{T} \mid \chi_1(t) = \chi_2(t) = 1\}.$$

Since χ_1, χ_2 are linearly independent, the homomorphism $\mathbb{T} \to (\mathbb{C}^*)^2$ given by (χ_1, χ_2) is surjective with finite kernel Γ_0, hence is finite étale. Let $\overline{X}_{y_1 y_2} \to (\mathbb{C}^*)^2$ be the morphism given by (y_1, y_2). The morphism

$$\mathbb{T} \times V_0 \to \mathbb{T} \times_{(\mathbb{C}^*)^2} \overline{X}_{y_1 y_2}, \quad (t, v) \to (t, t \cdot v)$$

is an isomorphism, with inverse $(t, x) \to (t, t^{-1} \cdot x)$. Therefore, the morphism

$$\mu_0 : \mathbb{T} \times V_0 \to \overline{X}_{y_1 y_2}, \quad (t, v) \to t \cdot v$$

is finite étale, and the fibers of μ_0 are the orbits of the action of Γ_0 on $\mathbb{T} \times V_0$ given by $\zeta \cdot (t, v) := (t\zeta, \zeta^{-1} \cdot v)$. Since $\overline{X}_{y_1 y_2}$ is normal (Construction-Theorem 5.2(1)), $\mathbb{T} \times V_0$ is normal, and so is V_0. Hence the connected components of V_0 are irreducible. Let V_1 be a connected component of V_0, $\Gamma_1 \subset \Gamma_0$ the stabilizer of V_1, and $\mu := \mu_0|_{\mathbb{T} \times V_1}$. Then μ is finite étale, so the image of μ is open and closed in the irreducible variety $\overline{X}_{y_1 y_2}$. Thus μ is surjective. If $\mu(t, v) = \mu(t', v')$ for $(t, v), (t', v') \in \mathbb{T} \times V_1$, then $(t', v') = \zeta \cdot (t, v)$ for some $\zeta \in \Gamma_0$, so $(\zeta^{-1} \cdot V_1) \cap V_1 \neq \emptyset$. Since Γ_0 permutes the connected components of V_0, $\zeta \in \Gamma_1$. Thus $\mu : \mathbb{T} \times V_1 \to \overline{X}_{y_1 y_2}$ is the quotient by Γ_1 (the induced morphism from $(\mathbb{T} \times V_1)/\Gamma_1$ to $\overline{X}_{y_1 y_2}$ is finite, étale and bijective, hence an isomorphism).

Let $W := \overline{X}_{y_1y_2} \cap \widehat{X}$ and $V := V_1 \cap \widehat{X}$. Since $\widehat{X}$ is $\mathbb{T}$-invariant, $\mu^{-1}(W) = \mathbb{T} \times V$, and the restriction of μ to $\mathbb{T} \times V$ is the quotient $\mathbb{T} \times V \to W$ by Γ_1. Since h is a geometric quotient by $\mathbb{T}$ (Construction-Theorem 5.2(2)) and W is $\mathbb{T}$-invariant, $X^\circ := h(W)$ is an open subset of X and $W = h^{-1}(X^\circ)$. Since $\bar{x} \in W$, X° contains $h(\bar{x}) \in \mathrm{center}_X P$, hence contains the generic point of $\mathrm{center}_X P$. Let $q : V \to X^\circ$ be the restriction of h. Then q is Γ_1-invariant and $h \circ \mu(t, v) = q(v)$ for any $(t, v) \in \mathbb{T} \times V$. For any open subset $X_1 \subset X^\circ$, the regular functions on X_1 are the $\mathbb{T}$-invariant regular functions on $h^{-1}(X_1)$, that is, the regular functions on $\mu^{-1}(h^{-1}(X_1)) = \mathbb{T} \times q^{-1}(X_1)$ which are invariant under $\mathbb{T}$ and Γ_1, that is, the Γ_1-invariant regular functions on $q^{-1}(X_1)$. The morphism q is the composite of the closed embedding $V \to \mathbb{T} \times V$, $v \to (1, v)$, the finite morphism $\mu|_{\mathbb{T}\times V}$, and the affine morphism $h|_W$ (for any $g \in R_{jL}$, the preimage under h of the affine open subset of X defined by g is $\{g \neq 0\} \subset \overline{X}$). Hence q is affine, and q induces an isomorphism $V/\Gamma_1 \to X^\circ$. Since Γ_1 is finite, for any affine open subset $X_1 \subset X^\circ$, the coordinate ring of $q^{-1}(X_1)$ is integral over its Γ_1-invariant subring, which is the coordinate ring of X_1, so q is finite.

We show that q is quasi-étale. Let $V_s \subset V$ and $W_s \subset W$ be the subsets of points with non-trivial stabilizer in $\mathbb{T}$. Since $\mathbb{T}$ is commutative, the stabilizers of $t \cdot v$ and v in $\mathbb{T}$ are equal, so $\mu^{-1}(W_s) = \mathbb{T} \times V_s$. By Construction-Theorem 5.2(2), W_s is of codimension at least 2 in W. Since μ is étale, $\mathbb{T} \times V_s$ is of codimension at least 2 in $\mathbb{T} \times V$, so V_s is of codimension at least 2 in V. If $\zeta \cdot v = v$ for some $v \in V$ and $1 \neq \zeta \in \Gamma_1$, then $v \in V_s$. Thus Γ_1 acts freely on $V \backslash V_s$, and q is étale over $q(V \backslash V_s)$. Hence q is quasi-étale.

Step 2. In this step, we construct $\widehat{w}$ and prove (1) and the equality $a(\widehat{w}, \overline{X}, 0) = a(P, X, 0)$.

Let $X' \to X^\circ$ be a proper birational morphism from a normal variety X' such that P is a prime divisor on X', and let V' be the normalization of X' in the function field of V. Since V is normal and finite over X°, the morphism $V' \to X^\circ$ factors through a proper birational morphism $V' \to V$. Since $V' \to X'$ is finite and surjective, there exists a prime divisor P' on V' which is mapped onto P. Let k be the multiplicity of the pullback of P along P', and $w' := k^{-1} v_{P'}$, where $v_{P'}$ is the valuation defined by P'. Then $q_* w' = v_P$. Since q is quasi-étale, $K_V = q^* K_{X^\circ}$ [Kol13, 2.41], so Lemma 6.3 applied to q with $\Delta_1 = \Delta_2 = 0$ gives

$$a(w', V, 0) = a(P, X^\circ, 0) = a(P, X, 0).$$

Choose a basis u_1, u_2 of Λ, and regard u_1, u_2 as coordinates on $\mathbb{T}$ via $u_i(t) := t(u_i)$. Then $\mathbb{T} = \{u_1 u_2 \neq 0\} \subset \mathbb{A}^2$, and $\mathbb{T} \times V$ is an open subset of $\mathbb{A}^2 \times V$. Let w'' be the Gauss extension of w' with weights $0, 0$ (Definition 2.6, with the factors of $V \times \mathbb{A}^2$ written in the order $\mathbb{A}^2 \times V$). The restrictions of u_1, u_2 to $\mathbb{A}^2 \times P'$ are algebraically independent over the function field of P', so $w'' = k^{-1} v_{\mathbb{A}^2 \times P'}$, where $v_{\mathbb{A}^2 \times P'}$ is the valuation defined by the prime divisor $\mathbb{A}^2 \times P'$ on $\mathbb{A}^2 \times V'$. The center of w'' on $\mathbb{A}^2 \times V$ is $\mathbb{A}^2 \times \mathrm{center}_V w'$, whose intersection with $\mathbb{T} \times V$ is non-empty. Thus w'' is a divisorial valuation over $\mathbb{T} \times V$. Since $\mathbb{A}^2 \times V' \to \mathbb{A}^2 \times V$ is the product of the identity of $\mathbb{A}^2$ and $V' \to V$,

$$a(w'', \mathbb{T} \times V, 0) = a(w', V, 0) = a(P, X, 0).$$

As a valuation over $\mathbb{T} \times V_1$, $w'' = k^{-1} v_{\mathbb{T}\times P'}$, where $v_{\mathbb{T}\times P'}$ is the valuation defined by the prime divisor $\mathbb{T} \times P'$ on $\mathbb{T} \times V'$. Let $\widehat{w} := \mu_* w''$, a valuation over $\overline{X}$. Since μ is finite, the image of $\mathbb{T} \times P'$ under μ is not dense in $\overline{X}$, so by Lemma 6.1, μ maps $\mathbb{T} \times P'$ onto a prime divisor $\widehat{P}$ on a smooth variety $\widehat{Y}$ with a projective birational morphism $\widehat{Y} \to \overline{X}$, and $\mu_* v_{\mathbb{T}\times P'} = m v_{\widehat{P}}$, where $v_{\widehat{P}}$ is the valuation defined by $\widehat{P}$ and m is the multiplicity of the pullback of $\widehat{P}$ along $\mathbb{T} \times P'$. Thus $\widehat{w} = (m/k) v_{\widehat{P}}$ is a divisorial valuation. The center of $\widehat{w}$ on $\overline{X}$ is the closure of the image of the center of w'' on $\mathbb{T} \times V$, and the generic point of $\mathrm{center}_{\overline{X}}(\widehat{w})$ belongs to $W \subset \widehat{X}$. This implies (1). Since the restriction $\mathbb{T} \times V \to W$ of μ is finite étale, $K_{\mathbb{T}\times V}$ is the pullback of K_W, and K_W is Cartier since $K_{\overline{X}}$ is. Hence Lemma 6.3 applied to the finite étale morphism $\mathbb{T} \times V \to W$ with $\Delta_1 = \Delta_2 = 0$ implies

$$a(\widehat{w}, \overline{X}, 0) = a(\widehat{w}, W, 0) = a(w'', \mathbb{T} \times V, 0) = a(P, X, 0).$$

Step 3. In this step, we prove that $\widehat{w}$ is $\mathbb{T}$-invariant, $h_*\widehat{w} = v_P$, and (3).

For any $s \in \mathbb{T}$, let s act on $\mathbb{T} \times V_1$ by $(t, v) \to (st, v)$. Then $\mu(st, v) = s \cdot \mu(t, v)$, so $\mu^*(s^*\phi) = s^*(\mu^*\phi)$ for any non-zero rational function ϕ on $\overline{X}$. For any non-zero polynomial $\psi = \sum_\beta a_\beta u_1^{\beta_1} u_2^{\beta_2}$ whose coefficients a_β are rational functions on V_1,

$$s^*\psi = \sum_\beta u_1(s)^{\beta_1} u_2(s)^{\beta_2} a_\beta u_1^{\beta_1} u_2^{\beta_2},$$

and $u_1(s)^{\beta_1} u_2(s)^{\beta_2} \in \mathbb{C}^*$. Hence $w''(s^*\psi) = w''(\psi)$ by Definition 2.6. Since any non-zero rational function on $\mathbb{T} \times V_1$ is a quotient of two such polynomials, w'' is $\mathbb{T}$-invariant. Thus

$$\widehat{w}(s^*\phi) = w''(s^*(\mu^*\phi)) = w''(\mu^*\phi) = \widehat{w}(\phi)$$

for any $s \in \mathbb{T}$ and any non-zero rational function ϕ on $\overline{X}$, so $\widehat{w}$ is $\mathbb{T}$-invariant.

Let ϕ be a non-zero rational function on X. Since $h \circ \mu(t, v) = q(v)$ for $(t, v) \in \mathbb{T} \times V$, the rational function $\mu^*h^*\phi$ on $\mathbb{T} \times V$ is the pullback of $q^*\phi$ under the projection $\mathbb{T} \times V \to V$, that is, a polynomial of degree 0 in u_1, u_2 with coefficient $q^*\phi$. By Definition 2.6,

$$(h_*\widehat{w})(\phi) = w''(\mu^*h^*\phi) = w'(q^*\phi) = (q_*w')(\phi) = v_P(\phi).$$

Thus $h_*\widehat{w} = v_P$. Together with **Step 2**, this implies (2).

Let j be a positive integer divisible by j_0 and $0 \neq g \in R_{jL}$. Let ξ be the generic point of $\operatorname{center}_X P$ and $\hat{\xi}$ the generic point of $\operatorname{center}_{\widehat{X}}(\widehat{w})$. Since $h_*\widehat{w} = v_P$, $h(\hat{\xi}) = \xi$. By Construction-Theorem 5.2(4), $-jK_X$ is Cartier, so there exists a local equation ϕ of D_g on an open neighborhood X_2 of ξ. Since $h^*D_g = \operatorname{div}(g|_{\widehat{X}})$ (Construction-Theorem 5.2(4)), $\operatorname{div}(g/h^*\phi) = 0$ on $h^{-1}(X_2)$. Since $\widehat{X}$ is normal, $g/h^*\phi$ is a unit on $h^{-1}(X_2)$, which contains $\hat{\xi}$. Therefore,

$$\widehat{w}(g) = \widehat{w}(h^*\phi) = v_P(\phi) = \operatorname{mult}_P D_g.$$

This implies (3). □

In Subsection 6.3, we apply the next lemma to the pushforward of $\widehat{w}$ under $\overline{X} \to \mathbb{A}^4$. The number $\lambda(L)$ enters the estimate there.

Lemma 6.5. *Notation as in Construction-Theorem 5.2, and let $\mathbb{T}_0 = \operatorname{Hom}(\operatorname{Pic}(Z), \mathbb{C}^*)$ act on $\mathbb{A}^4 = \operatorname{Spec} \mathcal{S}$ via the $\operatorname{Pic}(Z)$-grading of $\mathcal{S}$. Let $\bar{v}$ be a $\mathbb{T}_0$-invariant valuation over $\mathbb{A}^4$, and let v be the pushforward of $\bar{v}$ under the rational map $\mathbb{A}^4 \dashrightarrow Z$ given by the geometric quotient of Lemma 2.10(3). Then the following hold.*

1. *There exists a unique additive function $\lambda : \operatorname{Pic}(Z)_\mathbb{Q} \to \mathbb{R}$ such that $\bar{v}(g) = v(\operatorname{div}_Z(g)) + \lambda(Q)$ for any $Q \in \operatorname{Pic}(Z)$ and any $0 \neq g \in \mathcal{S}_Q$, where $\operatorname{div}_Z(g)$ is the divisor of g regarded as a section of $\mathcal{O}_Z(Q)$.*
2. *If $\bar{v}$ is a divisorial valuation, then v is either trivial or divisorial, and*

$$a(\bar{v}, \mathbb{A}^4, 0) = a(v, Z, 0) + \lambda(-K_Z),$$

where $a(v, Z, 0) := 0$ if v is trivial.

Proof. For any $Q \in \operatorname{Pic}(Z)$, $s \in \mathbb{T}_0 = \operatorname{Hom}(\operatorname{Pic}(Z), \mathbb{C}^*)$, and $g \in \mathcal{S}_Q$, we have $g(s \cdot y) = s(Q)g(y)$ for any $y \in \mathbb{A}^4$. Let $f : \mathbb{A}^4 \dashrightarrow Z$ be the rational map given by the geometric quotient.

We first show the following. Let $\phi_1, \dots, \phi_m$ be rational functions on $\mathbb{A}^4$ and $Q_1, \dots, Q_m \in \operatorname{Pic}(Z)$ pairwise distinct classes such that $\phi_i(s \cdot y) = s(Q_i)\phi_i(y)$ for any i and any $s \in \mathbb{T}_0$. Then

$$\bar{v}\Big(\sum_{i=1}^m \phi_i\Big) = \min\{\bar{v}(\phi_i) \mid \phi_i \neq 0\} \tag{6.1}$$

whenever $\sum_i \phi_i \neq 0$. Write $\phi := \sum_i \phi_i$. The inequality $\geq$ holds for any valuation. Since $Q_1, \dots, Q_m$ are pairwise distinct, the characters $s \to s(Q_i)$ of $\mathbb{T}_0$ are pairwise distinct. By the linear independence of characters, there exist $s_1, \dots, s_m \in \mathbb{T}_0$ such that the matrix $(s_l(Q_i))_{i,l}$ is invertible. Since $s_l^*\phi = \sum_i s_l(Q_i)\phi_i$ for any l, each ϕ_i is a $\mathbb{C}$-linear combination of $s_1^*\phi, \dots, s_m^*\phi$.

Since $\bar{v}$ is $\mathbb{T}_0$-invariant, $\bar{v}(s_l^*\phi) = \bar{v}(\phi)$ for any l, so $\bar{v}(\phi_i) \geq \bar{v}(\phi)$ for any i such that $\phi_i \neq 0$. This implies (6.1).

(1) Let $Q \in \mathrm{Pic}(Z)$ and $0 \neq g, g' \in \mathcal{S}_Q$. Since $\mathcal{S}_Q = H^0(Z, \mathcal{O}_Z(Q))$ (Lemma 2.10(3)), g/g' is the pullback under f of a rational function on Z whose divisor is $\mathrm{div}_Z(g) - \mathrm{div}_Z(g')$. Hence

$$\bar{v}(g) - \bar{v}(g') = v(\mathrm{div}_Z(g)) - v(\mathrm{div}_Z(g')).$$

Thus $\lambda_0(Q) := \bar{v}(g) - v(\mathrm{div}_Z(g))$ does not depend on the choice of $0 \neq g \in \mathcal{S}_Q$. Let M be the set of $Q \in \mathrm{Pic}(Z)$ with $\mathcal{S}_Q \neq 0$. Since

$$\mathrm{div}_Z(gg') = \mathrm{div}_Z(g) + \mathrm{div}_Z(g')$$

for $g \in \mathcal{S}_Q$ and $g' \in \mathcal{S}_{Q'}$, M is closed under addition and $\lambda_0(Q+Q') = \lambda_0(Q) + \lambda_0(Q')$ for any $Q, Q' \in M$. Since M contains $H = \deg x_0$ and $E = \deg e$, which form a basis of $\mathrm{Pic}(Z)$ (Lemma 2.10(1)), M generates $\mathrm{Pic}(Z)$. Thus λ_0 extends uniquely to an additive function on $\mathrm{Pic}(Z)$, and then uniquely to an additive function λ on $\mathrm{Pic}(Z)_{\mathbb{Q}}$. The function λ satisfies (1), and any additive function satisfying (1) is equal to λ_0 on M, hence to λ.

(2) Write $\bar{v} = cv_G$ for a real number $c > 0$, where v_G is the valuation defined by a prime divisor G over $\mathbb{A}^4$. By Lemma 6.1 applied to f and G, either v is trivial, or $v = ckv_F$, where v_F is the valuation defined by a prime divisor F on a smooth variety Z' with a projective birational morphism $Z' \to Z$ and k is the multiplicity of the pullback of F along G. In particular, v is divisorial in the second case. Let $c_v := 0$ if v is trivial and $c_v := ck$ if v is non-trivial.

Let

$$z := \frac{ex_1}{x_0} \quad \text{and} \quad t := \frac{x_2}{x_1}.$$

The rational functions z and t on $\mathbb{A}^4$ are $\mathbb{T}_0$-invariant, and they are the pullbacks under f of coordinates of the open subset $\{x_0x_1 \neq 0\} \cong \mathbb{A}^2$ of Z, which we also denote by z and t. For $i = 0, 1, 2$, let $D_i := \mathrm{div}_Z(x_i)$. The open subsets $\{x_0x_2 \neq 0\}$ and $\{x_1e \neq 0\}$ of Z are isomorphic to $\mathbb{A}^2$ with coordinates $(z', t') := (ex_2/x_0, x_1/x_2)$ and $(z'', t'') := (x_0/(x_1e), x_2/x_1)$ respectively, and

$$dz \wedge dt = -\frac{1}{t'} dz' \wedge dt' = -\frac{1}{z''^2} dz'' \wedge dt'',$$

since $z = z't'$, $t = 1/t'$, $z = 1/z''$, and $t = t''$. The complement of $\{x_0x_1 \neq 0\}$ in Z is $D_0 \cup D_1$, and $D_1 = \{t' = 0\}$ on $\{x_0x_2 \neq 0\}$ and $D_0 = \{z'' = 0\}$ on $\{x_1e \neq 0\}$. Hence

$$\mathrm{div}_Z(dz \wedge dt) = -2D_0 - D_1 = -\mathrm{div}_Z(x_0^2x_1), \tag{6.2}$$

where $x_0^2x_1 \in \mathcal{S}_{3H-E}$ is regarded as a section of $\mathcal{O}_Z(3H-E) = \mathcal{O}_Z(-K_Z)$. On $\{x_0x_1 \neq 0\} \subset \mathbb{A}^4$, we have $x_2 = tx_1$ and $e = zx_0/x_1$, so $dx_2 \equiv x_1 dt$ and $de \equiv (x_0/x_1)dz$ modulo dx_0, dx_1. A direct calculation shows

$$dx_0 \wedge dx_1 \wedge dx_2 \wedge de = -x_0^2x_1 dz \wedge dt \wedge \frac{dx_0}{x_0} \wedge \frac{dx_1}{x_1}. \tag{6.3}$$

Let $b_0 := \bar{v}(x_0)$ and $b_1 := \bar{v}(x_1)$. Since x_0, x_1 are regular on $\mathbb{A}^4$ and $\bar{v}$ is a valuation over $\mathbb{A}^4$, $b_0, b_1 \geq 0$. By (1), $b_0 = v(D_0) + \lambda(H)$ and $b_1 = v(D_1) + \lambda(H-E)$. Let u_1, u_2 be the coordinates of $\mathbb{A}^2$ and v' the Gauss extension of v with weights b_0, b_1 (Definition 2.6). Let

$$\Phi := (f, x_0, x_1) : \mathbb{A}^4 \dashrightarrow Z \times \mathbb{A}^2.$$

Since $x_0 = u_1$, $x_1 = u_2$, $x_2 = tu_2$, and $e = zu_1/u_2$ on $\{x_0x_1 \neq 0\}$, Φ is birational. We show that $\bar{v}(\Phi^*\psi) = v'(\psi)$ for any non-zero rational function ψ on $Z \times \mathbb{A}^2$. It suffices to consider $\psi = \sum_\beta a_\beta u_1^{\beta_1} u_2^{\beta_2}$, whose coefficients a_β are rational functions on Z. Then $\Phi^*\psi = \sum_\beta f^*a_\beta x_0^{\beta_1} x_1^{\beta_2}$, and the term $f^*a_\beta x_0^{\beta_1} x_1^{\beta_2}$ satisfies the condition of (6.1) with the class $\beta_1 H + \beta_2(H-E)$. The classes $\beta_1 H + \beta_2(H-E)$ are pairwise distinct for distinct β, since H and $H-E$ form a basis of $\mathrm{Pic}(Z)$. By (6.1),

$$\bar{v}(\Phi^*\psi) = \min\{v(a_\beta) + b_0\beta_1 + b_1\beta_2 \mid a_\beta \neq 0\} = v'(\psi).$$

In particular, v' is a divisorial valuation over $Z \times \mathbb{A}^2$.

For any normal variety Y birational to $\mathbb{A}^4$, any divisorial valuation $w = c'v_{G'}$ over Y, where G' is a prime divisor on a smooth variety Y' birational to Y, and any non-zero rational 4-form ω, we let $w(\omega) := c' \operatorname{mult}_{G'} \operatorname{div}_{Y'}(\omega)$. It depends only on w and ω, since $\mathcal{O}_{Y',G'}$ is the valuation ring of w. If (Y, Δ) is a pair, then choosing $K_Y = \operatorname{div}_Y(\omega)$ and computing on a log resolution of (Y, Δ) on which G' is a prime divisor, the definition of log discrepancies implies

$$a(w, Y, \Delta) = c' + w(\omega) - w(\operatorname{div}_Y(\omega) + \Delta). \tag{6.4}$$

Let $\omega_4 := dx_0 \wedge dx_1 \wedge dx_2 \wedge de$, so that $\operatorname{div}_{\mathbb{A}^4}(\omega_4) = 0$. Let $D := Z \times \{u_1u_2 = 0\}$ and

$$\theta := dz \wedge dt \wedge \frac{du_1}{u_1} \wedge \frac{du_2}{u_2}.$$

By (6.2),

$$\operatorname{div}_{Z\times\mathbb{A}^2}(\theta) + D = -\operatorname{pr}_Z^* \operatorname{div}_Z(x_0^2x_1),$$

where $\operatorname{pr}_Z : Z \times \mathbb{A}^2 \to Z$ is the projection. By (6.3), $\omega_4 = -\Phi^*(u_1^2u_2\theta)$. Since Φ is birational and $\bar{v}(\Phi^*\psi) = v'(\psi)$ for any non-zero rational function ψ on $Z \times \mathbb{A}^2$, $\bar{v}(\omega_4) = \bar{v}(x_0^2x_1) + v'(\theta)$. Since $v'(\operatorname{pr}_Z^* \phi) = v(\phi)$ for any non-zero rational function ϕ on Z,

$$v'(\operatorname{pr}_Z^* \operatorname{div}_Z(x_0^2x_1)) = v(\operatorname{div}_Z(x_0^2x_1)).$$

By (6.4) for $(\mathbb{A}^4, 0)$ with ω_4 and for $(Z \times \mathbb{A}^2, D)$ with θ,

$$a(\bar{v}, \mathbb{A}^4, 0) = c + \bar{v}(x_0^2x_1) + v'(\theta) \quad \text{and} \quad a(v', Z \times \mathbb{A}^2, D) = c + v'(\theta) + v(\operatorname{div}_Z(x_0^2x_1)). \tag{6.5}$$

By (1),

$$\bar{v}(x_0^2x_1) - v(\operatorname{div}_Z(x_0^2x_1)) = \lambda(3H - E) = \lambda(-K_Z).$$

Thus

$$a(\bar{v}, \mathbb{A}^4, 0) = a(v', Z \times \mathbb{A}^2, D) + \lambda(-K_Z).$$

It remains to show that $a(v', Z \times \mathbb{A}^2, D) = a(v, Z, 0)$. If v is trivial, let $g_Z : Z' \to Z$ be the identity and $\xi \in Z$ any closed point. If v is non-trivial, possibly replacing Z' with a log resolution of $(Z', \operatorname{Exc}(Z' \to Z))$ which is an isomorphism near the generic point of F, by [Har77, Theorem V.3.9], which only blows up closed points, we may assume that the morphism $g_Z : Z' \to Z$ is a log resolution of Z, and let $\xi \in F$ be a closed point which is not contained in any other g_Z-exceptional prime divisor. Let σ_1, σ_2 be a regular system of parameters of $\mathcal{O}_{Z',\xi}$, where σ_1 is a local equation of F if v is non-trivial, and let $\xi' := (\xi, 0) \in Z' \times \mathbb{A}^2$. For any $a_\beta \in \mathcal{O}_{Z',\xi}$, $v(a_\beta)$ is the value at a_β of the quasi-monomial valuation at ξ with weights $c_v, 0$ in σ_1, σ_2 [JM12, Proposition 3.1]. Hence, if v'' is the quasi-monomial valuation at ξ' with weights $c_v, 0, b_0, b_1$ in $\sigma_1, \sigma_2, u_1, u_2$, then $v'(\psi) = v''(\psi)$ for any polynomial $\psi = \sum_\beta a_\beta u_1^{\beta_1} u_2^{\beta_2}$ with $a_\beta \in \mathcal{O}_{Z',\xi}$, so $v' = v''$. The morphism $g_Z \times \mathrm{id}_{\mathbb{A}^2}$ is a log resolution of $(Z \times \mathbb{A}^2, D)$, and near ξ', the union of the strict transform of D and the exceptional locus is contained in $\{\sigma_1u_1u_2 = 0\}$. The prime divisors $Z' \times \{u_1 = 0\}$ and $Z' \times \{u_2 = 0\}$ have log discrepancy 0 with respect to $(Z \times \mathbb{A}^2, D)$, and if v is non-trivial, then $a(F \times \mathbb{A}^2, Z \times \mathbb{A}^2, D) = a(F, Z, 0)$. By Definition 2.2,

$$a(v', Z \times \mathbb{A}^2, D) = c_v a(F, Z, 0) = a(v, Z, 0)$$

if v is non-trivial, and $a(v', Z \times \mathbb{A}^2, D) = 0 = a(v, Z, 0)$ if v is trivial. This implies (2). □

6.2. **The Fermat hypersurface.** In this subsection, we bound the values of a divisorial valuation over $\overline{X}$ on elements of R in terms of their normal forms in Construction-Theorem 5.2(6). The method, a finite base change after which a finite cover of $\overline{X}$ is birational to a product with a Fermat hypersurface, on which we use the bound of [Che01, Theorem 1.3], is explained in part (d) of the sketch of the proof in Section 1.

Lemma 6.6. *Notation as in Construction-Theorem 5.2. Let $\widehat{w}$ be a divisorial valuation over $\overline{X}$, let $\bar{v}$ be the pushforward of $\widehat{w}$ under the morphism $\overline{X} \to \mathbb{A}^4 = \operatorname{Spec} \mathcal{S}$ induced by $\mathcal{S} \subset R$, and let $c := \bar{v}(F_C)$. Then there exist real numbers $\eta \geq \zeta \geq 0$ satisfying the following.*

(1) We have

$$a(\widehat{w}, \overline{X}, 0) = a(\bar{v}, \mathbb{A}^4, 0) - \frac{c}{N} + \eta,$$

where $a(\bar{v}, \mathbb{A}^4, 0) := 0$ *if* $\bar{v}$ *is trivial.*

(2) Let $0 \neq g \in R$*, let* a_I *be the elements of Construction-Theorem 5.2(6) for* g*, and let* $b(g) := \min\{\bar{v}(a_I) + |I|c/N \mid a_I \neq 0\}$*. Then*

$$b(g) - \deg_U(g)\zeta \leq \widehat{w}(g) \leq b(g) + \deg_U(g)(\eta/\delta_N - \zeta).$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we assume that $\bar{v}$ is non-trivial, and we construct a finite base change along the center of $\bar{v}$ together with an extension of $\widehat{w}$.

Let $\widehat{P}$ be a prime divisor over $\overline{X}$ such that $\widehat{w}$ is a positive multiple of the valuation defined by $\widehat{P}$. Since $\bar{v}$ is non-trivial, the image of $\widehat{P}$ in $\mathbb{A}^4$ is not dense. By Lemma 6.1, applied to the morphism $\overline{X} \to \mathbb{A}^4$ and $\widehat{P}$, there exist a projective birational morphism $W \to \mathbb{A}^4$ from a smooth variety W and a prime divisor G on W such that the induced rational map $\overline{X} \dashrightarrow W$ maps $\widehat{P}$ onto G. By Lemma 6.1, the pushforward of the valuation defined by $\widehat{P}$ is a positive integer multiple of the valuation defined by G. Since $\widehat{w}$ is a positive multiple of the valuation defined by $\widehat{P}$, there exists a real number $\mu > 0$ such that

$$\bar{v}(\phi) = \mu \operatorname{mult}_G \operatorname{div}(\phi)$$

for any non-zero rational function ϕ on $\mathbb{A}^4$, and $\operatorname{center}_W(\bar{v}) = G$. Let $m := \operatorname{mult}_G \operatorname{div}(F_C)$. Then $m \geq 0$, as F_C is a regular function on $\mathbb{A}^4$.

Choose an affine open subset $W_1 \subset W$ intersecting G, a regular function t on W_1, and a nowhere vanishing regular function θ on W_1, such that $G \cap W_1$ is smooth, $\operatorname{div}(t) = G \cap W_1$, and $F_C = t^m\theta$ on W_1. Let

$$W_0 := \{(y, s_0, \theta_0) \in W_1 \times \mathbb{A}^2 \mid s_0^N = t(y),\ \theta_0^N = \theta(y)\}.$$

Since t is part of a regular system of parameters at any point of $G \cap W_1$, the subvariety of $W_1 \times \mathbb{A}^1$ defined by $s_0^N = t(y)$ is smooth, and the divisor of s_0 on it is isomorphic to $G \cap W_1$. Since $N\theta_0^{N-1}$ is nowhere vanishing on W_0, the projection from W_0 to $\{s_0^N = t\}$ is étale. Thus W_0 is smooth, the divisor of s_0 on W_0 is smooth, and the projection $W_0 \to W_1$ is finite and surjective. Let W' be a connected component of W_0, let $f_1 : W' \to W_1$ be the projection, and let $f : W' \to \mathbb{A}^4$ be the composite of f_1 and $W \to \mathbb{A}^4$. Then $f_1^*t = s_0^N$, $f^*F_C = s_0^{Nm}\theta_0^N$, and θ_0 is nowhere vanishing on W'.

Let

$$V_N := \{Y_0^N + Y_1^N + \cdots + Y_{N-1}^N = 0\} \subset \mathbb{P}^{N-1} \quad \text{and} \quad V_N^0 := V_N \cap \{Y_0 = 0\}.$$

Since $N \geq 3$, V_N is a smooth hypersurface of degree N in $\mathbb{P}^{N-1}$, V_N^0 is a smooth hypersurface of degree N in $\{Y_0 = 0\} \cong \mathbb{P}^{N-2}$, and $K_{V_N} = 0$ by adjunction. Let

$$\psi : W' \times V_N \dashrightarrow \overline{X}$$

be the rational map given by f on x_0, x_1, x_2, e and by $U_i = s_0^m\theta_0 Y_i/Y_0$ for $1 \leq i \leq N-1$. It is well defined, since $F_C + \sum_i U_i^N$ pulls back to $s_0^{Nm}\theta_0^N Y_0^{-N} \sum_{i=0}^{N-1} Y_i^N = 0$. Since f is generically finite and $Y_i/Y_0 = U_i/(s_0^m\theta_0)$ where $s_0 \neq 0$, ψ is generically finite. As $\dim(W' \times V_N) = N + 2 = \dim \overline{X}$ and $\overline{X}$ is irreducible, ψ is dominant. We regard rational functions on W' as rational functions on $W' \times V_N$ via the projection.

Let $Y \to \overline{X}$ be a projective birational morphism from a normal variety Y on which $\widehat{P}$ is a prime divisor, and let $\nu : Y' \to Y$ be the normalization of Y in the function field of $W' \times V_N$, which contains the function field of Y via ψ. Then ν is finite and surjective (the normalization in a finite extension of the function field is finite [Har77, Theorem I.3.9A] and surjective by lying over), and Y' is birational to $W' \times V_N$. Choose a prime divisor P' on Y' with $\nu(P') = \widehat{P}$. For

any non-zero rational function ϕ on Y, $\operatorname{mult}_{P'}\operatorname{div}(\nu^*\phi)$ is the multiplicity of $\nu^*\widehat{P}$ along P' times $\operatorname{mult}_{\widehat{P}}\operatorname{div}(\phi)$. Thus there exists a positive multiple $\widetilde{w}$ of the valuation defined by P' such that

$$\widetilde{w}(\psi^*\phi) = \widehat{w}(\phi) \tag{6.6}$$

for any non-zero rational function ϕ on $\overline{X}$. In particular, $\widetilde{w}(f^*\phi) = \bar{v}(\phi)$ for any non-zero rational function ϕ on $\mathbb{A}^4$.

We show that $\widetilde{w}$ has a center on $W' \times V_N$. Since $\operatorname{center}_W(\bar{v}) = G$ intersects W_1, $\bar{v} \geq 0$ on the regular functions on W_1. Since $W' \to W_1$ is finite, any regular function on W' satisfies a monic equation with coefficients regular on W_1, so $\widetilde{w} \geq 0$ on the regular functions on W'. Moreover, $\widetilde{w}(s_0) = \bar{v}(t)/N = \mu/N > 0$. Thus the regular functions ϕ on W' with $\widetilde{w}(\phi) > 0$ form the ideal of an irreducible closed subset $S' \subset \{s_0 = 0\}$ of W' that maps onto $G \cap W_1$. As $W' \to W_1$ is finite, $\dim S' = \dim G$, so S' is a connected component of the smooth divisor $\{s_0 = 0\}$ of W'. Since V_N is proper, the valuative criterion of properness implies that $\widetilde{w}$ has a center on $W' \times V_N$, and $\operatorname{center}_{W' \times V_N}(\widetilde{w})$ dominates S'. Thus $\widetilde{w}$ is a divisorial valuation over $W' \times V_N$. Hence we may shrink W' to any open subset containing the generic point of S'. Removing the other connected components of $\{s_0 = 0\}$, we may assume that $\operatorname{div}(s_0) = S'$. The local ring of W' at the generic point of S' is a discrete valuation ring with uniformizer s_0, and $\widetilde{w}$ is non-negative on it, so

$$\widetilde{w}(\phi) = \frac{\mu}{N}\operatorname{mult}_{S'}\operatorname{div}(\phi) \tag{6.7}$$

for any non-zero rational function ϕ on W'. By (6.7) and $\widetilde{w}(f^*\phi) = \bar{v}(\phi)$, we have

$$\operatorname{mult}_{S'}\operatorname{div}(f^*\phi) = N\operatorname{mult}_G\operatorname{div}(\phi) \tag{6.8}$$

for any non-zero rational function ϕ on $\mathbb{A}^4$.

Step 2. In this step we define η and ζ when $\bar{v}$ is non-trivial and prove (1).

Let

$$\eta := a(\widetilde{w}, W' \times V_N, S' \times V_N) \quad \text{and} \quad \zeta := \widetilde{w}(W' \times V_N^0).$$

Since S' and V_N^0 are smooth, $S' \times V_N + W' \times V_N^0$ is a reduced simple normal crossing divisor on the smooth variety $W' \times V_N$, so $(W' \times V_N, S' \times V_N + W' \times V_N^0)$ is lc [KM98, Corollary 2.31(3)]. Thus

$$\eta - \zeta = a(\widetilde{w}, W' \times V_N, S' \times V_N + W' \times V_N^0) \geq 0.$$

Moreover, $\zeta \geq 0$, as $\widetilde{w}$ has a center on $W' \times V_N$. Hence $\eta \geq \zeta \geq 0$.

Let

$$\Omega := \frac{dx_0 \wedge dx_1 \wedge dx_2 \wedge de \wedge dU_1 \wedge \cdots \wedge dU_{N-2}}{U_{N-1}^{N-1}}.$$

Since Ω is N times the Poincaré residue of

$$\frac{dx_0 \wedge dx_1 \wedge dx_2 \wedge de \wedge dU_1 \wedge \cdots \wedge dU_{N-1}}{F_C + \sum_i U_i^N}$$

along the hypersurface $\overline{X} \subset \mathbb{A}^{N+3}$, Ω is a nowhere vanishing section of $\omega_{\overline{X}}$ by adjunction, and $\operatorname{div}(\Omega) = 0$ on $\overline{X}$. Possibly shrinking W', let $\varpi_{W'}$ be a nowhere vanishing section of $\omega_{W'}$, and let ϖ_{V_N} be the nowhere vanishing section of ω_{V_N} equal to $dy_1 \wedge \cdots \wedge dy_{N-2}/y_{N-1}^{N-1}$ on $V_N \backslash V_N^0$, where $y_i := Y_i/Y_0$. Here ϖ_{V_N} is N times the Poincaré residue along V_N of

$$\frac{\sum_i (-1)^i Y_i dY_0 \wedge \cdots \wedge \widehat{dY_i} \wedge \cdots \wedge dY_{N-1}}{\sum_i Y_i^N},$$

which is nowhere vanishing since $\deg V_N = N$, cf. adjunction. The coefficient of G in K_W minus the pullback of $K_{\mathbb{A}^4}$ is $a(G, \mathbb{A}^4, 0) - 1$, $f_1^*t = s_0^N$, θ_0 is nowhere vanishing, and dU_i pulls back to $s_0^m\theta_0 dy_i$ plus y_i times a 1-form pulled back from W'. A direct calculation shows that, possibly shrinking W',

$$\psi^*\Omega = u s_0^{Na(G,\mathbb{A}^4,0)-1-m}\varpi_{W'} \wedge \varpi_{V_N} \tag{6.9}$$

for a nowhere vanishing regular function u on W'. Let

$$\Delta' := \big(1 + m - Na(G, \mathbb{A}^4, 0)\big)(S' \times V_N).$$

By (6.9), $\operatorname{div}(\psi^*\Omega) = -\Delta'$ on $W' \times V_N$.

We compare log discrepancies under the finite morphism $\nu : Y' \to Y$ of **Step 1**. Let $\Delta_Y := -\operatorname{div}_Y(\Omega)$ and $\Delta_{Y'} := -\operatorname{div}_{Y'}(\nu^*\Omega)$, where Ω and $\nu^*\Omega$ are regarded as rational sections of ω_Y and $\omega_{Y'}$. With $K_Y := \operatorname{div}_Y(\Omega)$ and $K_{Y'} := \operatorname{div}_{Y'}(\nu^*\Omega)$, we have

$$K_Y + \Delta_Y = 0, \qquad K_{Y'} + \Delta_{Y'} = \nu^*(K_Y + \Delta_Y),$$

and, by $\operatorname{div}(\Omega) = 0$ on $\overline{X}$ and $\operatorname{div}(\psi^*\Omega) = -\Delta'$ on $W' \times V_N$,

$$a(\widehat{w}, \overline{X}, 0) = a(\widehat{w}, Y, \Delta_Y) \quad \text{and} \quad a(\widetilde{w}, W' \times V_N, \Delta') = a(\widetilde{w}, Y', \Delta_{Y'}).$$

By (6.6), $\nu_*\widetilde{w} = \widehat{w}$, so Lemma 6.3 applied to ν implies $a(\widetilde{w}, Y', \Delta_{Y'}) = a(\widehat{w}, Y, \Delta_Y)$. Thus

$$a(\widehat{w}, \overline{X}, 0) = a(\widetilde{w}, W' \times V_N, \Delta').$$

Since

$$\Delta' = S' \times V_N - \big(Na(G, \mathbb{A}^4, 0) - m\big)(S' \times V_N),$$

$\operatorname{div}(s_0) = S'$, and $\widetilde{w}(s_0) = \mu/N$, we obtain

$$a(\widetilde{w}, W' \times V_N, \Delta') = \eta + \frac{\mu}{N}\big(Na(G, \mathbb{A}^4, 0) - m\big) = \eta + a(\bar{v}, \mathbb{A}^4, 0) - \frac{c}{N},$$

because $a(\bar{v}, \mathbb{A}^4, 0) = \mu a(G, \mathbb{A}^4, 0)$ and $c = \bar{v}(F_C) = \mu m$. This proves (1).

Step 3. In this step we prove (2) when $\bar{v}$ is non-trivial. The term $-\deg_U(g)\zeta$ is the value of $\widetilde{w}$ on $-\deg_U(g)(W' \times V_N^0)$, where $W' \times V_N^0$ is the divisor of the denominator Y_0 that appears when $U_1, \dots, U_{N-1}$ are homogenized with respect to Y_0, and we prove the upper bound by [Che01, Theorem 1.3] and inversion of adjunction.

Let $0 \neq g \in R$, write $g = \sum_I a_I U_1^{I_1} \cdots U_{N-1}^{I_{N-1}}$ as in Construction-Theorem 5.2(6), and let $d := \deg_U(g)$. Let

$$k := \min\{N \operatorname{mult}_G \operatorname{div}(a_I) + m|I| \mid a_I \neq 0\}.$$

Since

$$\bar{v}(a_I) + \frac{|I|c}{N} = \frac{\mu}{N}\big(N \operatorname{mult}_G \operatorname{div}(a_I) + m|I|\big)$$

for any I with $a_I \neq 0$, we have $b(g) = k\mu/N$. For any I with $a_I \neq 0$, let

$$c_I := s_0^{-k} f^*(a_I)(s_0^m \theta_0)^{|I|}.$$

Since $f^*(a_I)$ is regular on W', $\operatorname{div}(s_0) = S'$, and (6.8) holds for $\phi = a_I$, the function c_I is regular on the normal variety W', and c_I does not vanish identically on S' if and only if $N \operatorname{mult}_G \operatorname{div}(a_I) + m|I| = k$. Let

$$s_g := \sum_{I : a_I \neq 0} c_I Y_0^{d-|I|} Y_1^{I_1} \cdots Y_{N-1}^{I_{N-1}},$$

a section of the pullback of $\mathcal{O}_{V_N}(d)$ to $W' \times V_N$, and let D be the divisor of s_g. Then $D \geq 0$ and $\psi^* g = s_0^k s_g / Y_0^d$, so

$$\operatorname{div}(\psi^* g) = kS' \times V_N + D - d(W' \times V_N^0).$$

By (6.6) and $\widetilde{w}(s_0) = \mu/N$,

$$\widehat{w}(g) = \widetilde{w}(\psi^* g) = b(g) + \widetilde{w}(D) - d\zeta. \tag{6.10}$$

Since $\widetilde{w}$ has a center on $W' \times V_N$, $\widetilde{w}(D) \geq 0$, and the lower bound in (2) follows from (6.10).

For the upper bound, it suffices to show that $\widetilde{w}(D) \leq (d/\delta_N)\eta$. If $d = 0$, then $s_g = c_I$ for $I = (0, \dots, 0)$, a regular function on W' which does not vanish identically on S', so $\widetilde{w}(D) = 0$ by (6.7). Suppose that $d > 0$. Let $q \in S'$ be a closed point at which $c_I(q) \neq 0$ for some I with $N \operatorname{mult}_G \operatorname{div}(a_I) + m|I| = k$. The restriction of s_g to $\{q\} \times V_N$ is the polynomial $\sum_I c_I(q) Y_0^{d-|I|} Y_1^{I_1} \cdots Y_{N-1}^{I_{N-1}}$. Distinct I correspond to distinct monomials, and $I_{N-1} < N$

whenever $a_I \neq 0$, so the polynomial is non-zero and of degree less than N in Y_{N-1}. Any non-zero multiple of $\sum_{i=0}^{N-1} Y_i^N$ is of degree at least N in Y_{N-1}, so the restriction of s_g to $\{q\} \times V_N$ is non-zero, and D restricts to $D_q \in |\mathcal{O}_{V_N}(d)|$ on $V_N \cong \{q\} \times V_N$. By [Che01, Theorem 1.3], which holds for $\mathbb{R}$-divisors with positive coefficients, $(V_N, \delta_N D')$ is lc for any $\mathbb{R}$-divisor $D' \geq 0$ on V_N numerically equivalent to a hyperplane section. With $D' = D_q/d$, the pair $(V_N, (\delta_N/d)D_q)$ is lc. Since $\operatorname{div}(s_0) = S'$ is smooth, there exists a regular system of parameters $z_1 = s_0, z_2, z_3, z_4$ of W' at q. The closed points of W' at which some c_I does not vanish form an open neighborhood W'_q of q, and $D|_{W'_q \times V_N}$ contains no fiber of the projection $W'_q \times V_N \to W'_q$, since the restriction of s_g to each such fiber is non-zero by the argument above. Shrinking W' to W'_q, the projection is smooth with fibers V_N and $(\delta_N/d)D \geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor supporting no fiber, so $(W' \times V_N/W', (\delta_N/d)D)$ is a family of log varieties in the sense of [Amb16, Section 2.3] by [Amb16, Lemma 2.5]. By [Amb16, Lemma 2.7], applied to the projection $W' \times V_N \to W'$, the pair

$$\Big(W' \times V_N, \frac{\delta_N}{d} D + \sum_{i=1}^{4} \operatorname{div}(z_i) \times V_N\Big)$$

is lc near $\{q\} \times V_N$. Removing the components $\operatorname{div}(z_i) \times V_N$ for $i \geq 2$, the pair

$$\Big(W' \times V_N,\ S' \times V_N + \frac{\delta_N}{d} D\Big)$$

is lc near $\{q\} \times V_N$.

The closed points $q \in S'$ at which some c_I with $N \operatorname{mult}_G \operatorname{div}(a_I) + m|I| = k$ does not vanish form a dense open subset S'_0 of S'. The non-lc locus $\Sigma \subset W' \times V_N$ of the pair $(W' \times V_N, S' \times V_N + (\delta_N/d)D)$ is closed, being the image of finitely many divisors with negative log discrepancy on a log resolution. The image of Σ in W' is closed, as V_N is proper, and is disjoint from S'_0 by the previous paragraph, so it does not contain S'. Since $\operatorname{center}_{W' \times V_N}(\widetilde{w})$ dominates S', the pair is lc near the generic point of $\operatorname{center}_{W' \times V_N}(\widetilde{w})$. Thus

$$0 \leq a\Big(\widetilde{w}, W' \times V_N, S' \times V_N + \frac{\delta_N}{d} D\Big) = \eta - \frac{\delta_N}{d} \widetilde{w}(D),$$

so $\widetilde{w}(D) \leq (d/\delta_N)\eta$. By (6.10), $\widehat{w}(g) \leq b(g) + d(\eta/\delta_N - \zeta)$, which proves (2).

Step 4. In this step we treat the case when $\bar{v}$ is trivial.

We only indicate the differences from **Steps 1–3**. Let

$$W_0 := \{(y, \theta_0) \in (\mathbb{A}^4 \backslash \{F_C = 0\}) \times \mathbb{A}^1 \mid \theta_0^N = F_C(y)\},$$

which is smooth and étale over $\mathbb{A}^4 \backslash \{F_C = 0\}$, let W' be a connected component of W_0, let $f : W' \to \mathbb{A}^4$ be the projection, and let $\psi : W' \times V_N \dashrightarrow \overline{X}$ be given by f and $U_i = \theta_0 Y_i / Y_0$. The functions t and s_0 do not occur here. As in **Step 1**, $\widehat{w}$ extends to a divisorial valuation $\widetilde{w}$ satisfying (6.6). Since $\bar{v}$ is trivial and $W' \to \mathbb{A}^4 \backslash \{F_C = 0\}$ is finite, $\widetilde{w}$ is trivial on the non-zero rational functions on W', and $\operatorname{center}_{W' \times V_N}(\widetilde{w})$ dominates W'. Let

$$\eta := a(\widetilde{w}, W' \times V_N, 0) \quad \text{and} \quad \zeta := \widetilde{w}(W' \times V_N^0).$$

Since $(W' \times V_N, W' \times V_N^0)$ is lc, $\eta \geq \zeta \geq 0$. As f is étale and θ_0 is nowhere vanishing, $\psi^*\Omega$ is a nowhere vanishing section of $\omega_{W' \times V_N}$, so $a(\widehat{w}, \overline{X}, 0) = \eta$ by Lemma 6.3, applied as in **Step 2**. This is (1), since $a(\bar{v}, \mathbb{A}^4, 0) = 0$ and $c = 0$. For (2), $b(g) = 0$. Let

$$s_g := \sum_I f^*(a_I) \theta_0^{|I|} Y_0^{d-|I|} Y_1^{I_1} \cdots Y_{N-1}^{I_{N-1}},$$

and let D be the divisor of s_g. Then $\widehat{w}(g) = \widetilde{w}(D) - d\zeta$. If $d = 0$, then $\widetilde{w}(D) = 0$. If $d > 0$, then for a general closed point $q \in W'$, the restriction of s_g to $\{q\} \times V_N$ is non-zero, and the argument of **Step 3**, with a regular system of parameters $z_1, \ldots, z_4$ of W' at q and all the components $\operatorname{div}(z_i) \times V_N$ removed, shows that $(W' \times V_N, (\delta_N/d)D)$ is lc near the generic point of $\operatorname{center}_{W' \times V_N}(\widetilde{w})$. Thus $\widetilde{w}(D) \leq (d/\delta_N)\eta$, and (2) follows. □

6.3. **The estimate.** In this subsection, we prove Proposition 6.7, which bounds $T_{-K_X}(P)$ for any prime divisor P over X. Part (d) of the sketch of the proof in Section 1 explains why the pushforward to Z of the valuation defined by P does not suffice and why the slope is γ/δ_N.

Proposition 6.7. *Notation as in Construction-Theorem 5.2. Let P be a prime divisor over X. Then one of the following holds.*

(1) The image of P under ρ is dense in Z, and $T_{-K_X}(P) \le (\gamma/\delta_N)a(P,X,0)$.

(2) There exist a projective birational morphism $Z' \to Z$ from a smooth variety Z', a prime divisor F on Z', and a positive integer k such that the induced rational map $X \dashrightarrow Z'$ maps P onto F, the pullback of F has multiplicity k along P, $a(P,X,0) \ge ka(F,Z,B)$, and

$$T_{-K_X}(P) \le kT_L(F) + \frac{\gamma}{\delta_N}\big(a(P,X,0) - ka(F,Z,B)\big).$$

Proof. By Lemma 6.4, there exists a $\mathbb{T}$-invariant divisorial valuation $\widehat{w}$ over $\overline{X}$ with center on $\widehat{X}$ such that $h_*\widehat{w}$ is the valuation defined by P, $a(\widehat{w},\overline{X},0) = a(P,X,0)$, and $\widehat{w}(g) = \operatorname{mult}_P D_g$ for any positive integer j divisible by j_0 and any $0 \ne g \in R_{jL}$. Let $\bar{v}$ be the pushforward of $\widehat{w}$ under $\overline{X} \to \mathbb{A}^4$, let $c := \bar{v}(F_C)$, and let $\eta \ge \zeta \ge 0$ be real numbers satisfying (1) and (2) of Lemma 6.6 for $\widehat{w}$. Let v be the pushforward under ρ of the valuation defined by P. Since $\rho \circ h$ coincides with the composite of $\widehat{X} \to \mathbb{A}^4$ and the rational map $\mathbb{A}^4 \dashrightarrow Z$ of Lemma 2.10(3) on a dense open subset of $\widehat{X}$, and pushforwards compose, v is the pushforward of $\bar{v}$ under $\mathbb{A}^4 \dashrightarrow Z$. If v is trivial, then we let $a(v,Z,B) := 0$ and $T_L(v) := 0$.

We show that $\bar{v}$ is $\mathbb{T}_0$-invariant (Definition 6.2). Since $\operatorname{Pic}(Z) \subset \Lambda$ and $\mathbb{C}^*$ is divisible, the restriction

$$\mathbb{T} = \operatorname{Hom}(\Lambda,\mathbb{C}^*) \to \mathbb{T}_0 = \operatorname{Hom}(\operatorname{Pic}(Z),\mathbb{C}^*)$$

is surjective, and $\mathcal{S} \subset R$ is equivariant with respect to it. For $t_0 \in \mathbb{T}_0$, choose $t \in \mathbb{T}$ mapping to t_0. Then

$$\bar{v}(t_0^*\phi) = \widehat{w}(t^*\phi) = \widehat{w}(\phi) = \bar{v}(\phi)$$

for any non-zero rational function ϕ on $\mathbb{A}^4$.

Let $\lambda : \operatorname{Pic}(Z)_{\mathbb{Q}} \to \mathbb{R}$ be the additive function of Lemma 6.5(1) for $\bar{v}$, and let $\sigma := \lambda(L)$. By Lemma 6.1, applied to $\overline{X} \to \mathbb{A}^4$ and a prime divisor over $\overline{X}$ whose valuation is a positive multiple of $\widehat{w}$, $\bar{v}$ is either trivial or divisorial. If $\bar{v}$ is divisorial, then Lemma 6.5(2) implies $a(\bar{v},\mathbb{A}^4,0) = a(v,Z,0) + \lambda(-K_Z)$. If $\bar{v}$ is trivial, then v is trivial and $\lambda = 0$ by the uniqueness in Lemma 6.5(1), so $a(\bar{v},\mathbb{A}^4,0) = a(v,Z,0) + \lambda(-K_Z)$ holds with both sides zero. Since $F_C \in \mathcal{S}_{[C]}$ and the divisor of F_C is C, Lemma 6.5(1) implies $c = v(C) + \lambda([C])$. As $a(v,Z,B) = a(v,Z,0) - v(C)/N$ (Lemma 2.3) and $\lambda(L) = \lambda(-K_Z) - \lambda([C])/N$, Lemma 6.6(1) and Lemma 6.4 imply

$$a(P,X,0) = a(v,Z,B) + \sigma + \eta. \tag{6.11}$$

Let j be a positive integer divisible by j_0, let $0 \ne g \in R_{jL}$, and let a_I be the elements of Construction-Theorem 5.2(6) for g. By Construction-Theorem 5.2(6), $a_I \in \mathcal{S}_{jL-|I|\tau}$ for any I. If $a_I \ne 0$, then $jL - |I|\tau \in \operatorname{Pic}(Z)$, and $\operatorname{div}_Z(a_I) + (|I|/N)C \ge 0$ is a $\mathbb{Q}$-divisor $\mathbb{Q}$-linearly equivalent to jL. Since $\lambda(\tau) = \lambda([C])/N$, Lemma 6.5(1) and $c = v(C) + \lambda([C])$ imply

$$\bar{v}(a_I) + |I|c/N = j\sigma + v\big(\operatorname{div}_Z(a_I) + (|I|/N)C\big),$$

and $0 \le v(\operatorname{div}_Z(a_I) + (|I|/N)C) \le jT_L(v)$. Thus, with $b(g) := \min\{\bar{v}(a_I) + |I|c/N \mid a_I \ne 0\}$,

$$j\sigma \le b(g) \le j\big(\sigma + T_L(v)\big). \tag{6.12}$$

We show that $\deg_U(g) \le j\gamma$. By Lemma 2.10(1), $L = -K_Z - B$, and $\tau = [C]/N$, we have $\tau \equiv (3-\ell_1)H - (1-\ell_2)E$. If $a_I \ne 0$, then $\operatorname{div}_Z(a_I) \ge 0$ is a divisor in the class $jL - |I|\tau$, and H is nef, so

$$0 \le \big((jL - |I|\tau)\cdot H\big) = j\ell_1 - |I|(3-\ell_1).$$

Moreover, $3-\ell_1=(C\cdot H)/N>0$, as noted after Set-up 5.1. Thus $|I|\le j\ell_1/(3-\ell_1)=j\gamma$ whenever $a_I\neq 0$, so

$$\deg_U(g)\le j\gamma. \tag{6.13}$$

We show that $0\le\sigma\le\gamma\zeta$. Choose a positive integer j divisible by j_0 such that $\mathcal{O}_Z(jL)$ and $\mathcal{O}_X(-jK_X)$ are very ample. Choose $s_1\in\mathcal{S}_{jL}=H^0(Z,\mathcal{O}_Z(jL))$ such that $\operatorname{div}_Z(s_1)$ does not contain $\operatorname{center}_Z v$. Then $v(\operatorname{div}_Z(s_1))=0$, so Lemma 6.5(1) implies $\bar v(s_1)=j\sigma$. Since $s_1\in R$ and $\widehat{w}$ has a center on $\overline{X}$, $j\sigma=\bar v(s_1)=\widehat{w}(s_1)\ge 0$, so $\sigma\ge 0$. Choose $0\neq g_0\in R_{jL}$ such that D_{g_0} does not contain $\operatorname{center}_X P$. Then $\widehat{w}(g_0)=\operatorname{mult}_P D_{g_0}=0$. By Lemma 6.6(2), (6.12), (6.13), and $\zeta\ge 0$,

$$0=\widehat{w}(g_0)\ge b(g_0)-\deg_U(g_0)\zeta\ge j\sigma-j\gamma\zeta,$$

so $\sigma\le\gamma\zeta$.

Now let j be a positive integer divisible by j_0 and $0\neq g\in R_{jL}$. Since $0<\delta_N<1$ and $\eta\ge\zeta\ge 0$, we have $\eta/\delta_N-\zeta\ge 0$. By Lemma 6.6(2), (6.12), (6.13), and $\sigma\le\gamma\zeta$,

$$\operatorname{mult}_P D_g=\widehat{w}(g)\le b(g)+\deg_U(g)\Big(\frac{\eta}{\delta_N}-\zeta\Big)\le j\big(\sigma+T_L(v)\big)+j\gamma\Big(\frac{\eta}{\delta_N}-\zeta\Big)\le jT_L(v)+\frac{j\gamma}{\delta_N}\eta.$$

Let $D\ge 0$ be an $\mathbb{R}$-divisor on X with $D\sim_{\mathbb{Q}}-K_X$. Then D is a $\mathbb{Q}$-divisor, and $jD\sim -jK_X$ for some positive integer j divisible by j_0, so $jD=D_g$ for some $0\neq g\in R_{jL}$ by Construction-Theorem 5.2(4). Thus $\operatorname{mult}_P D\le T_L(v)+(\gamma/\delta_N)\eta$, and

$$T_{-K_X}(P)\le T_L(v)+\frac{\gamma}{\delta_N}\eta.$$

By (6.11) and $\sigma\ge 0$, we have $0\le\eta\le a(P,X,0)-a(v,Z,B)$. Hence $a(P,X,0)\ge a(v,Z,B)$ and

$$T_{-K_X}(P)\le T_L(v)+\frac{\gamma}{\delta_N}\big(a(P,X,0)-a(v,Z,B)\big). \tag{6.14}$$

By Lemma 6.1, applied to ρ and P, either the image of P under ρ is dense in Z and v is trivial, in which case (6.14) is (1), or there exist a projective birational morphism $Z'\to Z$ from a smooth variety Z' and a prime divisor F on Z' such that the induced rational map $X\dashrightarrow Z'$ maps P onto F and $v=kv_F$, where v_F is the valuation defined by F and k is the multiplicity of the pullback of F along P, a positive integer. In the second case, $a(v,Z,B)=ka(F,Z,B)$ and $T_L(v)=kT_L(F)$, and (6.14) implies (2). □

7. Proofs of the main theorems

In this section, we prove Theorem 1.3 and Corollary 1.4 by substituting the pair of Section 4 into Sections 5 and 6, as outlined in the sketch of the proof in Section 1.

Theorem 7.1. *Notation as in Construction-Proposition 4.4, and let X and $\rho: X\dashrightarrow Z$ be as in Construction-Theorem 5.2 for an element $F_C\in\mathcal{S}_{[C]}$ whose divisor is C. Then X is a klt Fano variety of dimension $N=10^8n$, $\alpha(X)=\alpha_0>1$, and $a(P,X,0)>\alpha_0T_{-K_X}(P)$ for any prime divisor P over X. Moreover, there exists a quasi-monomial valuation w over X of rational rank 2 such that $\rho_*w=w_{r^2-1}$ and w computes $\alpha(X)$.*

Proof. **Step 1.** In this step we check the conditions of Set-up 5.1 and show that X is a klt Fano variety of dimension N.

Since $n\ge 1$, we have $N=10^8n\ge 3$. By Construction-Proposition 4.4(1)(2), C is a reduced divisor and $C\sim 299982354nH-99994565nE$, so $C\neq 0$. By Construction-Proposition 4.4(4), (Z,B) is klt. By Construction-Proposition 4.4(2), $-(K_Z+B)\sim_{\mathbb{Q}}\ell_1H-\ell_2E$, where

$$\ell_1=\frac{17646}{10^8}>3\ell_2=\frac{16305}{10^8}>0.$$

Thus the $\mathbb{Q}$-divisor $L=-(K_Z+B)$ of Set-up 5.1 and the $\mathbb{Q}$-divisor $L=\ell_1H-\ell_2E$ of Construction-Proposition 4.4 are $\mathbb{Q}$-linearly equivalent. Since T_L (Definition 2.4) and R_{jL} depend only on the

class of L in $\mathrm{Pic}(Z)_{\mathbb{Q}}$, we do not distinguish them. Together with the choice of F_C, the conditions of Set-up 5.1 hold. By Construction-Theorem 5.2(3), X is a klt Fano variety of dimension N.

Step 2. In this step we show that $a(P,X,0) > \alpha_0 T_{-K_X}(P)$ for any prime divisor P over X, and that $\alpha(X) \geq \alpha_0$.

Recall that $\gamma = \ell_1/(3-\ell_1)$ and $\delta_N = (N-2)/N$ (Set-up 5.1). Since $3-\ell_1 > 0$, Lemma 4.6(4) implies that $\alpha_0\gamma = \ell_1\alpha_0/(3-\ell_1) < 1/4$. Since $N \geq 3$, we have $\delta_N \geq 1/3$. Hence

$$\frac{\alpha_0\gamma}{\delta_N} < \frac{3}{4} < 1.$$

Let P be a prime divisor over X. By Proposition 6.7, one of the following two cases occurs.

Suppose that Proposition 6.7(1) holds for P. Since X is klt, $a(P,X,0) > 0$. Thus

$$a(P,X,0) - \alpha_0 T_{-K_X}(P) \geq \Big(1 - \frac{\alpha_0\gamma}{\delta_N}\Big) a(P,X,0) > 0.$$

Suppose that Proposition 6.7(2) holds for P, and let $Z' \to Z$, F, and k be as there. Then F is a prime divisor over Z, so $a(F,Z,B) - \alpha_0 T_L(F) > 0$ by Proposition 4.7(1). Since $k > 0$ and $a(P,X,0) - ka(F,Z,B) \geq 0$, we have

$$a(P,X,0) - \alpha_0 T_{-K_X}(P) \geq k\big(a(F,Z,B) - \alpha_0 T_L(F)\big) + \Big(1 - \frac{\alpha_0\gamma}{\delta_N}\Big)\big(a(P,X,0) - ka(F,Z,B)\big) > 0.$$

Thus $a(P,X,0) > \alpha_0 T_{-K_X}(P)$ in both cases. Since $-K_X$ is ample, for a sufficiently divisible positive integer j, some member of $|-jK_X|$ contains the center of P on X, hence $T_{-K_X}(P) > 0$. Therefore $a(P,X,0)/T_{-K_X}(P) > \alpha_0$ for any prime divisor P over X, and $\alpha(X) \geq \alpha_0$ by Lemma 2.5(1).

Step 3. In this step we construct w and show that $\alpha(X) \leq \alpha_0$.

Let $v := w_{r^2-1}$. By Proposition 4.7(2), v is a quasi-monomial valuation over Z of rational rank 2, and $a(v,Z,B) = \alpha_0 T_L(v)$. In particular, v is non-trivial. Since (Z,B) is klt (Construction-Proposition 4.4(4)), $a(v,Z,B) > 0$ by Definition 2.2. By Proposition 5.3, there exists a quasi-monomial valuation w over X of rational rank 2 such that $\rho_* w = v$,

$$a(w,X,0) = a(v,Z,B) \quad \text{and} \quad T_{-K_X}(w) = T_L(v).$$

Hence

$$T_{-K_X}(w) = T_L(v) = \frac{a(v,Z,B)}{\alpha_0} > 0.$$

By Lemma 2.5(2),

$$\alpha(X) T_{-K_X}(w) \leq a(w,X,0) = a(v,Z,B) = \alpha_0 T_{-K_X}(w).$$

Dividing by $T_{-K_X}(w)$, we obtain $\alpha(X) \leq \alpha_0$.

By **Steps 2 and 3**, $\alpha(X) = \alpha_0$, and $\alpha_0 > 1$ by Lemma 4.6(4). Since $\rho_* w = v$ is non-trivial, w is non-trivial. Since $a(w,X,0) = \alpha_0 T_{-K_X}(w) = \alpha(X) T_{-K_X}(w)$, w computes $\alpha(X)$. □

Proof of Theorem 1.3. Let n_0 be the positive integer of Construction-Proposition 4.4, let $n \geq n_0$ be an integer, and let C be the divisor of Construction-Proposition 4.4. Since $C \geq 0$ is a divisor on Z and $\mathcal{S}_{[C]} = H^0(Z, \mathcal{O}_Z(C))$ (Lemma 2.10(3)), the section of $\mathcal{O}_Z(C)$ defining C is an element $F_C \in \mathcal{S}_{[C]}$ whose divisor is C. Let X and $\rho : X \dashrightarrow Z$ be as in Construction-Theorem 5.2 for F_C. By Theorem 7.1, X is a klt Fano variety of dimension $10^8 n$ and $\alpha(X) = \alpha_0 > 1$. By Lemma 4.6(3), α_0 is irrational. This proves (1).

If a prime divisor P over X computes $\alpha(X)$, then $a(P,X,0) = \alpha_0 T_{-K_X}(P)$, which contradicts Theorem 7.1. This proves (2). By Theorem 7.1, there exists a quasi-monomial valuation over X of rational rank 2 which computes $\alpha(X)$. This proves (3). □

Proof of Corollary 1.4. Let $D \geq 0$ be an $\mathbb{R}$-divisor on X such that $D \sim_{\mathbb{R}} -K_X$. Since $-K_X$ is ample, $D \neq 0$. Let $f : Y \to X$ be a log resolution of (X,D) with Y projective. For a real number $t \geq 0$, write $K_Y + \Delta_t := f^*(K_X + tD)$. Then the coefficient of a prime divisor E on

Y in Δ_t is $1 - a(E, X, 0) + t \operatorname{mult}_E D$, and $\operatorname{Supp} \Delta_t$ is snc. Hence (X, tD) is lc if and only if $a(E, X, 0) - t \operatorname{mult}_E D \geq 0$ for any prime divisor E on Y: the "only if" part follows from the definition of lc pairs. For the "if" part, assume that each coefficient of Δ_t is at most 1, and let D^+ be the sum of the prime divisors on Y whose coefficients in Δ_t are positive. Then D^+ is a reduced simple normal crossing divisor, so (Y, D^+) is lc [KM98, Corollary 2.31(3)]. Let F be a prime divisor over X, and let $g : Y' \to Y$ be a log resolution of $(Y, \operatorname{Supp}(f_*^{-1}D) \cup \operatorname{Exc}(f))$ on which F is a prime divisor, so that $f \circ g$ is a log resolution of (X, tD). As $\operatorname{Supp} D^+ \subset \operatorname{Supp} \Delta_t$ and $\operatorname{Supp} \Delta_t \subset \operatorname{Supp}(f_*^{-1}D) \cup \operatorname{Exc}(f)$, g is also a log resolution of (Y, Δ_t) and of (Y, D^+). Since Y is smooth, $g^*(K_Y + \Delta_t) = g^*K_Y + g^*\Delta_t$, and since the coefficients of Δ_t are at most 1, $\operatorname{mult}_F g^*\Delta_t \leq \operatorname{mult}_F g^*D^+$. Hence, by Definition 2.1 and $K_Y + \Delta_t = f^*(K_X + tD)$,

$$a(F, X, tD) = a(F, Y, \Delta_t) = a(F, Y, 0) - \operatorname{mult}_F g^*\Delta_t \geq a(F, Y, 0) - \operatorname{mult}_F g^*D^+ = a(F, Y, D^+) \geq 0,$$

so (X, tD) is lc. Since X is klt, we have

$$\operatorname{lct}(X; D) = \min\left\{ \frac{a(E, X, 0)}{\operatorname{mult}_E D} \;\middle|\; E \text{ is a prime divisor on } Y \text{ and } \operatorname{mult}_E D > 0 \right\},$$

where the set on the right is finite and non-empty because $D \neq 0$. Let E be a prime divisor on Y such that $\operatorname{mult}_E D > 0$ and $\operatorname{lct}(X; D) = a(E, X, 0)/\operatorname{mult}_E D$. Since Y is smooth, Lemma 2.5(3) implies that $0 < \operatorname{mult}_E D \leq T_{-K_X}(E)$. By Lemma 2.5(1), $a(E, X, 0) \geq \alpha(X) T_{-K_X}(E)$. Since E does not compute $\alpha(X)$, $a(E, X, 0) > \alpha(X) T_{-K_X}(E)$. Therefore

$$\operatorname{lct}(X; D) = \frac{a(E, X, 0)}{\operatorname{mult}_E D} \geq \frac{a(E, X, 0)}{T_{-K_X}(E)} > \alpha(X).$$

As D is arbitrary, this proves Corollary 1.4. □

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